Class 7
MATHEMATICS

Table of Content

1.	Number System	1 – 27
2.	Exponents and Powers	28 – 30
3.	Surds	31 – 37
4.	Logarithms	38 – 42
5.	Polynomials	43 – 59
6.	Linear Equations	60 – 82

1. NUMBER SYSTEM

Worksheet – 1

Level – I

1. Answer) C
2. $\{22,23,24,25,26,27,28,29,30,31\} = 10$

   Answer) **B**
3. Answer) **D**
4. Multiplication

   Answer) **B**
5. Not defined

   Answer) **D**
6. Odd

   Answer) **A**
7. $2n$

   Answer) **B**
8. $|x| = -x$

   Answer) **B**
9. Number of prime numbers between 1 to 100 are

   $\{2,3,5,7,11,13,17,19,23,29,31,37,41,43,47,53,59,61,67,71,73,79,83,89,97\}$

   Answer) **A**
10. (5,7)

    Answer) **A**

Level – II

1. Largest two digit prime number = 97

   Answer) **C**
2. Least 2 digit composite number = 10

   Answer) **C**
3. predecessor of a prime is odd or even predecessor of 2 = 1

   Answer) **C**
4. Successor of $a + 3 = 5$

   $\therefore a + 3 = 4$

   $a = 1$

   Answer) **A**

5. Sum of first ten odd numbers

$= 1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 = (10)^2 = 100$

Answer) **C**

6. 10,24,25 Because $(10)^2 + (24)^2 \neq (25)^2$

Answer) **D**

7. $(100)^2 - (99)^2 = (100 + 99)(100 - 99) = (100 + 99)(1) = 199.$

Answer) **D**

8. conceptual

Answer) **A**

9. (9,11)

Answer) **C**

10. $(a, b) = (3,5)$

$(b, c) = (5,7)(b, c) = (5,7)$

$\therefore a + c = 3 + 7 = 10$

Answer) **A**

11. Number of natural no's from a to b $= b - a + 1$

Answer) **D**

12. Successor of $n = n + 1$

Predecessor of $n = n - 1$

Difference $= (n + 1) - (n - 1) = n + 1 - n + 1 = 2$

Answer) **A**

13. Successor of 9 + predecessor of $9 = 10 + 8 = 18$

Answer) **D**

14. As $m, p < n \Rightarrow n^2 = m^2 + p^2$

Answer) **B**

15. Product of first 50 odd natural numbers $= 1 \times 3 \times 5 \times \ldots.99$ Always ends with 5 because product of odd number with 5 always ends with 5 .

Answer) **B**

16. $2 \times 4 \times 6 \times 8 \times \ldots\ldots\times 70 \Rightarrow$ unitplace $= 0$

Multiplying any number with 10 gives unit place ' 0 ')

Answer) **C**

17. $a - b$ and $a + b$ are twin primes

$(a + b) - (a - b) = 2 \Rightarrow a + b - a + b = 2$

$2b = 2 \Rightarrow b = 1$

$a-1$ and $a+1$

$a-1$ and $a+1$ should be odd numbers

∴ a should be an even number ∴ a,b are coprimes

Answer) **B**

18. Odd composite numbers are 9,15,21,25,27,33....

Least number that can be written as sum of two odd composite number is

$9+15=24$

Answer) **C**

19. a and b are twin primes and a > b

$\therefore a-b=2 \Rightarrow a-2=b \Rightarrow a=b+2$

$a^3-b^3=(a-b)(a^2+ab+b^2)=(a-b)[(a-b)^2+3ab]$

$=2[(2)^2+3(b+2)(b)]$

$=2[4+3b^2+6b]=6b^2+12b+8$

Answer) **D**

20. a and b are twin primes

$a-b=2$

Given

$a^2-b^2=120$

$(a+b)(a-b)=120$

$(a+b)(2)=120$

$a+b=60---(2)$

Avg of $a,b=\frac{a+b}{2}=\frac{60}{2}=30$

Answer) **B**

21. 21. $n^2+9n+20=(n+4)(n+5)$

$n+4, n+5$ are prime numbers

$n+4=2, n+5=3 \Rightarrow n=-2$

$\therefore n^2+1=(-2)^2+1=4+1=5$

Answer) **C**

22. 22. a^2-b^2 is prime $a^2-b^2=(a-b)(a+b)$ any prime number can be written as product of 1 and the number it self

$\therefore a-b=1 \ \therefore a^2-b^2=(1)(a+b)=a+b$

Answer) **B**

23. . 23. $3n-4, 4n-5, 5n-3$ are primes.

$(3n-4)+5n-3=8n-7$

$8n-7$ is necessarily odd that is one of the numbers $3n-4,5n-3$ should be even only even prime is $2.3n-4=2$

$3n=6;\ n=2$

$4n-5=4(2)-5=8-5=3$

$5n-3=5(2)-3=10-3=7$

$\therefore$ only for $n=2$

$3n-4,4n-5,5n-3$ are prime numbers

Answer) **A**

24. Sum of the three numbers = 100

Let the primes be a, b, c $a+b+c=100$

Diff of two primes = 36

$b-c=36---(2)\Rightarrow b=36+c$ From (1) & (2)

$a+(3b+c)+c=100$

$a+2c=64\Rightarrow 2c=64-a$

$c=\frac{64-a}{2}$

If $a=2, c=\frac{64-2}{2}=31$

From (1) $a+b+c=100\Rightarrow b=67$

Answer) **D**

Level – III

25. Every 2 prime numbers have 1 as common factor and any two consecutive numbers have 1 as common factor.

Answer) **A, D**

26. Smallest odd prime = 3.

Answer) **B**

27. $2n+1$ is an odd number smallest odd number which is not a prime is 9 .

$\therefore 2n+1=9$

$2n=8$

$n=4$

Answer) **B**

28. 373 is a prime number 29.conceptual

Answer) **C**

29. a-p,s,t; b-q,r; c-p,s,t; d-p,t

30.

31. $30.\ 54 = 2^1 \times 3^3$

No. of prime factors = 2

Answer) **2**

31. $n^3 - 1 = (n-1)(n^2 + n + 1)$

$n - 1 = 1 \Rightarrow n = 2$

$n^2 + n + 1 = (2)^2 + 2 + 1 = 4 + 2 + 1 = 7$

$\therefore n^2 + n + 1$ is a prime

Answer) **A, B, C**

32. a-q; b-q; c-q; d-s

32. For no value of P belongs to prime numbers, $p + 2, p + 5$ are prime

Answer) **0**

33. Prime lying between 60 and 75 are 61,67,71,73

Answer) **D**

Worksheet – 2

Level – I

1. 3 2. 1 3. 2 4. 1 5. 2 6. 3 7. 2
8. 3 9. 1 10. 1

Level – II

1. $8400 = 2^4 \times 3^1 \times 5^2 \times 7^1$

Number of divisors $= (4+1)(1+1)(2+1)(1+1) = (5)(2)(3)(2) = 60$

Number of divisors except $1 = 60 - 1 = 59$.

Answer) **B**

2. $N^{\frac{1}{2}d(N)}$

Answer) **A**

3. $484 = 2^2 \times 11^2 \Rightarrow d(N) = (a+1)(b+1)$

Here 11 is odd prime factor

So, no.of odd factors $= (2 + 1) = 3$.

Answer) **A**

4. If $p = 2$ then $p^3 = 8$

Factors of 8 – 1,2,4,8

Number of factors = 4

Answer) **D**

5. $144 = 12 \times 12 = 2^2 \times 3 \times 2^2 \times 3 = 2^4 \times 3^2$

Total number of factors $= (4+1)(2+1) = 15$

Number of odd factors $= (2 + 1) = 3$

Number of even factors $= (15 - 3) = 12$

Answer) **B**

6. 2011 is prime ⇒ Factors of 2011 are 1,2011

Other than 2011 factors are 1

Sum = 1

Answer) **B**

7. LCm of fractions = LCM of numerators /HCF of denominators

$$\frac{\text{LCM of } (1,5,2,4)}{\text{HCF of } (3,6,9,27)} = \frac{20}{3}$$

Answer) **C**

8. $2923 = 37 \times 79; 3239 = 41 \times 79; \text{HCF} = 79.$

Answer) **D**

9. $36 = 2^2 \times 3^2$

No.of factors $= (2 + 1)(2 + 1) = 9$

number of ways of expressing 36 as a product of two factors $= \frac{9+1}{2} = 5$

Answer) **B**

10. LCM of 2,4,6 = 12 and are in the ratio $1:2:3$. $11.15 \times 13, 11 \times 13 = 195,143$

Answer) **D**

11. Divisors of 16

$16 = 2^4$

Sum of divisors $= \frac{2^5-1}{2-1}$

Sum of divisors of 16 = 31

Sum of divisors other than unity $= 31 - 1 = 30$

Answer) **A**

12. Answer) **C**

13. $6^3 \times 5^3 \times 10^3 = 2^3 \times 3^3 \times 5^3 \times 2^3 \times 5^3$

$= 2^6 \times 3^3 \times 5^6$

Number of distinct factors = 3

Answer) **A**

14. $36 = 2^2 \times 3^2$

Number of factors $= (2 + 1)(2 + 1) = 9$

Product of factors of $36 = 36^{9/2} = (6^2)^{9/2} = 6^9 \therefore n = 9$

Answer) **C**

15. Factors of 21 are 1,3,7,21

Factors of 35 are 1,5,7,35

So we found three more factors 3,5,7

Let's collect all factors we get

{1,3,5,7,21,35,105}

∴ The number is 105

Answer) **B**

16. $512 = 2^9 \Rightarrow$ Number of factors $= (9 + 1) = 10$

Product of factor of $512 = (512)^{\frac{10}{2}} = (2^9)^5 = 2^{45}$

Answer) **A**

17. $1080 = 2^3 \times 3^3 \times 5$

$\therefore$ Number of factors of $\therefore 1080 = (3 + 1)(3 + 1)(1 + 1) = 4 \times 4 \times 2 = 32$

Answer) **A**

18. $270 = 2 \times 3^3 \times 5$

Sum of factors of $270 = \left(\frac{2^{1+1}-1}{2-1}\right)\left(\frac{3^{3+1}-1}{3-1}\right)\left(\frac{5^{1+1}-1}{5-1}\right) = \frac{3\times80\times24}{1\times2\times4} = 720$

Answer) **B**

19. $576 = 2^6 \times 3^2 \Rightarrow \therefore$ Total number of factors $= (6 + 1)(2 + 1) = 21$

So number of ways of expressions 576 as a product of two distinct factors $= \frac{(21-1)}{2} = 10$

Answer) **B**

20. $13^4 \times 7^5 \times 6^2 = 2^2 \times 3^2 \times 7^5 \times 13^4$

$\therefore$ number of factors $= (2 + 1)(2 + 1)(5 + 1)(4 + 1) = 3 \times 3 \times 6 \times 5 = 270$

Answer) **C**

21. $360 = 2^3 \times 3^2 \times 5^1 \Rightarrow$ Sum of divisors of $360 = \left(\frac{2^{3+1}-1}{2-1}\right)\left(\frac{3^{2+1}-1}{3-1}\right)\left(\frac{5^{1+1}-1}{5-1}\right)$

$= \frac{15}{1} \times \frac{26}{2} \times \frac{24}{4} = 15 \times 13 \times 6 = 1170$

Answer) **A**

Level – III

22. Answer) **A**

23. Answer) **A, C**

24. $16 = 2^4 \Rightarrow$ Sum of divisors $= \left(\frac{2^{4+1}-1}{2-1}\right) = \left(\frac{2^5-1}{1}\right) = 31$

Answer) **B**

Level – III

25. $26 = 2^1 \times 13^1 \Rightarrow$ Sum of divisors $26 = \left(\frac{2^{1+1}-1}{2-1}\right)\left(\frac{13^{1+1}-1}{13-1}\right) = \frac{3}{1} \times \frac{168}{12} = 42$

Answer) **B**

26. $18 = 2^1 \times 3^2 \Rightarrow$ Sum of divisor of $18 = \left(\frac{2^{1+1}-1}{2-1}\right)\left(\frac{3^{2+1}-1}{3-1}\right) = \frac{3}{1} \times \frac{26}{2} = 39$

Answer) **A**

27. a-s; b-t; c-p; d-q

28. The numbers are 12,24,36. Their HCF = 12, their sum = 72

Answer) **A, B, C**

29. a-q; b-r; c-p; d-s

32. LCM of $\left(\frac{6}{10}, \frac{96}{10}, \frac{12}{100}\right) = \frac{96}{10}$

Answer) **B**

33. HCF of $\left(\frac{5}{7}, \frac{25}{3}\right) = \frac{5}{21}$

Answer) **A**

32. LCM of $\left(\frac{2}{3}, \frac{7}{3}\right) = \frac{14}{3}$

Answer) **B**

33. $786 = 2^1 \times 3^1 \times 131^1; d(786) = (1+1)(1+1)(1+1) = 8$

Answer) **B**

34. $120 = 2^3 \times 3^1 \times 5^1$

Sum of divisors of $120 = \left(\frac{3^{1+1}-1}{3-1}\right)\left(\frac{2^{3+1}-1}{2-1}\right)\left(\frac{5^{1+1}-1}{5-1}\right) = \frac{8}{2} \times \frac{15}{1} \times \frac{24}{4} = 360$

Answer) **B**

35. $18 = 2^1 \times 3^2$

Product of divisors of $18 = 18^{\frac{(1+1)(2+1)}{2}} = 18^{\frac{1}{2} \times 2 \times 3} = 18^3$

Answer) **B**

Worksheet – 3

Level – I

1. A	2. D	3. B	4. A	5. C	6. B
7. A	8. B	9. C	10. A		

Level – II

1. Answer) **C**

2. $1! + 2! + 3! + 4! = 1 + 2 + 6 + 24 = 23$

$33 \equiv 3 (\text{mod} 10)$

Answer) **A**

3. $(n+1)! = 90(n-1)\ !$

   $(n+1)(n)(n-1)! = 90(n-1)!$

   $n(n+1) = 90$

   n = 9

   Answer) **D**

4. Answer) **A**

5. The numbers that are divisible by 3 are visible by 9

   Answer) **B**

6. Cyclicity of 5^{2010} is 1

   Answer) **A**

7. $e = \sum_{i=1}^{\infty} \left[\frac{100!}{5!}\right] + \left[\frac{100}{5}\right] + \left[\frac{100}{5^2}\right] + \left[\frac{100}{5^3}\right] = 20 + 4 + 0 = 24$

   Answer) **B**

8. Cyclicity of 6 is 1

   Answer) **C**

9. $AB \equiv p(\text{mod}10)$

   $AB = 2^2 \times 3^2 \times 2^3 \times 3^3 = 2^5 \times 3^5 = 6^5 = 7776$

   $7776 \equiv 6(\text{mod}10)$

   Answer) **A**

10. $e = \left[\frac{8}{5}\right] + \left[\frac{8}{5^2}\right] \ldots = [1 \cdot 6] + 0 = 1$

    Answer) **D**

11. $(n+2 =)(n+1)! = 60(n+1)!; n = 58$

    Answer) **C**

12. cyclicity of 3 is 4

    $43 \equiv 3(\text{mod}10); (43)^{17} \equiv 3^{17}\text{mod}10i \cdot e$

    last digit of 43^{17} is same as 3^{17}

    $3^4 = 81 \equiv 1(\text{mod}10) \Rightarrow 3^{17} = 3^{4\times4+1} = 1 \times 1 \times 1 \times 1 \times 3;\ 3^{17} \equiv 3(\text{mod}10)$

    Answer) **D**

13. $3^{2005} = 3^{4(501)+1}$; units place $= 3^1 = 3$

    Answer) **A**

14. Answer) **B**

15. $2 \equiv 2\text{mod}3 \therefore a \equiv a\text{mod}m$

    raising power to 5

$2^5 \equiv 32 \bmod 3$ or $32 \equiv 2 \bmod 3$

$2 \equiv 16 \bmod 3$ $2^{100} = 1 \bmod 3$

1 is remainder when 2^{100} is divided by 3

Answer) **B**

16. unit digit of

$1^1 = 1$

$2^2 = 2$

$3^3 = 3$

$4^4 = 4$

$5^5 = 5$

$6^6 = 6$

$7^7 = 7$

$8^8 = 8$

$9^9 = 9$

$10^{10} = 0$

unit's digit of given expression will be 7($\therefore 1 + 4 + 7 + 6 + 5 + 6 + 3 + 6 + 9 + 0 = 47$).

Answer) **C**

17. Let $s = 1! + 2! + 3! + \cdots .. 1997$!

from 5 ! all the numbers will have units digit ' 0 'and also $1! + 2! + 3! + 4! = 33$

$\therefore$ unit digit of S is 3

Answer) **B**

18. $21 = 3 \times 7$ so 21 , divides n ! for $n \geq 7$

The remainder when $1! + 2! + \cdots .6! = 873 = 21(41) + 12$

is divided by 21

Remainder is 12

Answer) **C**

19. Answer) **B**

20. Let $n = 2^{\frac{36}{10}}$

Remainder is 6,

Answer) **B**

21. $4! \equiv 0 \bmod 24$

$5! \equiv 0 \bmod 24 = 24$

$100! \equiv 0 \bmod 24$

$4! + 5! + \cdots .. 100! \equiv 0 \text{mod} 24$

$1! + 2! + 3! \equiv 9 \text{mod} 24$

$1! + 2! + 3! + 4! + 5! + 100! \equiv 9 \text{mod} 24$

9 is remainder

Answer) **B**

22. By taking x values 1,2,3,4,5

Answer) **D**

Level – III

23. $e = \sum_{i=1}^{\infty} \left[\frac{5^n - 1}{5^i}\right] = \left[5^{n-1} - \frac{1}{5}\right] + \left[5^{n-2} - \frac{1}{5^2}\right] + \cdots .. \left[5 - \frac{1}{5^{n-1}}\right]$

$= (5^{n-1} + 5^{n-2} + \cdots .5) - (n-1) = \frac{5\left(5^{n-1}-1\right)}{5-1} - (n-1) = \frac{(5^n - 4n - 1)}{5-1}$ = Total No of zeros.

Answer) **D**

24. $1! + 2! + 3! + 4! + 5! = 1 + 2 + 6 + 24 + 120 = 150 + 30 = 153$

It is composite, odd number and multiple of 3.

Answer) **A, B**, **D**

25. $m = 9$

Answer) **C**

26. $81 + 9 = 90$

Answer) **C**

27. $729 + 27 = 756$

Answer) **A**

28. a) $(n-1)! = (n-2)\ !$

$(n-1)(n-2)! = (n-2)\ !$

$n = 2$

b) $\frac{(n-2)!}{(n-3)!} = \frac{(n-2)!}{(n-1)!} \times 2$

c) 15^{550}

cyclicity of 5 is 1

d) 45672 is divisible by $-3,6$

Answer) **a-q; b-p; c-r; d-p,s,q**

29. Last digit of expression wil be same as the

last digit of $222^{888} + 888^{222}$

Now last digit of 222^{888} is 6

last digit of 888^{222} is 4

$222^{888} + 888^{222} = 10$

since $6 + 4 = 10$

Answer) **A, B**

30. $(n-2)! = 20(n-4)!$

$(n-2)(n-3)(n-4)! = 20(n-4)!$ $n = 7$

Answer) **B**

31. $n!\,(n+1)! = (n+4)(n+3)(n+2)(n+1)\,!$

$n = 6$

Answer) **C**

32. $\frac{(n+2!)}{(n+1)!} - \frac{10\times 9!}{9!} = 6$

$16(n+1)! = (n+2)(n+1)!$

$n = 14$

Answer) **B**

33. Unit digit depends of 7^{777}

The order of repetetion in the power of is $4i \cdot e7,9,3,1$

divide 777 by 4 Quantient = 194

Remainder

Unit's digit is 7

Answer) **A**

34. units digit in 53^{53} is ' 3 '

unit digit in 33^{33} is ' 3 '

$\therefore$ given expression is divisible by 10

Answer) **C**

35. $11^2 = 121 = 12 \times 10 + 1$

$(11^2)^{66} \equiv 1^{66}(\text{mod}10)$

$11^{132} \equiv 1\text{mod}10$

The digit in the unit's place of 11^{132} is 1.

Answer) **A**

Worksheet – 4

Level – I

1. Answer) **D**

2. $4\frac{4}{7} = \frac{4\times 7+4}{7} = \frac{32}{7}$ 4. $\frac{15}{25} = \frac{3}{5}$

Answer) **D**

3. Answer) **D**

4. Answer) **D**

5. Answer) **D**

6. Answer) **A**

7. $\frac{1+2}{2+3}=\frac{3}{5}$

   Answer) **D**

8. $\frac{2}{3}$ and $\frac{3}{4}$

   $\frac{2}{3},\frac{2+3}{3+4},\frac{3}{4}=\frac{2}{3},\frac{5}{7},\frac{3}{7}=\frac{2}{3},\frac{2+5}{3+7},\frac{5}{7},\frac{5+3}{7+4},\frac{3}{4}=\frac{2}{3},\frac{7}{10},\frac{5}{7},\frac{8}{11},\frac{3}{4}$

   Answer) **D**

9. Answer) **C**

10. $\frac{2+3}{5}=\frac{5}{5}=1$

    Answer) **D**

11. $\frac{5}{8}\times x=75\Rightarrow x=\frac{75\times 8}{5}=120$

    Answer) **C**

12. $2\frac{1}{2}\div\frac{5}{2}$ of $\frac{2}{15}\Rightarrow\frac{5}{2}\div\frac{1}{3}\Rightarrow\frac{5}{2}\times 3=\frac{15}{2}$

    Answer) **A**

Level – II

1. Answer) **B**

2. $\frac{50}{80}=\frac{5}{8}$

   Answer) **C**

3. Answer) **B**

4. $\frac{25}{100}=\frac{1}{4}$

   Answer) **B**

5. LCM of 3,2,6 = 6

   Now $\frac{2}{3}=\frac{2\times 2}{3\times 2}=\frac{4}{6},\frac{1}{2}=\frac{1\times 3}{2\times 3}=\frac{3}{6},\frac{1}{6}$

   Now compare the numerators of like fractions $\frac{4}{6},\frac{3}{6},\frac{1}{6}$

   Arranging them in ascending order $\frac{1}{6}<\frac{3}{6}<\frac{4}{6}$ so, $\frac{1}{6}<\frac{1}{2}<\frac{2}{3}$

   Answer) **A**

6. $\frac{-7}{8} - x = \frac{5}{12}$

$x = \frac{-7}{8} - \frac{5}{12} = \frac{-21-10}{24} = \frac{-31}{24}$

Answer) **D**

7. $\left(11\frac{2}{5} + 7\frac{1}{3}\right) - \left(9\frac{3}{4} + 5\frac{5}{6}\right)$

$\left(\frac{57}{5} + \frac{22}{3}\right) - \left(\frac{39}{4} + \frac{35}{6}\right) = \left(\frac{171+110}{15}\right) - \left(\frac{117+70}{12}\right) = \left(\frac{281}{15}\right) - \left(\frac{187}{12}\right)$

$= \frac{1124-935}{60} = \frac{189}{60} = \frac{63}{20} = 3\frac{3}{20}$

Answer) **B**

8. $\frac{3}{4}x - \frac{2}{3}x = 6$

$\frac{9x-8x}{12} = 6 \Rightarrow \frac{x}{12} = 6 \rightarrow x = 72$

Answer) **C**

9. $\frac{26}{3} \times x = \frac{16}{3}$

$x = \frac{16}{3} \times \frac{3}{26} = \frac{8}{13}$

Answer) **A**

10. $\frac{3}{5}$ of $\frac{2}{3} = \frac{2}{5}$

Answer) **C**

11. L.C.M of 2,4,2 = 4

Now $\frac{3}{2} = \frac{3\times2}{2\times2} = \frac{6}{4}$,

$\frac{11}{4}, \frac{5}{2} = \frac{5\times2}{2\times2} = \frac{10}{4}$

Now greatest one is 11 , so $\frac{11}{4}$

Answer) **B**

12. Answer) **D**

13. $\frac{5}{2} + \frac{7}{2} + \frac{9}{2} = x \Rightarrow x = \frac{21}{2} \Rightarrow x - \frac{11}{2} = \frac{21}{2} - \frac{11}{2} = \frac{10}{2} = 5$

Answer) **B**

14. By option verification

Answer) **A**

15. Sum is 3 sides = $\frac{83}{5}$

$$\frac{57}{10}+\frac{27}{4}+x=\frac{83}{5}$$
$$x=\frac{83}{5}-\left(\frac{57}{10}+\frac{27}{4}\right)$$
$$x=\frac{83}{5}-\left(\frac{228+270}{40}\right)=\frac{83}{5}-\frac{498}{40}=\frac{664-498}{40}=\frac{166}{40}=4\frac{6}{40}=4\frac{3}{20}$$

Answer) **A**

16. Total flour $=\frac{85}{7}$ Flour last $=\frac{2}{5}\times\frac{85}{7}=\frac{34}{7}$ kg

Answer) **D**

17. $\frac{1}{2},\frac{2}{3},\frac{3}{4},\frac{4}{5}$

LCM of 2,3,4,5 = 60

$$\frac{1}{2}\times\frac{30}{30}=\frac{30}{60},\frac{2}{3}\times\frac{20}{20}=\frac{40}{60},\frac{3}{4}\times\frac{15}{15}=\frac{45}{60}\Rightarrow\frac{4}{5}\times\frac{12}{12}=\frac{48}{60}$$

Answer) **B**

18. $\frac{286}{2}=143$

In 143 pears $\frac{4}{13}\times 143$ are rotten

$= 44$ are rotten

$143-44=99$ pears are good

$$99\times\frac{45}{11}=Rs.405$$

Answer) **D**

19. $D=AB, Q=CD$

$A=12, B=8, C=6, D=4$

$$P=12\times 8=96\frac{9}{5Q}=\frac{96}{5\times 24}=\frac{4}{5}\Rightarrow \mathrm{Q}=24$$

Answer) **C**

20. $\frac{5}{9}=0.555\ldots\ldots\frac{11}{18}>\frac{5}{9}$

$$\frac{13}{24}<\frac{5}{9}\ \frac{17}{36}<\frac{5}{9}\ \frac{5}{8}>\frac{5}{9}$$

Answer) **B, C**

Level – III

21. $\frac{5}{10}=\frac{m-2}{30}\Rightarrow m-2=15\Rightarrow m=17$

Answer) **B**

22. $\frac{2\times 17+3}{19}=\frac{34+3}{19}=\frac{37}{19}=1\cdot\frac{18}{19}$

Answer) **D**

23. $\frac{m-10}{21} = \frac{17-10}{21} = \frac{7}{21} = \frac{1}{3}$

Answer) **A**

24. a) $\frac{35}{100} = \frac{7}{20}$

b) $\frac{75}{200} = \frac{3}{8}$

c) $\frac{16}{24} = \frac{2}{3}$

d) $\frac{250}{3000} = \frac{1}{12}$

Answer) **a-s; b-q,t; c-r; d-p**

25. $\frac{7}{20} \times \frac{3}{3} = \frac{21}{60} \Rightarrow 53 + x = 60 \Rightarrow x = 7$

Answer) **7**

26. Answer) **C, D**

27. $\frac{2}{5} \times 56000 = 22400$

Answer) **D**

28. $\frac{3}{8} \times 33600 = 12600$

Answer) **D**

29. Wife's share = 21000

Answer) **C**

30. a) $\frac{4}{15} \times 180 = 48$ b) $180 - 48 = 132$

c) $\frac{3}{4} \times 132 = 99$

d) $\frac{1}{4} \times 132 = 33$

Answer) **a-s; b-p; c-q; d-t**

31. a) $\frac{25}{4} \div \frac{7}{3} = \frac{25}{4} \times \frac{3}{7} = \frac{75}{28}$

b) let the number = x

$\frac{5}{6} \times x = 45 = x = 45 \times \frac{6}{5} = 9 \times 6 = 54$

c) $x\left(\frac{1}{4}\right) = \frac{15}{4} = x = \frac{15}{4} \times \frac{4}{1} = 15$

d) $\frac{7}{4} \times \frac{5}{2} = \frac{35}{8} = 4\frac{3}{8}$

Answer) **a-r; b-s; c-q; d-p**

32. Total bus capacity = 42 passangers

No. of passengers drops at first station $= \frac{1}{3} \times 42 = 14$

No.of remaining passengers in a bus $= 42 - 14 = 28$

No.of passengers it takes at first station $= \frac{1}{7} \times 42 = 6$ passangers

Answer) **B**

33. New total $= 28 + 6 = 34$

No. of passengers drops at second station $= \frac{1}{2} \times 34 = 17$

Answer) **B**

34. No.of passengers drop at last station $= 34 - 17 = 17$

Answer) **B**

Worksheet – 5

Level – I

1. C	2. B	3. C	4. D	5. C	6. C	7. D
8. B	9. C	10. C	11. B	12.B	13.C	

Level – II

1. $\frac{10}{-35} = \frac{-2}{7}$

   Answer) **C**

2. $\frac{2+3}{2} = \frac{5}{2}$

   Answer) **B**

3. Answer) **C**

4. Answer) **D**

5. Answer) **C**

6. Terminating

   Answer) **C**

7. $\frac{5}{210} = 0.02380952$ non-terminating

   Answer) **D**

8. Answer) **A**

9. $\frac{289}{13} = 22.\overline{230769}, \therefore$ length Q the period $= 6$

   Answer) **6**

10. $(\sqrt{5}+1)(\sqrt{5}-1) = 5 - 1 = 4$

    Answer) **4**

11. $\frac{3}{4} \times \frac{5}{3} \times \frac{9}{5} \times \frac{a}{b} = \frac{9}{4} \Rightarrow \frac{9}{4} \times \frac{a}{b} = \frac{9}{4} \Rightarrow \frac{a}{b} = 1 \Rightarrow$ Additive invence of $1 = -1$

Answer) **C**

12. $\frac{6}{11} \times \frac{-11}{18} = \frac{-6}{18} = \frac{-1}{3}$

Answer) **A**

13. $\frac{5}{12} \times -\frac{36}{10} = \frac{-3}{2} = -1\frac{1}{2}$

Answer) **B**

14. $0.\overline{09} = x$(let)

$100x = 9.090909$ ....

$x = 0.090909$ ...

$99x = 9$

$x = \frac{1}{11}$

Answer) **B**

15. Answer) **C**

16. $\frac{p}{q} = \frac{-r}{s} = \frac{p}{q} + \frac{r}{s} = \frac{-r}{s} + \frac{r}{s} = 0$

Answer) **C**

17. $x \times \left(\frac{-4}{9}\right) = \frac{28}{27} \Rightarrow x = \frac{28}{27} \times \frac{-9}{4} = \frac{-7}{3}$

Answer) **D**

18. $\frac{\frac{-44}{8}}{p} = \frac{-11}{3} \Rightarrow p = \frac{\frac{-44}{9}}{\frac{-4}{3}} = \frac{4}{3}$

Answer) **A**

19. $\frac{22}{7} = 3.\overline{142857}$

Answer) **C**

Period = 142857

20. $\frac{\frac{81}{2}}{\frac{9}{4}} = 9 \times 2 = 18$

Answer) **C**

21. $\frac{583}{\frac{11}{4}} = 53 \times 4 = \text{Rs. } 212$

Answer) **C**

22. Answer) **C**

23. $\frac{3}{10}100 + \frac{1}{10}100 + 17 + x = 100 \Rightarrow 30 + 10 + 17 + x = 100 \Rightarrow x = 100 - 57 = 43$

Answer) **B**

24. Answer) **D**

Level – III

25. $\frac{69}{216}=\frac{23}{72},\frac{138}{108}=\frac{46}{36}=\frac{161}{54}\neq\frac{29}{54}\Rightarrow\frac{207}{270}=\frac{69}{90}=\frac{23}{30}$, 1,2,3,4 are not equal to $\frac{23}{54}$

Answer) **A, B, C**, **D**

26. $\frac{-1-2}{2}=\frac{-3}{2}$

Answer) **B**

27. $\frac{0+1}{2}=\frac{1}{2}$

Answer) **B**

28. $\left(\frac{3}{4}+\frac{4}{4}+\frac{4}{4}\right)=\frac{3}{4}+1+1=\frac{3}{4}+2=\frac{3+8}{4}=\frac{11}{4}$

Answer) **B**

29. (a) Non terminating non repeating

(b) $\sqrt{97}$ - irrational number

c) $-\sqrt{\frac{9}{81}}=-\sqrt{\frac{1}{9}}=-\frac{1}{3}$ rational number non-terminating repeacting

d) $\sqrt{0.0001}$ = 0.01-rational number terminating decimals

Answer) **a-q,b-t,c-p,r;d-p,s**

30. $\frac{7}{3}$ = 2.333 ….2.3 (rational)

$\frac{22}{7}=3.\overline{143857}$ (rational)

Answer) **C, D**

31. Answer) **a-r; b-p; c-q; d-t**

32. $\frac{22+24}{23+25}=\frac{46}{48}$

Answer) **C**

33. 33. $\frac{1+2}{2+3}=\frac{3}{5}$

Answer) **A**

34. 34. $\frac{21}{20}$ = 1.05

Answer) **C**

Worksheet – 6

Level – I

1. $a^2=b\Rightarrow\sqrt{b}=\sqrt{a^2}=a$ 2. $\sqrt{625}=b\Rightarrow b=25$

Answer) **C**

2. Answer) **B**

3. $\sqrt{0.9} \times \sqrt{1.6} = \sqrt{0.9 \times 1.6} = \sqrt{\frac{9}{10} \times \frac{16}{10}} = \sqrt{\frac{144}{100}} = \frac{12}{10} = 1.2$

   Answer) **C**

4. $(4)^3 = 4 \times 4 \times 4 = 64$

   Answer) **C**

5. $\sqrt{169} = 13 \Rightarrow$ Successor of $13 = 14$

   Answer) **D**

6. 1234 is not a perfect cube.

   Answer) **D**

7. Predecessor of prime is odd or even.

   Answer) **C**

8. $\sqrt{0.0009} = \sqrt{\frac{9}{10000}} = \frac{3}{100} = 0.03$

   Answer) **D**

9. Composite Number

   Answer) **A**

10. $\sqrt[3]{3375} = \sqrt[3]{(15)^3} = 15$

    Answer) **C**

Level – II

1. $28\sqrt{x} + 1426 = 3 \times 718 \Rightarrow 28\sqrt{x} = 2154 - 1426 \Rightarrow \sqrt{x} = 26$

   $\Rightarrow x = (26)^2 = 676.$

   Answer) **B**

2.

2	2744
2	1372
2	686
7	343
7	49
	7

   $\therefore 2744 = 2^3 \times 7^3 \ \ \therefore \sqrt[3]{2744} = 2 \times 7 = 14$

   Answer) **A**

3. $(0.000216)^{1/3} = \left(\frac{216}{10^6}\right)^{\frac{1}{3}} = \left(\frac{6\times6\times6}{10^2\times10^2\times10^2}\right)^{\frac{1}{3}} = \frac{6}{10^2} = \frac{6}{100} = 0.06.$

   Answer) **B**

4. $3087 = 3 \times 3 \times 7 \times 7 \times 7$ with 3 it should be multiplied

   Answer) **A**

5. Clearly $4320 = 2^3 \times 3^3 \times 2^2 \times 5$.

   To make it perfect cube, it must be multiplied by $2 \times 5^2 = 25 \times 2 = 50$.

   Answer) **C**

6. $\sqrt{272^2 - 128^2} = \sqrt{(272 + 128)(272 - 128)}$

   $= \sqrt{400 \times 144} = \sqrt{57600} = 240.$

   Answer) **C**

7. $\sqrt[3]{2} \times \sqrt[3]{4} = x$

   $\sqrt[3]{2 \times 4} = x \Rightarrow x = \sqrt[3]{8} \Rightarrow x = 2$

   Answer) **B**

8. $\sqrt{1.1664} = \sqrt{\frac{11664}{10000}} = \sqrt{\left(\frac{108}{100}\right)^2} = \frac{108}{100} = 1.08$

   Answer) **A**

9. $\frac{\sqrt{243}}{\sqrt{363}} = x \Rightarrow x = \sqrt{\frac{243}{363}}$ (divide by 3)

   $\Rightarrow x = \sqrt{\left(\frac{9}{11}\right)^2} = \frac{9}{11} = 0.8181 \ldots\ldots = 0.\overline{81}$

   Answer) **A**

10. $\sqrt{45} = \sqrt{9} \times \sqrt{5} = 3\sqrt{5} = 3(2.23607) = 6.708$

    Answer) **A**

11. Given 5x is a perfect cube. x become a $25,25 = 5 \times 5$.

    $\therefore 5x = 5 \times 5 \times 5 = 125$ is a perfect cube.

    $\therefore$ 2 fives that must be contained in the prime factorization of x.

    Answer) **B**

12. $3\sqrt{5} + \sqrt{125} = 17.88$

    $3\sqrt{5} + 5\sqrt{5} = 17.88$

    $8\sqrt{5} = 17.88 \Rightarrow \sqrt{5} = \frac{17.88}{8}$

    $\sqrt{80} + 6\sqrt{5} = \sqrt{16 \times 5} + 6\sqrt{5} = 4\sqrt{5} + 6\sqrt{5} = 10\sqrt{5}$

    $= 10 \times \frac{17.88}{8} = 22.35$

    Answer) **D**

13. $41^2 = 1681$, which is smaller than 1752

    and $42^2 = 1764$, which is slightly bigger than 1752

    $\therefore$ required number is $1764 - 1752 = 12$

    Answer) **C**

14. We take all options one by one, concentrating on units digit

    7 : units digit is $1300 - 7 = 3$ (7 is not possible at units place)

    4 : units digit is $1300 - 4 = 6$ (6 is not possible at units place)

    8 : units digit is $1300 - 8 = 2$ (2 is not possible at units place)

    10: units digit is $1300 - 10 = 0$ (0 is not possible at units place)

    $\therefore$ 4 is the only possible number out of the given options.

    Answer) **B**

15. $882 = 2 \times 21 \times 21$

    here except for 2 , other factors are in pairs.

    $\therefore$ we have to multiply 882 by 2 to make it a perfect square number.

    Answer) **B**

16. $\left(5 - \sqrt{y^2 - 25}\right)^2$

    $= (5)^2 + \left(\sqrt{y^2 - 25}\right)^2 - 2 \times 5 \times \sqrt{y^2 - 25}$ [$\because (a - b)^2 = a^2 + b^2 - 2ab$]

    $= 25 + y^2 - 25 - 10\sqrt{y^2 - 25} = y^2 - 10\sqrt{y^2 - 25}$

    Answer) **C**

17. $a = 0.1039$

    $= \sqrt{4a^2 - 4a + 1} + 3a = \sqrt{(2a - 1)^2} + 3a$

    $= \pm(2a - 1) + 3a$ $\therefore 2a - 1 + 3a = 5a - 1$ or $1 - 2a + 3a = 1 + a$

    $\therefore 1 + a = 1 + 0.1039 = 1.1039$

    Answer) **C**

18. 160 should be subtracted

    Answer) **D**

19. $\sqrt{\frac{196}{x}} \times \sqrt{\frac{900}{x}} = 4 \Rightarrow \frac{14}{\sqrt{x}} \times \frac{30}{\sqrt{x}} = 4 \Rightarrow \frac{14 \times 30}{x} = 4 \rightarrow x = 105$

    Answer) **C**

20. $\frac{18 \times 72 \times 105}{48 \times 315} = \sqrt{x}$

    On cancelation $\sqrt{x} = 9 \Rightarrow (\sqrt{x})^2 = (9)^2 \Rightarrow x = 81$ $x = 729$

    Answer) **A**

21. $\sqrt{110\frac{1}{4}} = \sqrt{\frac{441}{4}} = \frac{21}{2} = 10.5$

Answer) **B**

22. $[27 \times 2744]^{\frac{1}{3}} = [3^3 \times (14)^3]^{1/3} = [(3 \times 14)^3]^{1/3} = 42$

Answer) **D**

23. $\sqrt{41} - \sqrt{21 + \sqrt{19 - \sqrt{9}}} = \sqrt{41 - \sqrt{21 + \sqrt{19 - 3}}} = \sqrt{41 - \sqrt{21 + \sqrt{16}}}$

$= \sqrt{41 - \sqrt{21 + 4}} = \sqrt{41 - \sqrt{25}} = \sqrt{41 - 5} = \sqrt{36} = 6$

Answer) **C**

24. $3a = 4b = 6c$

$3a = 6c, 4b = 6c$

$a = 2c, 2b = 3c \Rightarrow b = \frac{3c}{2}$

$a + b + c = 27\sqrt{29} \Rightarrow 2c + \frac{3c}{2} + c = 27\sqrt{29} \Rightarrow \frac{9c}{2} = 27\sqrt{29} \Rightarrow c = 6\sqrt{29}$

$a = 12\sqrt{29} \Rightarrow b = 9\sqrt{29} \Rightarrow \mathrm{c} = 6\sqrt{29}$

$\sqrt{a^2 + b^2 + c^2} = \sqrt{(12\sqrt{29})^2 + (9\sqrt{29})^2 + (6\sqrt{29})^2}$

$= \sqrt{144 \times 29 + 81 \times 29 + 36 \times 29} = \sqrt{29(144 + 81 + 36)}$

$= \sqrt{29 \times 261} = \sqrt{29 \times 29 \times 9} = 29 \times 3 = 87$

Answer) **C**

25. $a^3 + b^3 = (a + b)(a^2 - ab + b^2) \Rightarrow \frac{a^3 + b^3}{a^2 - ab + b^2} = a + b = 73 + 53 = 126$

Answer) **A**

Level – III

26. If n is odd then n^2 is odd, n^3 is odd and n^4 is odd and soon.

Answer) **A, B**

27. $x = \sqrt[3]{a} = \sqrt[3]{\sqrt{4096}} = \sqrt[3]{64} = 4; \mathrm{y} = \sqrt[3]{\sqrt{729}} = \sqrt[3]{8} = 2; \mathrm{z} = \sqrt[3]{\sqrt{729}} = \sqrt[3]{27} = 3$

Answer) **A**

28. $x + 2y + 2011z \Rightarrow 4 + 2(2) + 2011(3) = 4 + 2(2) + 2011(3) = 4 + 4 + 6033 = 6041$

Answer) **C**

29. $\sqrt[3]{(2001)^{x+y+z}} = ((2001)^{4+2+3})^{1/3} = (2001)^3$

Answer) **B**

30. Answer) **a-p,q,s; b-p; c-s,r; d-s,r**

31. $n^2 + n + 1$ is a prime

    Answer) **A, B, C**

32. 32. $\sqrt[3]{2} \times \sqrt[3]{4} = \sqrt[3]{8} = 2$

    Answer) **A,**

33. a) $\sqrt{x} = \frac{18\times72\times105}{48\times315} \Rightarrow \sqrt{x} = 9 \Rightarrow x = 81$

    b) $\sqrt{\frac{25}{81} - \frac{1}{9}} = \sqrt{\frac{25-9}{81}} = \sqrt{\frac{16}{81}} = \frac{4}{9}$

    c) $\sqrt{0.000441} = 0.021$

    d) $\sqrt{0.01} + \sqrt{0.81} + \sqrt{1.21} + \sqrt{0.0009} = 0.1 + 0.9 + 1.1 + 0.03 = 2.13$

    Answer) **a-q; b-r; c-p; d-s**

Worksheet – 7

Level – I

1. Answer) **A**
2. Answer) **B**
3. Answer) **C**
4. Answer) **B**
5. Answer) **D**
6. Obviously option (2) is required answers since base must be greater than any integer of the number

   Answer) **B**

7. $(25)_x = (85)_{10} \Rightarrow 2 \times x^1 + 5 \times x^0 = (85)_{10} \Rightarrow 2x + 5 \times x^0 = 85 \Rightarrow 2x = 85 - 5 \Rightarrow 2x = 80 \Rightarrow x = 40$

   Answer) **C**

8. Answer) **B**
9. Answer) **B**
10. Answer) **C**

Level – II

1. $|x + 1| \leq 6 \Rightarrow -7 \leq x \leq 5$

   Answer) **B**

2. $3x - 2 = 10 \Rightarrow x = 4;\ -3x + 2 = 10 \Rightarrow -3x = 8$

   $x = -8/3$

   Answer) **A**

3. $x-2 \geq 12-x+2 \geq 12; x \geq 14$ or $x \leq -10$

   Answer) **B**

4. $i^{4n+2} \Rightarrow$ let n = 1 $\Rightarrow i^{4+2} = i^6 = -1$

   Answer) **A**

5. $5 = -x \Rightarrow x = -5$

   Answer) **B**

6. $\frac{i^2+1}{i^2}$; $i^2 = -1$

   Answer) **C**

7. $\sqrt{-5} \times \sqrt{-125} = -25$

   Answer) **B**

8. 000

   Answer) **A**

9. 1001

   Answer) **B**

10. $i^{120} = (i^2)^{60} = (-1)^{60} = 1$

    Answer) **A**

11. $9-4x \geq 4 \Rightarrow 5 \geq 4x\ x \leq \frac{5}{4} \Rightarrow -9+4x \geq 4 \Rightarrow x \geq \frac{13}{4}$

    Answer) **A**

12. $|x-2| < 6$

    x-2<6 -x+2<6

    x<8 x>-4

    -14 < x < 8

    Answer) **B**

13. $(1+i)(2+i) = 2+3i-i^2 = 1+3i$

    Answer) **C**

14. $i^{2020} \times i^{-2020} = 1$

    Answer) **A**

15. Since last three digit ($i \cdot e$ 956) is not divisible by 2^3(= 8)956 is divisible by 4 .

    Hence last two digits will be 00

    Answer) **B**

16. $(765)_{16} = 7 \times 16 \times 26 \times 16^1 + 5 \times 16^0 = 1792 + 96 + 5 = (1893)_{10}$

    Answer) **C**

17. $(1+2i)(3-4i) = 3-4i+6i-8i^2 = 11+2i$ additive inverse is $-11-2i$

    Answer) **B**

18. $(1+i)^6 + (1-i)^3 = -2-10i$

    Answer) **C**

19. $(2n+5)(3n+5)n^3+n+5 \Rightarrow n = 8$ or $n = -2$

    Answer) **C**

20. $5 < x < 9 \Rightarrow |x-9| + |5-x| + |x+9| + |5+x| = 28$

    Answer) **B**

21. $\left|\frac{x-1}{2}\right| \leq 2 \Rightarrow -3 \leq x \leq 5$

    Answer) **D**

22. Put n = 1 $\Rightarrow (1+i)^6 + (1-i)^6 = 0$

    Answer) **A**

23. $(1+i)^n = (1-i)^n \Rightarrow$ n = 4

    Answer) **B**

Level – III

24. $(4+2i)(1+2i) = A+iB = 4+10i-4 = 10i; A = 0, B = 10$

    Answer) **A, B**

25. $i^{4n} = 1,$

    $i^{4n+2} = -1,$

    $\frac{i^{49}+i^{52}+i^{35}}{i^{39}+i^{40}+i^{37}} - 1 = 0$

    Answer) **B**

26. $i^{19} = -i, i^{25} = i$

    $\left(-i+\frac{1}{i}\right)^2 = \left(\frac{-i^2+1}{i}\right)^2 = \left(\frac{2}{i}\right)^2 = \frac{4}{i^2} = -4$

    Answer) **A**

27. $i^{17} = i,\ i^{34} = i^2 = -1 \Rightarrow (i+1)^2 = 2i$

    Answer) **C**

28. Answer) **a-t; b-p; c-s; d-q**

29. Put $x = 0,1$ and $\frac{1}{2}$ $-1 \leq c \leq 1, -1 \leq a+b+c \leq 1$

    $-4 \leq a+2b+4c \leq 4 - 8 \leq 3a+2b \leq 8$
    $-8 \leq a+2b \leq 8 \Rightarrow -32 \leq 4b \leq 32 \Rightarrow -8 \leq b \leq 8$

    Answer) **A, B, C, D**

30. a) ${}_{1}^{5}32_{1110}^{4}{}^{0} = 1 \times 2^5 + 0 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0$

b) $1101110 = 1 \times 2^6 + 1 \times 2^5 + 0 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0$

c) ${}^{6\,5\,4\,3\,2\,1\,0}1011101 = 1\times2^6+0+1\times2^4+1\times2^3+1\times2^2+0\times2^0+1\times2^1$

d) 1111110

Answer) **a-q; b-s; c-p; d-r**

31. As $a, b > 0 \Rightarrow a^3 + b^3 > 0 \Rightarrow a - b > 0 \Rightarrow a > b$;

$a^3 + b^3 - a + b = 0 \Rightarrow a(a^2 - 1) + b(b^2 + 1) = 0$

$\Rightarrow a^2 - 1 > 0 \Rightarrow -1 < a < 1$, as $a > 0 \Rightarrow 0 < a < 1$

Answer) **A, D**

32. $\left|\frac{3x}{4} - 2\right| < 8$ $\quad\quad$ $\frac{3x}{4} - 2 < 8$

$\frac{3x}{4} - 2 > -8$ $\quad\quad$ $\frac{3x}{4} < 10$

$\frac{3x}{4} > -6$ $\quad\quad$ $x < \frac{40}{3}$

$x > -8$

Answer) **B**

33. $|3 - 4x| \leq 20$

$-20 \leq 3 - 4x \leq 20$

$-23 \leq -4x \leq 17$

$\frac{-17}{4} \leq x \leq \frac{23}{4}$

Answer) **A**

34. $|x + 3| \geq 30$

$x + 3 \leq -30$ or $x + 3 \geq 30$

$x \leq -33$ or $x \geq 27$

$\therefore x(-\infty, -33] \cup [27, \infty)$

Answer) **A**

2. EXPONENTS & POWERS

WORKSHEET-1

Level – I

1.C	2.A	3.D	4.B	5.A	6.A	7.B
8.A	9.C	10.A	11. A	12.A		

1. $2^x \times 2^2 = 13 \times 4 = 52$
2. $2^{2(x-1)} = 2^{x+2} \Rightarrow 2x - 2 = x + 2 \Rightarrow x = 4$
3. $a^{4+5+1} = a^{10}$
4. 1
5. $2^3 \times 1 = 8$
6. 2^9
7. 4.65×10^8
8. $2^{3\times2} = 2^6$
9. $x = 35$
10. -3
11. $\sqrt[15]{9}$
12. $(2^4)^{\frac{3}{4}} = 8$

Level – II

1.A	2.B	3.A	4.A	5.A	6.C	7.B
8.A	9.B	10.D	11.B	12.B	13.C	14.A
15.C	16. D	17. C	18.C	19.A	20. A	21.D
22.A						

1. $\frac{5^{2x}\times5}{5^x} = 25$
2. $3^4 + 4^3 = 81 + 64 = 145$

   $5^{2x-x} = \frac{25}{5} \Rightarrow 5^x = 5^1 \Rightarrow x = 1$
3. $(625)^{0.16+0.09} = (5^4)^{0.25} = (5^4)^{\frac{1}{4}} = 5$ 4. $(2^4)^{\frac{-2}{4}} = \frac{1}{4} \Rightarrow (2^6)^{\frac{-2}{4}} = 2^{\frac{-12}{4}} = 2^{-3} = \frac{1}{2^3} = \frac{1}{8}$
4. $2^x \times 2^4 - 2^x \times 2^2 = 3 \Rightarrow 2^x(16-4) = 3 \Rightarrow 2^x = \frac{3}{12} \Rightarrow 2^x = \frac{1}{4} \Rightarrow 2^x = 2^{-2} \Rightarrow x = -2$
5. $\frac{(7^3)^3}{(7^2)^3} = \left(\frac{7^3}{7^2}\right)^3 = 7^3 = 343$
6. $(5^3)^{-7} = 5^{-21}$
7. $\left(\frac{3}{2}\right)^{-2} \div x = 2 \Rightarrow \frac{4}{9} \times \frac{1}{x} = 2 \Rightarrow x = \frac{2}{9}$
8. $(3^{600})^{\frac{1}{3}} = 3^{200}$ 10. $\frac{x}{y} = \left(\frac{4}{3}\right)^{10} \times \left(\frac{4}{3}\right)^{10}$
9. $a^{x-y}b^{x-y} \cdot b^{y-z}c^{y-z}c^{z-x} \cdot a^{z-x}$

   $a^{x-y+z-x}b^{x-y+y-z}c^{y-z+z-x} \Rightarrow a^{z-y}b^{x-z}c^{y-x} \Rightarrow \frac{x}{y} = \left(\frac{4}{3}\right)^{20}$

12. $4^3 < 3^4$

13. $a + 5 = 2a$

$a = 5$

$2b + 3 = 4b + 1$

$2\ b = 2$

$b = 1$

$5^1 + 1^5 = 6$

14. $\frac{x^{4n+1+1-4n}}{x^{3n+2}} = x^{2-3n-2} = (x^{-3n})^2 = x^{-6n}$

15. $[1-\{1-(1-X^3)^{-1}\}^{-1}]^{-1/3} \Rightarrow \left[1-\left\{1-\frac{1}{1-X^3}\right\}^{-1}\right]^{-1/3}$

$\left[1-\left\{\frac{1-X^3-1}{1-X^3}\right\}^{-1}\right]^{-1/3} \Rightarrow \left[1-\left(\frac{1-x^3}{-x^3}\right)\right]^{-1/3} \Rightarrow \left(\frac{-x^3-1+x^3}{-x^3}\right)^{-1/3} \Rightarrow (x^3)^{\frac{1}{3}} = x$

16. $2^{\frac{31}{32}}$

17. $\frac{x^{a^2-b^2}x^{b^2-c^2}}{x^{a^2-c^2}} = x^{a^2-b^2+b^2-c^2-a^2+c^2} = 1$

18. $\frac{(3^5)^{\frac{n}{5}}\times 3^{2n}\times 3}{3^{2n}\times\frac{3^n}{3}} = \frac{3^n\times 3^{2n}\times 3\times 3}{3^{2n}\times 3^n} = 9$

19. $x = 2 + 1 + 2\sqrt{2} \Rightarrow x = (\sqrt{2})^2 + 2\sqrt{2} + (1)^2 = (\sqrt{2}+1)^2$

$\sqrt{x} = \sqrt{2}+1 \Rightarrow x^{1/2} - x^{-1/2} = \sqrt{x} - \frac{1}{\sqrt{x}} = \sqrt{2}+1-\frac{1}{\sqrt{2}+1} = (\sqrt{2}+1)-(\sqrt{2}-1) = 2$

20. $\frac{x^{ac}}{x^{bc}}\times\frac{x^{ba}}{x^{ca}}\times\frac{x^{cb}}{x^{ab}} = 1$

21. $m + n = mn$

$mn - 2m + mn - 2n$

$2mn - 2(m+n)$

$2mn - 2mn = 0$

22. $x - 2 = 3^{\frac{2}{3}} + 3^{\frac{1}{3}}$

Cubing on both sides $\Rightarrow x^3 - 3x^2 + 3\cdot x\cdot 2^2 - 2^3 = \left(3^{\frac{2}{3}}\right)^3 + \left(3^{\frac{1}{3}}\right)^3 + 3\cdot 3^{\frac{2}{3}}\cdot 3^{\frac{1}{3}}\cdot\left(3^{\frac{2}{3}}+3^{\frac{1}{3}}\right)$

$\Rightarrow x^3 - 6x^2 + 12x - 8 = 9 + 3 + 9(x-2)$

$x^3 - 6x^2 + 12x - 8 = 12 + 9x - 18 \Rightarrow x^3 - 6x^2 + 3x - 2 = 0$

23. $5^a = 5^5$ 24. $5^2 = 25$ 25. $5^0 = 1$

Level – III

23.B	24.A	25.B	26.a-s,q; b-t; c-p; d-r	27. B
28.B	29.A	30.a-r; b-p; c-t; d-q		31.1,2,3
32.B	33.A	34.C		

26. a) $\left(\frac{1}{1000}\right)^{\frac{2}{3}} = (10^{-3})^{\frac{2}{3}} = 10^{-2}$

b) $(8^3)^{\frac{-2}{3}} = 8^{-2} = (2^3)^{-2} = 2^{-6}$

c) $(64)^{\frac{-2}{3}} = (4^3)^{\frac{-2}{3}} = 4^{-2} = \frac{1}{4^2} = \frac{1}{16}$

d) $\left(\frac{27}{1000}\right)^{\frac{-4}{3}} = \left(\left(\frac{3}{10}\right)^3\right)^{\frac{-4}{3}} = (0.3)^{-4}$

27. $x^6 = 64 = 2^6 \Rightarrow x = 2$

28. $2x = 3y \Rightarrow x:y = 3:2$

29. $x = 30$

30. a-r ; b-p ; c-t ; d-q

a) $5^3 \cdot 2^3 = (5 \times 2)^3 = 10^3$

b) $a^x = a^y \Rightarrow x = y$

c) $(3+4)^4 = 7^4$

d) $(5xy)^3 = 5^3 \cdot x^3 \cdot y^3 = 125x^3y^3$

31. $4^x - \frac{4^x}{4} = 24 \Rightarrow 4^x\left(1 - \frac{1}{4}\right) = 24$

$4^x = 32 \Rightarrow 2^{2x} = 2^5 \Rightarrow 2x = 5 \Rightarrow x = \frac{5}{2} \Rightarrow (2x)^x = \left(2 \times \frac{5}{2}\right)^{\frac{5}{2}} = \sqrt{5^5}$

32. $1 + 2^x = 2^2 \cdot 2^x \Rightarrow 2^x(4-1) = 1 \Rightarrow 2^x = 1/3$

33. $\frac{(2^x)^2}{1+(2^x)^2} = \frac{1/9}{1+1/9} = \frac{1/9}{10/9} = 1/10$

34. $\frac{(2^x)^3}{1+(2^x)^3} = \frac{1/27}{1+1/27} = 1/28$

3. SURDS

WORKSHEET-1

Level - I

1. D	2. D	3. B	4. B	5. C	6. A	7.D
8. B	9. C	10. C	11. A	12. B	13. D	14. D

Level – II

1.C	2.A	3.B	4.B	5.D	6.D	7.C	8.C
9.C	10.B	11.B	12.10	13.C	14.A	15.A	16.C
17.C	18.C	19.D	20.C	21.C	22.C		

1. Surd of order 4 is called biquadratic surd
2. $3\sqrt[4]{5} = \sqrt[4]{3^4 \times 5} = \sqrt[4]{81 \times 5} = \sqrt[4]{405}$
3. $\sqrt[4]{2} \times \sqrt[4]{3} = \sqrt[4]{2 \times 3} = \sqrt[4]{6}$
4. $2.\sqrt[3]{5} = \sqrt[3]{2^3 \times 5} = \sqrt[3]{8 \times 5} = \sqrt[3]{40}$
5. $(K+1)\sqrt[4]{5}$ is entire surd then $K = 0$ because the coefficient of given surd is unity so it is entire surd
6. $\frac{2}{3}\sqrt{32} = \sqrt{\left(\frac{2}{3}\right)^2 \times 32} = \sqrt{\frac{4}{9} \times 32} = \sqrt{\frac{128}{9}} = \frac{\sqrt{128}}{3}$
7. $\sqrt[5]{288} = \sqrt[5]{2^5 \times 9} = \sqrt[5]{2^5} \times \sqrt[5]{9} = 2 \cdot \sqrt[5]{9}$
8. $\sqrt[6]{729} = \sqrt[6]{3^6} = 3$
9. $\sqrt[5]{x^3} = x^y \Rightarrow x^{\frac{3}{5}} = x^y \Rightarrow y = \frac{3}{5} \therefore \sqrt[4]{y} = \sqrt[4]{\frac{3}{5}}$
10. $\frac{\sqrt[3]{4} \times \sqrt[3]{8}}{\sqrt{4}} = \frac{\sqrt[3]{4} \times \sqrt[3]{2^3}}{2} = \frac{\sqrt[3]{4} \times 2}{2} = \sqrt[3]{4}$
11. $\sqrt[4]{\frac{625}{9}} = \sqrt{x} \Rightarrow \sqrt[4]{\frac{5^4}{3^2}} = \sqrt{x}$

    $\Rightarrow \frac{5}{\sqrt{3}} = \frac{\sqrt{x}}{1} \Rightarrow \sqrt{x} \cdot \sqrt{3} = 5 \Rightarrow \sqrt{x} = \frac{5}{\sqrt{3}} \ \therefore x = \frac{25}{3}$
12. $\sqrt[4]{625} \times \sqrt[3]{8} = 5 \times 2 = 10$
13. $x = 2 \times \sqrt{48} \times \sqrt{3} = 2 \times \sqrt{16 \times 3} \times \sqrt{3} = 2 \times 4\sqrt{3} \times \sqrt{3} \ \therefore x = 8 \times 3 = 24$

    $y = \sqrt{27} \times \sqrt{12} = \sqrt{3 \times 9} \times \sqrt{3 \times 4} = 3\sqrt{3} \times 2\sqrt{3} = 6 \times 3 = 18$

    $(x^2 + y^2)^{\frac{1}{2}} = (24^2 + 18^2)^{\frac{1}{2}} = (576 + 324) = (900)^{\frac{1}{2}} = 30$
14. $3\sqrt{45} - \sqrt{20} + 7\sqrt{5} \Rightarrow 3\sqrt{9 \times 5} - \sqrt{4 \times 5} + 7\sqrt{5}$

    $\Rightarrow 3\sqrt{(3)^2} \times \sqrt{5} - 2\sqrt{5} + 7\sqrt{5} \Rightarrow 9\sqrt{5} - 2\sqrt{5} + 7\sqrt{5} = \sqrt{5}(9 - 2 + 7) = 14\sqrt{5}$

15. $x = 3 + 2\sqrt{2} \Rightarrow x^2 - 6x + 1 = (3 + 2\sqrt{2})^2 - 6(3 + 2\sqrt{2}) + 1$

$\Rightarrow 9 + 4 \times 2 + 12\sqrt{2} - 18 - 12\sqrt{2} + 1 \Rightarrow 9 + 8 + 1 - 18 = 18 - 18 = 0$

16. Geometric mean of two surds $= 1 \Rightarrow \sqrt{ab} = 1$

$\sqrt{ab} = 1 \Rightarrow \sqrt{(2 + \sqrt{3}) \times b} = 1$

Squaring on both sides

$(2 + \sqrt{3}) \times b = 1$

$\therefore b = \frac{1}{2 + \sqrt{3}} = \frac{1}{2 + \sqrt{3}} \times \frac{2 - \sqrt{3}}{2 - \sqrt{3}} = \frac{2 - \sqrt{3}}{4 - 3} \ \therefore b = 2 - \sqrt{3}$

17. $2 \cdot \sqrt[3]{189} + 3 \cdot \sqrt[3]{875} - 7 \cdot \sqrt[3]{56}$

$= 2 \cdot \sqrt[3]{9 \times 21} + 3 \cdot \sqrt[3]{5 \times 175} - 7 \cdot \sqrt[3]{8 \times 7} = 2 \cdot \sqrt[3]{27 \times 7} + 3 \cdot \sqrt[3]{5 \times 5 \times 35} - 7 \cdot \sqrt[3]{8} \cdot \sqrt[3]{7}$

$= 2.3 \times \sqrt[3]{7} + 3 \cdot \sqrt[3]{125} \times \sqrt[3]{7} - 7 \times 2 \times \sqrt[3]{7} = 6 \times \sqrt[3]{7} + 3 \times 5 \times \sqrt[3]{7} - 14 \times \sqrt[3]{7}$

$= \sqrt[3]{7}(6 + 15 - 14) = \sqrt[3]{7}(21 - 14) = 7 \cdot \sqrt[3]{7}$

18. $x = \sqrt[8]{7} = 7^{\frac{1}{8}}, y = \sqrt[4]{7} = 7^{\frac{1}{4}}$ and $z = \sqrt[2]{7} = 7^{\frac{1}{2}}$

LCM of 8,4,2 = 8

$\Rightarrow x = \sqrt[8]{7} = 7^{\frac{1}{8} \times \frac{1}{1}} = \sqrt[8]{7};$

$\Rightarrow y = \sqrt[4]{7} = 7^{\frac{1}{4} \times \frac{2}{2}} = (7^2)^{\frac{1}{8}} = \sqrt[8]{7^2} = \sqrt[8]{49} \Rightarrow z = \sqrt[2]{7} = 7^{\frac{1}{2} \times \frac{4}{4}} = (7^4)^{\frac{1}{8}} = \sqrt[8]{7^4} = \sqrt[8]{2401}$

$\therefore$ Ascending order is x, y, z

19. $\sqrt[6]{729} = \sqrt[6]{3^6} = 3 \neq 1$

20. $a = 0, \sqrt{b}$ is similar with $\sqrt{d}, \sqrt{e}$

21.

a) $\sqrt[3]{343} = \sqrt[3]{7^3} = 7$

b) $\sqrt[3]{216} = \sqrt[3]{6^3} = 6$

c) $\sqrt[3]{64} = \sqrt[3]{4^3} = 4$

d) $\sqrt[3]{125} = \sqrt[3]{5^3} = 5$

$\therefore \sqrt[5]{a} = \sqrt[5]{7}, \sqrt[5]{b} = \sqrt[5]{6}, \sqrt[5]{c} = \sqrt[5]{4}, \sqrt[5]{d} = \sqrt[5]{5}$

$\therefore$ descending oredr is $\sqrt[5]{7}, \sqrt[5]{6}, \sqrt[5]{5}, \sqrt[5]{4}$

22. $125^{-\frac{1}{3}} \times \left(9^{\frac{-4}{3}} \div 27^{\frac{-2}{3}}\right) \times 3^{\frac{2}{3}} = \sqrt[3]{\frac{1}{125}} \times \left[\frac{\sqrt[3]{\frac{1}{9^4}}}{\sqrt[3]{\left(\frac{1}{27}\right)^2}}\right] \times \sqrt[3]{3^2} = \sqrt[3]{\left(\frac{1}{5}\right)^3} \times \frac{\sqrt[3]{\frac{1}{3^8}}}{\sqrt[3]{\frac{1}{3^6}}} \times \sqrt[3]{9}$

$= \frac{1}{5} \times \sqrt[3]{\frac{3^6}{3^8}} \times \sqrt[3]{9} = \frac{1}{5} \sqrt[3]{\frac{3^6 \times 9}{(3)^8}} = \frac{1}{5} \sqrt[3]{\frac{3^8}{3^8}} = \frac{1}{5}$

Level – III

23. B, C, D 24.A 26.C 27.B 28.A 29. a-t;b-p;c-q;d-r

30. A, B 31. a-p;b-t;c-s;d-r 32. B, D 33. A, B 34.B

23. $\sqrt[4]{81} - 8 \cdot \sqrt[3]{216} + \sqrt{225} + 15 \cdot \sqrt[5]{32} \Rightarrow \sqrt[3]{3^4} - 8 \cdot \sqrt[3]{6^3} + \sqrt{15^2} + 15 \cdot \sqrt[5]{2^5} \Rightarrow 3 - 8 \times 6 + 15 + 15 \times 2$

$\therefore 48 - 48 = 0$

24. $\sqrt[4]{1250} \times \sqrt[4]{8} = 5 \times \sqrt[4]{2} \times \sqrt[4]{8} = 5 \times 2 = 10$

$3\sqrt{75} \times 3\sqrt{48} = 3 \times 5\sqrt{3} \times 3 \times 4\sqrt{3} = 180 \times 3 = 540$

25. π and e are irrational numbers

26. $\sqrt{5} + \sqrt{20} + \sqrt{45} = \sqrt{5} + \sqrt{4 \times 5} + \sqrt{9 \times 5} \Rightarrow \sqrt{5} + 2\sqrt{5} + 3\sqrt{5} = 6\sqrt{5}$

27. $\sqrt{48} - \sqrt{72} - \sqrt{27} + 2\sqrt{18}$

$\Rightarrow \sqrt{16 \times 3} - \sqrt{36 \times 2} - \sqrt{9 \times 3} + 2\sqrt{9 \times 2} \Rightarrow 4\sqrt{3} - 6\sqrt{2} - 3\sqrt{3} + 2 \times 3\sqrt{2}$

$\therefore \sqrt{3}$

28. $A = 4\sqrt{2} + 2\sqrt{3}, B = 4\sqrt{2} - 2\sqrt{3}$

$\Rightarrow (A - B)^2 = (4\sqrt{2} + 2\sqrt{3} - 4\sqrt{2} + 2\sqrt{3})^2 \Rightarrow (4\sqrt{3})^2 = 16 \times 3 = 48$

29. a) $\sqrt[3]{2} \times \sqrt[3]{4} = \sqrt[3]{2 \times 4} = \sqrt[3]{8} = \sqrt[3]{2^3} = 2$ b) $\sqrt[5]{30} \div \sqrt[5]{6} = \sqrt[5]{\frac{30}{6}} = \sqrt[5]{5}$

c) $\sqrt[2]{\sqrt[3]{5}} = \sqrt[2\times3]{5} = \sqrt[6]{5}$

d) $\sqrt[5]{\sqrt[7]{[(2010)^7]^5}} = (2010)^{\frac{35}{35}} = 2010$

30. $\sqrt[6]{2187} = \sqrt[6]{3^6 \times 3} \Rightarrow \sqrt[6]{3^6} \times \sqrt[6]{3} \Rightarrow 3 \times \sqrt[6]{3}$

Its order = 6 is a rational number

31. a) $\sqrt{8} \times \sqrt{2} = \sqrt{4 \times 2} \times \sqrt{2} = 2 \times \sqrt{2} \times \sqrt{2} = 2 \times 2 = 4$

b) $\sqrt[3]{2} \times \sqrt[3]{2} \times \sqrt[3]{2} = \sqrt[3]{2^3} = 2$

c) $\sqrt[4]{2} \div \sqrt[8]{4} = 2^{\frac{1}{4}}, 4^{\frac{1}{8}}$

LCM 4,8 = 8

$\Rightarrow 2^{\frac{1}{4}} = 2^{\frac{1}{4} \times \frac{2}{2}} = 2^{\frac{2}{8}} = (2^2)^{\frac{1}{8}} = \sqrt[8]{4} \Rightarrow 4^{\frac{1}{8}} = 4^{\frac{1}{1}^{-\frac{1}{1}}} = \sqrt[8]{4}$

$\therefore \sqrt[4]{2} \div \sqrt[8]{4} = \frac{\sqrt[8]{4}}{\sqrt[8]{4}} = \sqrt[8]{\frac{4}{4}} = 1$

d) $(\sqrt{9} - \sqrt[3]{27}) \div (7\sqrt{4} \times 7\sqrt{2}) \Rightarrow \left(3 - \sqrt[3]{3^3}\right) \div (7\sqrt{8}) \Rightarrow (3 - 3) \div (\sqrt[7]{8}) = \frac{0}{\sqrt[7]{8}} = 0$

32. $A + B = 2\sqrt{8}, A - B = 2\sqrt{3} \Rightarrow (A + B)^2 + (A - B)^2 = 32 + 12 = 44$

33. $\sqrt[4]{5}, \sqrt[3]{4}, \sqrt[4]{7} \Rightarrow 5^{\frac{1}{4}}, 4^{\frac{1}{3}}, 7^{\frac{1}{4}} \Rightarrow$ LCM of 4,3,4 = 12

$\Rightarrow 5^{\frac{1}{4}} = 5^{\frac{1}{4} \times \frac{3}{3}} = 5^{\frac{3}{12}} = (5^3)^{\frac{1}{12}} = \sqrt[12]{125} \Rightarrow 4^{\frac{1}{3}} = 4^{\frac{1}{3} \times \frac{4}{4}} = (4^4)^{\frac{1}{12}} = \sqrt[12]{256}$

$\Rightarrow 7^{\frac{1}{4}} = 7^{\frac{1}{4} \times \frac{3}{3}} = 7^{\frac{3}{12}} = (7^3)^{\frac{1}{12}} = \sqrt[12]{343}$

Ascending order $\sqrt[4]{5}, \sqrt[3]{4}, \sqrt[4]{7}$

2. $\sqrt[3]{4}, \sqrt[4]{5}, \sqrt[4]{6} \Rightarrow 4^{\frac{1}{3}}, 5^{\frac{1}{4}}, 6^{\frac{1}{4}}$

LCM of 3,4,3 = 12

$\Rightarrow 4^{\frac{1}{3}} = 4^{\frac{1}{3}\times\frac{4}{4}} = 4^{\frac{4}{12}} = \sqrt[12]{4^4} = \sqrt[12]{256} \Rightarrow \sqrt[4]{5} = 5^{\frac{1}{4}\times\frac{3}{3}} = 5^{\frac{3}{12}} = \sqrt[12]{125}$

$\Rightarrow \sqrt[4]{6} = 6^{\frac{1}{4}\times\frac{3}{3}} = 6^{\frac{3}{12}} = \sqrt[12]{6^3} = \sqrt[12]{216}$

Ascending order $\sqrt[4]{5}, \sqrt[4]{6}, \sqrt[3]{4}$

3. $\sqrt[3]{4}, \sqrt[4]{7}, \sqrt[3]{8} \Rightarrow 4^{\frac{1}{3}}, 7^{\frac{1}{4}}, 8^{\frac{1}{3}}$

LCM of 3,4,4 = 12 $\Rightarrow \sqrt[3]{4} = 4^{\frac{1}{3}\times\frac{4}{4}} = 4^{\frac{4}{12}} = (4^4)^{\frac{1}{12}} = \sqrt[12]{4^4} = \sqrt[12]{256}$

$\Rightarrow \sqrt[4]{7} = 7^{\frac{1}{4}\times\frac{3}{3}} = 7^{\frac{3}{12}} = \sqrt[12]{7^3} = \sqrt[12]{343} \Rightarrow \sqrt[3]{8} = 8^{\frac{1}{3}\times\frac{4}{4}} = 8^{\frac{4}{12}} = \sqrt[12]{8^4} = \sqrt[12]{4096}$

Ascending order $\sqrt[3]{4}, \sqrt[4]{5}, \sqrt[4]{7}$

34. $13 - \sqrt[3]{5x-4} = 7 \Rightarrow 5x - 4 = 216 \Rightarrow x = \frac{220}{5} = 44$

WORKSHEET-2

Level – I

1.B	2.A	3.C	4.C	5.D	6.B	7.A
8.A	9.C	10.A				

Level – II

1.C	2.B	3.D	4.C	5.C	6.C	7.D
8.C	9.A	10.B	11.D	12.A	13.C	14.A
15.C	16.D	17.C	18.D	19.2		

1. $\sqrt[3]{36} = \sqrt[3]{6^2}; \sqrt[3]{6^2} \times \sqrt[3]{6} = \sqrt[3]{6^3} = 6$

∴ The rationalising factor of $\sqrt[3]{36}$ is $\sqrt[3]{6}$.

2. $(3+\sqrt{2})k$ = a ratinal number ⇒ k is R.F of $3+\sqrt{2} = 3-\sqrt{2}$

∵ $(3+\sqrt{2})(3-\sqrt{2}) = 3^2 - (\sqrt{2})^2 = 9 - 2 = 7$

3. $\sqrt{2} = 1.414 \Rightarrow \frac{10}{\sqrt{2}} = \frac{10}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{10\sqrt{2}}{2} = 5\sqrt{2} = 5 \times 1.414 = 7.07$

4. R.F of a given surd is not unique i.e. infinite

5. If $x > 0$ then $\sqrt{x\sqrt{x\sqrt{x \ldots . n \text{ roots}}}} = x^{2^{2^n - 1}}$

6. $\sqrt[3]{192} = \sqrt[3]{3 \times 2^6} \Rightarrow \sqrt[3]{3} \times \sqrt[3]{2^6} \Rightarrow \sqrt[3]{3} \times 2^2 \Rightarrow 4 \times \sqrt[3]{3}$

∴ simplest R.F = $\sqrt[3]{3^2} = \sqrt[3]{9}$

7. $A = \frac{1}{\sqrt{5}-\sqrt{3}} = \frac{1}{\sqrt{5}-\sqrt{3}} \times \frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}+\sqrt{3}} = \frac{\sqrt{5}+\sqrt{3}}{2}$

$B = \dfrac{1}{\sqrt{5}+\sqrt{3}} = \dfrac{1}{\sqrt{5}+\sqrt{3}} \times \dfrac{\sqrt{5}-\sqrt{3}}{\sqrt{5}-\sqrt{3}} = \dfrac{\sqrt{5}-\sqrt{3}}{2}$

$A \times B = \frac{\sqrt{5}+\sqrt{3}}{2} \times \frac{\sqrt{5}-\sqrt{3}}{2} = \frac{2}{4} = \frac{1}{2}$

8. $\sqrt{75}+\sqrt{32} = \sqrt{25\times 3}+\sqrt{16\times 2} = 5\sqrt{3}+4\sqrt{2}$

Its rationalising factor is $5\sqrt{3}-4\sqrt{2}$

9. $x = \frac{1}{2-\sqrt{3}} = \frac{1}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}} = \frac{2+\sqrt{3}}{2^2-(\sqrt{3})^2} = \frac{2+\sqrt{3}}{4-3} = 2+\sqrt{3}$

$\Rightarrow \frac{1}{x} = \frac{1}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}} = \frac{2-\sqrt{3}}{2^2-(\sqrt{3})^2} = \frac{2-\sqrt{3}}{4-3} = \frac{2-\sqrt{3}}{1} = 2-\sqrt{3}$

$\Rightarrow \left(x-\frac{1}{x}\right)^2 = [2+\sqrt{3}-(2-\sqrt{3})]^2 = (2+\sqrt{3}-2+\sqrt{3})^2 = (2\sqrt{3})^2 = 3\times 4 = 12$

10. $x = \sqrt{3}+\sqrt{2}$

$\Rightarrow \frac{1}{x} = \frac{1}{\sqrt{3}+\sqrt{2}} \times \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}-\sqrt{2}} = \frac{\sqrt{3}-\sqrt{2}}{3-2} = \sqrt{3}-\sqrt{2} \Rightarrow x+\frac{1}{x} = \sqrt{3}+\sqrt{2}+\sqrt{3}-\sqrt{2} = 2\sqrt{3}$

11. $\sqrt[3]{a}+\sqrt[3]{b}+\sqrt[3]{c} = 0 \Rightarrow a^{\frac{1}{3}}+b^{\frac{1}{3}}+c^{\frac{1}{3}} = 0$

If $a+b+c = 0$ then $a^3+b^3+c^3 = 3abc$

$\Rightarrow \left(a^{\frac{1}{3}}\right)^3+\left(b^{\frac{1}{3}}\right)^3+\left(c^{\frac{1}{3}}\right)^3 = 3a^{\frac{1}{3}}b^{\frac{1}{3}}c^{\frac{1}{3}} \Rightarrow a+b+c = 3(abc)^{\frac{1}{3}}$

Cubing on both sides $\Rightarrow (a+b+c)^3 = \left[(abc)^{\frac{1}{3}}\right]^3 = 27\cdot abc$

12. $\frac{1+\sqrt{2}}{3-2\sqrt{2}} = a+b\sqrt{2} \Rightarrow \frac{1+\sqrt{2}}{3-2\sqrt{2}} \times \frac{3+2\sqrt{2}}{3+2\sqrt{2}} = a+b\sqrt{2} \Rightarrow \frac{3+2\sqrt{2}+3\sqrt{2}+2(\sqrt{2})^2}{9-4\times 2} = a+b\sqrt{2}$

$\Rightarrow \frac{3+5\sqrt{2}+4}{1} = a+b\sqrt{2} \Rightarrow 7+5\sqrt{2} = a+b\sqrt{2}$

$\Rightarrow a = 7, b = 5 \ \therefore (a,b) = (7,5)$

13. $\sqrt{10+2\sqrt{21}} = \sqrt{7}+\sqrt{x} \Rightarrow 10+2\sqrt{21} = 7+x+2\sqrt{7x} \Rightarrow x+7 = 10 \Rightarrow x = 3$

14. $\frac{1+\sqrt{2}}{\sqrt{5}+\sqrt{3}}+\frac{1-\sqrt{2}}{\sqrt{5}-\sqrt{3}} = \frac{1+\sqrt{2}}{\sqrt{5}+\sqrt{3}} \times \frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}-\sqrt{3}}+\frac{1-\sqrt{2}}{\sqrt{5}-\sqrt{3}} \times \frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}+\sqrt{3}}$

$\frac{\sqrt{5}-\sqrt{3}+\sqrt{10}-\sqrt{6}}{5-3}+\frac{\sqrt{5}+\sqrt{3}-\sqrt{10}-\sqrt{6}}{5-3} = \frac{2(\sqrt{5}-\sqrt{6})}{2} = \sqrt{5}-\sqrt{6} = 2.236-2.449$

$= -0.213$

15. $x = \frac{5-\sqrt{21}}{2}; \frac{1}{x} = \frac{2}{5-\sqrt{21}} \times \frac{5+\sqrt{21}}{5+\sqrt{21}} = \frac{2(5+\sqrt{21})}{25-21} = \frac{(5+\sqrt{21})}{2}$

$x+\frac{1}{x} = \frac{5-\sqrt{21}}{2}+\frac{5+\sqrt{21}}{2} = \frac{10}{2} = 5$

16. $x = \sqrt[3]{2+\sqrt{3}} \Rightarrow x^3 = 2+\sqrt{3} \Rightarrow \frac{1}{x^3} = \frac{1}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}} = \frac{2-\sqrt{3}}{4-3} = 2-\sqrt{3}$

$\Rightarrow x^3+\frac{1}{x^3} = 2+\sqrt{3}+2-\sqrt{3} = 4$

17. $\frac{7+3\sqrt{5}}{3+\sqrt{5}}+\frac{7-3\sqrt{5}}{3-\sqrt{5}} = \frac{7+3\sqrt{5}}{3+\sqrt{5}} \times \frac{3-\sqrt{5}}{3-\sqrt{5}}+\frac{7-3\sqrt{5}}{3-\sqrt{5}} \times \frac{3+\sqrt{5}}{3+\sqrt{5}} =$

$$\frac{21-7\sqrt{5}+9\sqrt{5}-15+21+7\sqrt{5}-9\sqrt{5}-15}{9-5} = \frac{42-30}{4} = 3$$

18. $\sqrt{47+4\sqrt{33}}=\sqrt{47+2\sqrt{33\times4}}=\sqrt{(\sqrt{44}+\sqrt{3})^2}=\sqrt{44}+\sqrt{3}$

19. $\sqrt{13-\sqrt{160}}=\sqrt{13-\sqrt{4\times40}}=\sqrt{13-2\sqrt{40}}=\sqrt{(\sqrt{8}-\sqrt{5})^2}=\sqrt{8}-\sqrt{5}=2\sqrt{2}-\sqrt{5}$

Level – III

20.A, B, D 21.B 22.C 23.A 24.7 25.B 26.C

27.A 28.a-s;b-t;c-r;d-q 29.5 30.4 31.6 32.C

33.B 34.A

20. $\sqrt[3]{2}\times\sqrt[3]{4}=\sqrt[3]{8}=2\Rightarrow\sqrt[3]{2}\times\sqrt[3]{32}=\sqrt[3]{64}=4\Rightarrow\sqrt[3]{2}\times\sqrt[3]{256}=\sqrt[3]{512}=8$

21. $\sqrt[5]{343}=\sqrt[5]{7^3}$; R.F of $\sqrt[5]{7^3}$ is $\sqrt[5]{7^{5-3}}=\sqrt[5]{7^2}=\sqrt[5]{49}[\because R$ R.F of $\sqrt[m]{a^n}=\sqrt[m]{a^{m-n}}]$

22. $\frac{5+\sqrt{3}}{5-\sqrt{3}}=a+b\sqrt{3}\Rightarrow\frac{5+\sqrt{3}}{5-\sqrt{3}}\times\frac{5+\sqrt{3}}{5+\sqrt{3}}=a+b\sqrt{3}$

$\Rightarrow\frac{(5+\sqrt{3})^2}{25-3}=a+b\sqrt{3}\Rightarrow\frac{25+3+10\sqrt{3}}{22}=a+b\sqrt{3}$

$\Rightarrow\frac{28+10\sqrt{3}}{22}=a+b\sqrt{3}\Rightarrow\frac{14+5\sqrt{3}}{11}=a+b\sqrt{3}\Rightarrow a=\frac{14}{11};b=\frac{5}{11}$

23. $(\sqrt[3]{4}+\sqrt[3]{2}+1)(\sqrt[3]{2}-1)=\left(\sqrt[3]{2^2}+\sqrt[3]{2}+1^2\right)(\sqrt[3]{2}-1)=(\sqrt[3]{2})^3-(1)^3=2-1=1$

So, the R.F of $(\sqrt[3]{4}+\sqrt[3]{2}+1)$ is $(\sqrt[3]{2}-1)$.

24. $x=\sqrt{42+\sqrt{42+\sqrt{42+}}}\infty\Rightarrow x^2=42+x\Rightarrow x^2-x-42=0.$

$\Rightarrow(x-7)(x+6)=0\Rightarrow x=7.$

25. $\frac{\sqrt[3]{2}\times\sqrt[6]{6}}{\sqrt[6]{3}\times\sqrt[6]{8}}=\frac{\sqrt[6]{4}\times\sqrt[6]{6}}{\sqrt[6]{3}\times\sqrt[6]{8}}=\frac{\sqrt[6]{24}}{\sqrt[6]{24}}=1$

26. $\frac{\sqrt[6]{12}}{\sqrt{3}\sqrt[3]{2}}=\frac{\sqrt[6]{12}}{\sqrt[6]{27}\sqrt[6]{4}}=\sqrt[6]{\frac{12}{27\times4}}=\sqrt[6]{\frac{1}{9}}=3^{\frac{-1}{3}}=\frac{1}{\sqrt[3]{3}}$

27. $\frac{\sqrt[5]{2}\times\sqrt{5}}{\sqrt[3]{27}\times\sqrt{125}}=\frac{\sqrt[5]{2}\times\sqrt{5}}{\sqrt[3]{3^3}\times\sqrt{5\times25}}=\frac{\sqrt[5]{2}\times\sqrt{5}}{3\times5\sqrt{5}}=\frac{\sqrt[5]{2}}{15}$

28. a) $\frac{5}{\sqrt{3}}=\frac{5\sqrt{3}}{3}$

b) $\frac{7}{2\sqrt{13}}=\frac{7\sqrt{13}}{2\times13}=\frac{7\sqrt{13}}{26}$

c) $\frac{5\sqrt[5]{3}}{\sqrt[5]{45}}=\frac{5\sqrt[5]{3}}{\sqrt[5]{45}}=\sqrt[5]{\frac{3\times5^5}{45}}=\sqrt[5]{\frac{625}{3}}$

d) $\frac{3\sqrt{5}}{\sqrt{11}}=\frac{3\sqrt{5}}{\sqrt{11}}\times\frac{\sqrt{11}}{\sqrt{11}}=\frac{3\sqrt{55}}{11}$

29. $x=2\sqrt{6}+5;\frac{1}{x}=\frac{1}{5+2\sqrt{6}}\times\frac{5-2\sqrt{6}}{5-2\sqrt{6}}=\frac{5-2\sqrt{6}}{25-24}=5-2\sqrt{6}$

$x+\frac{1}{x}=5+2\sqrt{6}+5-2\sqrt{6}=10.\Rightarrow x+\frac{1}{x}=5+y\Rightarrow5+y=10\Rightarrow y=5$

30. $\sqrt[3]{20+14\sqrt{2}} = x + \sqrt{y}$

Cubing on both sides $\Rightarrow 20 + 14\sqrt{2} = (x^3 + 3xy) + \sqrt{y}(3x^2 + y)$

$\Rightarrow x^3 + 3xy = 20$

Let $\sqrt[3]{20-14\sqrt{2}} = x - \sqrt{y}$

$\Rightarrow (1) \times (2)$

$\Rightarrow \sqrt[3]{400 - 196(2)} = x^2 - y \Rightarrow \sqrt[3]{8} = x^2 - y$

$\Rightarrow x^2 - y = 2$

Now, $x^3 + 3x(x^2 - 2) = 20 \Rightarrow x^3 + 3x^3 - 6x - 20 = 0 \Rightarrow 4x^3 - 6x - 20 = 0 \Rightarrow 2x^3 - 3x - 10 = 0$

Let $f(x) = 2x^3 - 3x - 10 \Rightarrow f(2) = 16 - 6 - 10 = 0 \ \therefore x = 2$

$y = x^2 - 2 = 2$

$\therefore \sqrt[3]{20+14\sqrt{2}} = 2 + \sqrt{2}; \sqrt[3]{20-14\sqrt{2}} = 2 - \sqrt{2}$

$\Rightarrow \sqrt[3]{20+14\sqrt{2}} + \sqrt[3]{20-14\sqrt{2}} = 2 + \sqrt{2} + 2 - \sqrt{2} = 4$

31. $\frac{6-2\sqrt{3}}{33-19\sqrt{3}} = a + b\sqrt{3} \Rightarrow \frac{6-2\sqrt{3}}{33-19\sqrt{3}} \times \frac{33+19\sqrt{3}}{33+19\sqrt{3}} = a + b\sqrt{3}$

$\Rightarrow \frac{198 + 114\sqrt{3} - 66\sqrt{3} - 38 \times 3}{33^2 - (19\sqrt{3})^2} \Rightarrow \frac{84 + 48\sqrt{3}}{6} = a + 3\sqrt{3} \Rightarrow 14 + 8\sqrt{3} = a + b\sqrt{3}$

$\Rightarrow a - b = 14 - 8 = 6$

32. The rationalizing factor of $\sqrt[7]{9^3} = \sqrt[7]{9^{7-3}} = \sqrt[7]{9^4}$

33. The rationalizing factor of $\sqrt[11]{12^7 \times 5^{10}} = \sqrt[11]{12^{11-7} \times 5^{11-10}} = \sqrt[11]{12^4 \times 5^1}$

34. The rationalizing factor of s $\sqrt[15]{23^{11} \times 2^{10} \times 5^{13}} = \sqrt[15]{23^{15-11} \times 2^{15-10} \times 5^{15-13}} = \sqrt[15]{23^4 \times 2^5 \times 5^2}$

4. LOGARITHMS

WORKSHEET-1

Level – I

1. A	2. A	3. A	4. B	5. C	6. B
7. C	8. C	9. B	10. A		

4. $5 = \log_2^{32}$ (By using $a = b^p \Rightarrow \log_b^a = p$)

5. By using $a = b^p \Rightarrow \log_b^a = p$

6. $\log_a^{a^m} = m\log_a^a = m \Rightarrow \log_5^{5^2} = 2\log_5^5 = 2$

7. $\log_4^{64} = \log_4^{4^3} = 3\log_4^4 = 3$

8. $\log_x^{\sqrt{x}} = \log_x^{x^{\frac{1}{2}}} = \frac{1}{2}\log_x^x = \frac{1}{2}$

Level – II

1. C	2. D	3. D	4. C	5. D	6. A
7. A	8.D	9.16	10. C	11. A	12. B
13. B	14. B	15. B	16. A	17. B	18. B
19. D	20. A	21. A	22. B		

1. $a^{\log_a^N} = N$

   $5^{\log_5 27} = 27$

2. $\log_a 1 = 0$

3. $9^3 = 729$

4. $a = 5^2$

5. If $a^x = N \Rightarrow \log_a N = x$

6. $\log_a N = x \Rightarrow a^x = N$

7. By the defination of logarithm

8. By the defination

9. $y = 2^4$

10. $\log_9 27 - \log_{27} 9 = \log_{3^2} 3^3 - \log_{3^3} 3^2 = \frac{3}{2} - \frac{2}{3} = \frac{5}{6}$

11. $\log_a x^m = m\log_a x$

12. $\log_a \left(\frac{x}{y}\right) = \log_a x - \log_a y$

13. Let a = 1, b = 2, c = 3

14. $\log_{b^n} a^m = \frac{m}{n}\log_b a$

15. $0.01 = 10^{-2}$

16. $\log_{10} 9^{100} = 100\log_{10} 3^2 = 95.42$

17. $\log_2 \log_3 \log_5 x = 0 \Rightarrow \log_3 \log_5 x = 2^0 \Rightarrow \log_5 x = 3 \Rightarrow x = 5^3$

18. $\log_{10} x(x-9) = 1 \Rightarrow x^2 - 9x = 10$

    $x^2 - 9x - 10 = 0 \Rightarrow (x-10)(x+1) = 0 \Rightarrow x = 10$ or -1

19. $\log_a N = x \Rightarrow a^x = N$

20. $x = 100^a = 10^{2a}$

21. $3^a = 3^3 \Rightarrow a = 3$

22. $\log 7\sqrt{7} = \frac{3}{2}\log 7$

Level – III

23. A,C,D 24. C 25. C 26. C 27.a-s, b-r;c-t; d-s

28. C 29. C 30. a-r; b-q,t; c-t; d-q 31. D 32. C

33. C 34. D 35. a-r; b-p; c-s; d-q

23. $2^{\log_2 15^2} = 15^2 = 225$

24. $\frac{1}{\log_a a + \log_a bc} + \frac{1}{\log_b b + \log_b bc} + \frac{1}{\log_c c + \log_c ab} = \log_{abc} a + \log_{abc} b + \log_{abc} c = 1$

25. $\log_{xy} x + \log_{xy} y = 1$

26. $\log_2 8 = 3$

27. a) $\log_{10} 10^3 = 3$

    b) $\log_2 2^4 = 4$

    c) $\log_{10} 10^{-3} = -3$

    d) $\log_{10^{-1}} 10^{-3} = 3$

28. $a + b = ab \Rightarrow b = ab - a = a(b-1) \Rightarrow a = \frac{b}{b-1}$

29. $\log_2 x \left[1 + \frac{1}{2} + \frac{1}{4}\right] = \frac{21}{4} \Rightarrow \log_2 x = 3 \Rightarrow x = 2^3$

30. 1.$\log_{0.5}(0.5)^4$

    2.$\log_2 2^3$

    3.$\frac{\log 3}{\log 4} \times \frac{\log 4}{\log 5} \times \frac{\log 5}{\log 3}$

    4.$\log_3 \log_2 8 = 1$

31. $\log A = \log 2 + \log \pi + \log \sqrt{\frac{l}{g}} = \log 2 + \log \pi + \frac{1}{2}(\log l - \log g)$

32. $\log B = \log 2 + \log \pi$

33. $\log \mathrm{c} = \frac{1}{2}[\log l - \log \mathrm{g}]$

34. $\log_2 \frac{x^2+x}{x+1} = 2 \Rightarrow x = 2^2$

35.

    1.$\log_6 6^{3+\frac{1}{2}} = \frac{7}{2}$

    2. $\log_b a = \dfrac{\log a}{\log b}$

    3.$256 = 4^4 \Rightarrow \log_2 \log_2 \log_2 256 = 1$

4. $\log_3 \frac{3}{2} \cdot \frac{4}{3} \cdot \frac{5}{4} - - - - - \frac{54}{53} = \log_3 \frac{54}{2} = \log_3 27 = 3$

WORKSHEET-2

Level – I

1. B	2. A	3. C	4. A	5. C	6. A
7. A	8. A	9. D	10. A		

Level – II

1. B	2. B	3. A	4. B	5. B	6. C	7. D
8. D	9. B	10. B	11. A	12. B	13. 3	14. B
15. D	16. 13	17. A	18. C	19. C	20. C	21. A
22. 21	23. A,C	24. A	25. B	26. C	27. a-p,s;b-p,s;c-q,r;d-q,r	
28. 8	29. A,B,C	30. A,B,C	31. 4	32. a-q; b-r; c-p; d-t		33. B
34. B	35. A					

1. Integral part of a logarithm
2. $40\log_{10} 3 = 40(0.4771) = 19.084$

   Number of digits in $3^{40} = 20$
4. $\log(4.631) = 0.6657$

   mantissa (decimal part) $= 0.6657$
5. $\log_2 x + \log_4 x + \log_8 x = 11$

   If $x = 64, \log_2 64 + \log_4 64 + \log_8 64 = 6 + 3 + 2 = 11$
6. $\log_{0.2} x > 0 \Rightarrow x < (0.2)^0 \Rightarrow x < 1$
7. $\log_{0.5} x^2 < \log_{0.5} 25 \Rightarrow x^2 > 25$

   The possible values of x are 6,7
8. By using the values we have to check
9. $x^2 + y^2 = z^2 \Rightarrow x^2 = z^2 - y^2 \Rightarrow x^2 = (z + y)(z - y)$

   $\log_x(z - y) + \log_x(z + y) \Rightarrow \log_x(z^2 - y^2) \Rightarrow \log_x x^2 = 2\log_x x = 2$
10. $\log 0.0075 = \log \frac{75}{10000} = \log 75 - \log 10^4$

    $= 2\log 5 + \log 3 - 4\log 5 - 4\log 2 = \log 3 - 4\log 2 - 2\log 5 = -2.1249$
11. $\log(0.3274) = \bar{1}.5150$
12. $5^x = 10^3$

    $(0.5)^y = 10^3$

    $5 = 10^{3/x}$

    $0.5 = 10^{3/y}$

    $5 = 10^{\frac{3}{y}+1}$

    $10^{\frac{3}{x}} = 10^{\frac{3}{y}+1} \Rightarrow \frac{3}{x} = \frac{3}{y} + 1 \Rightarrow \frac{1}{x} - \frac{1}{y} = \frac{1}{3}$
13. $\log_{10} x = 1.9675$

    $\log_{10} 100x = 2\log_{10} 10 + \log_{10} x = 2 + 1.9675 = 3.9675$

14. $= \log_{10} 2 = 0.3010, \log 5^{20} = 20\log 5$

$= 200^{\log\left(\frac{10}{2}\right)} = 20[\log_{10} 10 - \log_{10} 2] = 20[1 - 0.3010] = 20 \times 0.699 = 13.98$

Number of digits in 5^{20} is 14

15. $\log_9 x - \log_9\left(\frac{x}{10} + \frac{1}{9}\right) = 1 = \log_9\left(\frac{x}{\frac{x}{10}+\frac{1}{9}}\right) = 1 \Rightarrow \frac{x}{\frac{x}{10}+\frac{1}{9}} = 9^1 \Rightarrow 10x = 9x + 10 \Rightarrow x = 10$

16. Let $y = 3^{3^3} \Rightarrow \log y = 3^3 \log 3$

$\log y = 27\log 3$

$\log y = 27(0.4771) = 12.88$

Number of digits in $3^{3^3} = 13$

17. $\log\frac{1}{2} = \bar{1}.6990$ Number of zeros are 30

18. $\log_5\left(\log_3(\log_2 x)\right) = 1 \Rightarrow \log_3(\log_2 x) = 5^1 \Rightarrow \log_2 x = 3^5 \Rightarrow x = 2^{243}$

19. $p = \log_3 5, q = \log_{17} 25 p = \frac{\log 5}{\log 3}, q = \frac{\log 25}{\log 17} \Rightarrow p > q$

20. $3^{1+x} = 7^{x/2} \Rightarrow \log 3^{1+x} = \log 7^{x/2} \Rightarrow (1+x)\log 3 = \frac{x}{2}\log 7 \Rightarrow (2+2x)\log 3 = x\log 7$

$2x\log 3 - x\log 7 = -2\log 3 \Rightarrow x[2\log 3 - \log 7] = -2\log 3 \Rightarrow x = \dfrac{2\log 3}{\log 7 - 2\log 3}$

$x = -8.7$

21. $\log_{10} 11 = 1.0414 \Rightarrow 10^{11} > 11^{10} \Rightarrow 11\log 10 > 10\log 11$

22. $\log 3 = 0.4771$

$\log 81^{11} = 11\log 81 = 11\log 3^4 = 44\log 3 = 44(0.4771) = 20.9924$

The number of digits in $(81)^{11}$ is 21

23. $\log_{10} x - \log_{10}\sqrt{x} = \frac{2}{\log_{10} x} \Rightarrow \log_{10} x - \frac{1}{2}\log_{10} x = \frac{2}{\log_{10} x}$

$\dfrac{1}{2}\log_{10} x = \dfrac{2}{\log_{10} x} \Rightarrow \log_{10} x = \dfrac{4}{\log_{10} x}$

$(\log_{10} x)^2 = 4 \Rightarrow \log_{10} x = 2 \Rightarrow x = 10^2 \Rightarrow x = 100$

24. $\log 15$, charecteristic is $2 - 1 = 1$

25. $25.\log 293$, charecteristic is $3 - 1 = 2$

26. $\log(0.00602)$ charecteristic is $\bar{3}$

27. a) $\log_3 0.2 \times \log_{20} 3 =$ (negative) $\Rightarrow [\log_3 2 - \log_3 10]\log_{20} 3$

b) $\log_{5/2}(1/5) \times \log_{10} 5$ (negative) $= -\log_{5/2} 5 \times \log_{10} 5$

c) $\log_{(3/2)} 2/3 \times \log_{10/3} 1/2$ is positive

d) $\log_5 5 = 1$ is positive

28. $\log_a(x-2) > \log_a 5$

$x - 2 > 5$

$x > 7$

Integer value greater than 7 is 8

29. $5^x = 400 \Rightarrow \log 5^x = \log 400 \Rightarrow \log_5 400$

$\Rightarrow x\log 5 = 4\log 2 + 2\log 5 \Rightarrow x = \dfrac{4\log 2 + 2\log 5}{\log 5} \Rightarrow$ x = 3.72

30. $= \log_{1/2}\frac{3}{2} \times \log_{1/2}\frac{5}{3} \times \log_{1/2} 0.9$

negative × negative × positive = positive

31. $2\log_{10} x - \log_x 0.01$

$2\log_{10} x + 2\log_x 10 = 2[\log_{10} x + \log_x 10] = 2(2) = 4$

32. $1) 10^{\log_x 83} = 4^{\log_9 3} + 9^{\log_2 4} \Rightarrow 10^{\log_x 83} = 2 + 81 \Rightarrow 10^{\log_x 83} = 83 \Rightarrow$ x = 10

$2) 1 + 3 + 5 + - - - - + 2n - 1 = n^2$

$3) 2^{\log_3 5} + 3^{\log_2 5} - 3^{\log_2 5} - 2^{\log_3 5} = 0$

$4) 0.16^{\log_{2.5}\left(\frac{1}{2}\right)} = 4$

33. $\frac{1}{\log_2 n} + \frac{1}{\log_3 n} + \frac{1}{\log_4 n} + - - - + \frac{1}{\log_{13} n}$

$= \log_n 2 + \log_n 3 + \log_n 4 + - - - - + \log_n 13 = \log_n(2 \times 3 \times 4 \times 5 \times - - - \times 13) = \log_n(13!)$

34. $\log_a x + \log_x a (AM \geq GM)$

35. $f(a) = \log\left(\frac{2+a}{2-a}\right) \Rightarrow \frac{1}{2} + \left(\frac{8a}{4+a^2}\right) = \frac{1}{2}\log\left(\frac{2+\frac{8a}{4+a^2}}{2-\frac{8a}{4+a^2}}\right) = \frac{1}{2}\log\left(\frac{8+2a^2+8a}{8+2a^2-8a}\right)$

$= \frac{1}{2}\log\left(\frac{4+a^2+4a}{4+a^2-4a}\right) = \frac{1}{2}\log\left(\frac{2+a}{2-a}\right)^2 \Rightarrow$ f(a)

5. POLYNOMIALS

WORKSHEET-1

Level – I

1. A	2. D	3. A	4. A	5. C	6. D	7. B
8. C	9. B	10. D	11. C	12. A	13. B	14. A

Level – II

1. A	2. D	3. D	4. C	5. A	6. B	7. C
8. A	9. B	10. A	11.B	12.C	13.B	14. D
15.C	16.A	17.A	18.A	19.A		

1. The highest power of the variable present in the polynomial is called its degree. So, the degree of the polynomial is n
2. $\frac{x^3}{x^2}+\frac{x^4}{x^2}-\frac{x^6}{x^2}=x+x^2-x^4$
3. $x^7+6x^4+3x^2-4x^{-1}+2$ is nota polynomial because variable contain negative power.
4. $25x^2+70x+49=25(-1)^2+70(-1)+49=4$
5. $x^2-2x=0\Rightarrow x(x-2)=0\Rightarrow x=0,2$
6. $6x^2-\left(\frac{1}{3}+\frac{2}{3}\right)x+\left(\frac{5}{2}+\frac{1}{2}\right)=6x^2-x+3$
7. $(-y^3-6x^2y-xy^2+x^3)-(x^3-xy^2+5x^2y-y^3)=-11x^2y$
8. Additive identity for any polynomial is zero.
9. $\frac{A}{B}=\frac{-126m^6n^2x}{6m^2nx}=-21m^4n$
10. $2a^3(3a+5b)=2a^3(3a)+2a^3(5b)=6a^4+10a^3b=6(1)^4+10(1)^3(2)=26$
11. x^2+2x+1 is the only polynomial
12. $p(x)=3x^2-1$ to find zeroes of polynomial, equate $p(x)=0$

    $3x^2-1=0\Rightarrow 3x^2=1\rightarrow x^2=\frac{1}{3}\Rightarrow x=\pm\frac{1}{\sqrt{3}}$
13. $(a^2-2b^2)(3a^2+b^2)=3a^4+a^2b^2-6a^2b^2-2b^4=3a^4-5a^2b^2-2b^4$
14. $(3x^3-5x^2+2x-3)-(2x^3-3x^2+x+1)=x^3-2x^2+x-4$
15. $x-1)2x^4-5x^2+5x-2(2x^3+2x^2-3x+2$

$2x^4 - 2x^3$
$-$
$2x^3 - 5x^2$
$2x^3 - 2x^2$
$+$
$-3x^2 - 5x$
$-3x^2 + 3x$
$+ \; - \; 2$
$2x - 2$
$+$

Quotient $2x^3 + 2x^2 - 3x + 2$

16. $\frac{-4}{27}xyz\left[\frac{9}{2}x^2yz - \frac{3}{4}xyz^2\right] \Rightarrow -\left(\frac{-4}{27}xyz\right)\left(\frac{9}{2}x^2yz\right) - \left(\frac{-4}{27}xyz\right)\left(\frac{3}{4}xyz^2\right) = \frac{-2}{3}x^3y^2z^2 + \frac{1}{9}x^2y^2z^3$

17. $(x + 3y) + (-2x + 2y) - (x - y) = x + 3y - 2x + 2y - x + y = 6y - 2x$

18. $a = 12, b = 6, c = 4, d = 3;\ \mathrm{P} = 12, \mathrm{Q} = 4$

$$A = 12x^{12}y^6, B = 4x^4y^3;\ \frac{A}{B} = \frac{12x^{12}y^6}{4x^4y^3} = 3x^8y^3$$

19. $A = 2x^2 - 3x + 5, B = 5x + 2, C = x^2 + 3x - 7$

$A \times B = (2x^2 - 3x + 5) \times (5x + 2) = 10x^3 - 15x^2 + 25x + 4x^2 - 6x + 10$

$= 10x^3 - 11x^2 + 19x + 10$

Level – III

20.B	21.C	22.a-1;b-2;c-(3,5);d-4	23. 3	24.A ,C	25.A
26. A,C	27. a-p,t; b-p,q; c-p,s,d-s		28.B	29. A	30. B

31.a-p;b-q;c-r;d-s

20. $B \times C = (x^2 + 3x - 7)(5x + 2) = 5x^3 + 15x^2 - 35x + 2x^2 + 6x - 14 = 5x^3 + 17x^2 - 29x - 14$

21. $A \times C = (2x^2 - 3x + 5)(x^2 + 3x - 7) = 2x^2(x^2 + 3x - 7) - 3x(x^2 + 3x - 7) + 5(x^2 + 3x - 7)$

$= 2x^4 + 6x^3 - 14x^2 - 3x^3 - 9x^2 + 21x + 5x^2 + 15x - 35 = 2x^4 + 3x^3 - 18x^2 + 36x - 35$

22. a) $(a^3 - 2a^2 + 4a - 5) - (-a^3 + 2a^2 - 8a + 5)$

$= a^3 - 2a^2 + 4a - 5 + a^3 - 2a^2 + 8a - 5 = 2a^3 - 4a^2 + 12a - 10$

b) $[b^3 - b^2 + b - 1] - [-b^3 - b^2 - b + 1] = b^3 - b^2 + b - 1 + b^3 + b^2 + b - 1 = 2b^3 + 2b - 2$

c) $(c^3 + 2c^2 - 3c - 4) - (-4c^3 - 3c^2 + 2c + 1) = 5c^3 + 5c^2 - 5c - 5$

d) $(d^4 - 3d^3 - 4d^2 - 5) - (-2d^4 - 5d^3 - 3d^2 - 4) = 3d^4 + 2d^3 - d^2 - 1$

23. $\frac{8m^2 - 6mn}{2m} = 4m - kn \Rightarrow \frac{8m^2}{2m} - \frac{6mn}{2m} = 4m - kn \Rightarrow 4m - 3n = 4m - kn$

$-3n = kn \Rightarrow k = 3$

24. $A = x^2 + x + 1$

$A = x^2 + x + 1$
$B = x^2 - x + 1$
$+$
$A - B = 2x$

$B = x^2 - x + 1$

$+$

$AB = 2x^2 + 2$

$= 2(x^2 + 1)$

25. Conceptual

26. $(3x^3y^2 - 2x^2y^3)(4x^2y^3 - 3x^3y^2) = 12x^5y^5 - 9x^6y^4 - 8x^4y^6 + 6x^5y^5 = 18x^5y^5 - 9x^6y^4 - 8x^4y^6$

27. Conceptual

28. Length $= 3a - b$, Breadth $= 6b - a$

Perimeter $= 2($ length + breadth $) = 2[3a - b + 6b - a] = 2(2a + 5b) = 4a + 10b$

29. Perimeter of triangle $= a + b + c = 2y + 3z + z - y + 4y - 2z = 5y + 2z$

30. Perimeter of rectangle $= 2(l + b)$

$16x^3 - 6x^2 + 12x + 4 = 2[(8x^2 + 3x) + b]$

$\frac{16x^3 - 6x^2 + 12x + 4}{2} = (8x^2 + 3x) + b$

$(8x^3 - 3x^2 + 6x + 2) - (8x^2 + 3x) = b$

$8x^3 - 11x^2 + 3x + 2 = b$

31. a) $[5a - 3b - 5c] - [3a - 2b + 4c] = 2a - b - 9c$

b) $[27 - 3y + 2x] - [x - 3y - 3z] = 27 + x + 3z$

c) $[7 - 2x - 3x^2] - [2x^2 - 5x - 3] = 7 - 2x - 3x^2 - 2x^2 + 5x + 3 = 3x - 5x^2 + 10$

d) $(y^2 + 2y - 7) - (5 - 3y - 2y^2)$

$= y^2 + 2y - 7 - 5 + 3y + 2y^2 = 3y^2 + 5y - 12$

WORKSHEET-2

Level – I

1.A	2.B	3.B	4.D	5.A	6.D	7.A
8.B	9.B					

Level – II

1.B	2.B	3.A	4.A	5.C	6.B	7.B
8.C	9.B	10.B	11.A	12.B	13.C	14.A
15.C	16.B	17.D	18.D	19.A	20.C	21.B

1. $(3\mathrm{P} - 1)(3\mathrm{P} + 5) = (3\mathrm{P})^2 + 3\mathrm{P}(5 - 1) + (-1)(5) = 9\mathrm{P}^2 + 12\mathrm{P} - 5$
2. $(3l + 4)^2 = (3l)^2 + 2(3L)(4) + (4)^2 = 9|^2 + 24I + 16$
3. $(x + 4)^2 + (x + 2)^2 = (x^2 + 8x + 16) + (x^2 + 4x + 4) = 2x^2 + 12x + 20$
4. $(y + 3)^2 - (y - 3)^2 = (y^2 + 6y + 9) - (y^2 - 6y + 9) = 12y$
5. $x^2 + y^2 = (x + y)^2 - 2xy = (7)^2 2(12) = 49 - 24 = 25$

6. If $x+\frac{1}{x}=\frac{10}{3}, \left(x-\frac{1}{x}\right)^2=\left(x+\frac{1}{x}\right)^2-4(x)\left(\frac{1}{x}\right)=\left(\frac{10}{3}\right)^2-4.1=\frac{100}{9}-4=\frac{64}{9}=\left(\frac{8}{3}\right)^2$

7. $\left(\frac{1}{2}+\frac{m}{3}\right)\left(\frac{1}{2}-\frac{m}{3}\right)\left(\frac{1^2}{4}+\frac{m^2}{9}\right)=\left[\left(\frac{1}{2}\right)^2-\left(\frac{m}{3}\right)^2\right]\left[\frac{1^2}{4}+\frac{m^2}{9}\right]=\left(\frac{1^2}{4}\right)^2-\left(\frac{m^2}{9}\right)^2=\frac{1^4}{16}-\frac{m^4}{81}$

8. $x-\frac{1}{x}=5$ then $x^3-\frac{1}{x^3}=\left(x-\frac{1}{x}\right)^3+3\cdot x\frac{1}{x}\left(x-\frac{1}{x}\right)=(5)^3 3(5)=125+15=140$

9. $a^3+b^3=(a+b)^3-3ab(a+b)=(10)^3-3(21)(10)=1000-630=370$

10. $a^2-5a+1=0$

    divide with "a" on both sides

    $\frac{a^2}{a}-\frac{5a}{a}+\frac{1}{a}=0\ a^2+\frac{1}{a^2}=\left(a+\frac{1}{a}\right)^2-2=(5)^2-2=23$

    $a+\frac{1}{a}-5=0\Rightarrow a+\frac{1}{a}=5$

11. $4x+5y=18, xy=4\Rightarrow 16x^2+25y^2=(4x)^2+(5y)^2=[4x+5y]^2-2(4x)(5y)$

    $=(18)^2-40(4)=324-160=164$

12. $16x^2+40x+P=(4x)^2+2(4x)(5)+P\Rightarrow\therefore P=(5)^2=25$

13. $4x^2+\frac{9}{x^2}-12=4(2)^2+\frac{9}{(2)^2}-12=16-12+\frac{9}{4}=\frac{25}{4}$

14. $P=(116)^2-(16)^2$

    $Q=(124)^2-(24)^2$

    $=(116-16)(116+16)=(124+24)(124-24)\ R=120$

    $=13200$

    $=14800$

    $\sqrt{\frac{13200}{100}+\frac{14800}{100}}+120=\sqrt{132+148+120}=\sqrt{400}=20$

15. $P=a^2-2ab+b^2, q=4ab\Rightarrow P+q=a^2-2ab+b^2+4ab=a^2+2ab+b^2=(a+b)^2$

16. $x+y=8, xy=12\Rightarrow x^2+y^2=(x+y)^2-2xy=(8)^2-2(12)=64-24=40$

17. $\left(2X+\frac{1}{3Y}\right)\left(2X-\frac{1}{3Y}\right)=(6)\left(-\frac{1}{2}\right)\Rightarrow 4X^2-\frac{1}{9Y^2}=-3$

18. If $a^{\frac{1}{3}}+b^{\frac{1}{3}}+c^{\frac{1}{3}}=0\Rightarrow\left(a^{\frac{1}{3}}\right)^3+\left(b^{\frac{1}{3}}\right)^3+\left(c^{\frac{1}{3}}\right)^3=3\cdot a^{\frac{1}{3}}\cdot b^{\frac{1}{3}}\cdot c^{\frac{1}{3}}\ \therefore a+b+c=3(abc)^{\frac{1}{3}}$

    cubing on both sides.

    $$(a+b+c)^3=\left[3(abc)^{\frac{1}{3}}\right]^3\Rightarrow 3^3\times(abc)^{\frac{1}{3}\times 3}\Rightarrow 27abc$$

19. $v^2-\frac{1}{v^2}=\left(v+\frac{1}{v}\right)\left(v-\frac{1}{v}\right)=16\frac{1}{16}=1$ multiply and divide by $a-b$

20. $\frac{(a-b)(a+b)(a^2+b^2)(a^4+b^4)(a^8+b^8)(a^{16}+b^{16})}{a-b}=\frac{a^{32}-b^{32}}{a-b}\cdot\therefore k=a-b$

21. $169a^2-121b^2=(13a)^2-(11b)^2=(13a+11b)(13a-11b)$

Level – III

22.A,B,C,D 23.A 24.A 25.A 26.A 27.a-p,b-q,r;c-s,d-t
28.2 29.B,D 30.a-s,b-q,c-r,d-p 31.A,B,D 32.A
33.a-t,b-q,c-p,d-s

22. $x-\frac{1}{x}=4 \Rightarrow \left(x+\frac{1}{x}\right)^2=\left(x-\frac{-1}{x}\right)^2+4=(4)^2+4=20$

23 Statement-1

$$A=16x^2+144y^2=(4x)^2\pm 2(4x)(12y)+(12y)^2=(4x)^2\pm 96xy+(12y)^2$$

24. $\mathrm{k}^2-3\mathrm{k}+1=0 \Rightarrow \mathrm{k}^2+1=3\mathrm{k} \Rightarrow \mathrm{k}+\frac{1}{\mathrm{k}}=3$

25. $\mathrm{k}^2+\frac{1}{\mathrm{k}^2}=\left(\mathrm{k}+\frac{1}{\mathrm{k}}\right)^2-2=(3)^2-2=7$

26. $=\left(\mathrm{k}^2+\frac{1}{\mathrm{k}^2}\right)^2-2=(7)^2-2=47$

27. Conceptual

28: $9\ell^2-k\ell m+4m^2=(3\ell)^2-k\ell m+(2m)^2$

$\therefore \mathrm{k}=2(3)(2)=12 \Rightarrow \mathrm{k}=10+\mathrm{x} \Rightarrow \mathrm{x}=2$

29. $x+\frac{1}{x}=3,$

2. $x^2+\frac{1}{x^2}=\left(x+\frac{1}{x}\right)^2-2x\cdot\frac{1}{x}=(3)^2-2=7$

3. $x^4+\frac{1}{x^4}=\left(x^2+\frac{1}{x^2}\right)^2-2=(7)^2-2=49-2=47$

30.

(a) $[3x+4y+z][3x-4y-z]=(3x-4y)^2-(z)^2$

(b) $[3x-4y+z][3x+4y-z]=[3x-(4y-z)][3x+(4y-z)]=(3x)^2-(4y-z)^2$

(c) $[5a+3b+2c][5a-(3b+2c)]=(5a)^2-(3b+2c)^2$

(d) $[2x+3y+5z][2x+3y-5z]=(2x+3y)^2-(5z)^2$

31. $pa^2+100ab+625b^2=pa^2+2\times 2a\times 25b+(25b)^2$

to make it perfect square $=(2a)^2+2\times 2a\times 25b+(25b)^2 \ \therefore p=4$

32. Conceptual

33. (a) $(2a+3b)(4a^2-6ab+ab^2)=(2a)^3+(3b)^3$

$-8a^3+27b^3$

(b) $[3x+4y][(3x)^2-(3x)(4y)+(4y)^2]=(3x)^3+(4y)^3=27x^3+64y^3$

(c) $[4a+3][(4a)^2-(4a)(3)+(3)^2]=(4a)^3+(3)^3=64a^3+27$

(d) $(5\mathrm{a}-3)[(5\mathrm{a})^2+(5\mathrm{a})(3)+(3)^2]=(5\mathrm{a})^3-(3)^3=125\mathrm{a}^3-27$

WORKSHEET-3

Level – I

1.A 2.C 3.B 4. A 5.B 6.B 7.A

8.A 9.B

4. $8x^2y^2 = 2 \times 2 \times 2 \times x \times x \times y \times y$

$12x^3y^2 = 2 \times 2 \times 3 \times x \times x \times x \times y \times y$

$24x^4y^3z^2 = 2 \times 2 \times 3 \times x \times x \times x \times x \times y \times y \times y \times z \times z$

$HCF = 2 \times 2 \times x \times x \times y \times y = 4x^2y^2$

5. $20x^2y(x^2 - y^2) = 2 \times 2 \times 5 \times x \times x \times y \times (x - y) \times (x + y)$

$35xy^2(x - y) = 5 \times 7 \times 3 \times x \times y \times y \times (x - y)$

$HCF = 5 \times x \times y \times (x - y) = 5xy(x - y)$

6. $9a(6a - 5b) = 3 \times 3a \times (6a - 5b)$

$12a^2(6a - 5b) = 3 \times 4a \times a \times (64 - 5b)$

$LCM = 3 \times a \times (6a - 5\ b) \times 3 \times 4a = 36a^2(6a - 5\ b)$

7. Product of = LCM × HCF; Polynomial = $36x^2y \times 3x = 108x^3y$

8. Factors are $3a(3\ b + 2)$; HCF = 1

9. $LCM = (5a + b)^2$

Level – II

1. A 2. C 3. C 4.A 5.C 6.A 7.A

8.B 9.C 10.A 11.A 12.D 13.B 14.D

15.C 16.C 17.A 18.A

1. $4a^4b^4c^3 = 2 \times 2 \times a^4 \times b^4 \times c^2 \Rightarrow 8a^3b^6c^3 = 2 \times 2 \times 2 \times a^3 \times b^6 \times c^3$

$16a^5b^3c^4 = 2 \times 2 \times 2 \times 2 \times a^5 \times b^3 \times c^4 \Rightarrow HCF == 4 \times a^3 \times b^3 \times c^2 = 4a^3b^3c^2$

2. $A = 384x^4y^5x^3 = 3 \times 128 \times x^4y^5z^3 = 3 \times 2^7 \times x^4 \times y^5 \times z^3$

$B = 2^8 \times x^2 \times y^3 \times z^3 \Rightarrow GCD = 2^7 \times x^2 \times y^3 \times z^3$

3. $24pm^2n^2 = 3^1 \times 2^3 \times p^1 \times m^2 \times n^2$

$9pmn^2 = 3^2 \times p^1 \times m^1 \times n^2 \Rightarrow 36p(mn)^3 = 2^2 \times 3^2 \times p^1 \times m^3 \times n^3$

HCF = Product of common terms with least powers = $3^1 \times p^1 \times m^1 \times n^2 = 3pmn^2$

4. $x^2 - 5x + 6 = (x - 2)(x - 3) \Rightarrow x^2 - 9 = (x - 3)(x + 3) \Rightarrow HCF = x - 3$

5. $LCM = (x - 2)(x - 3)(x + 3)$

6. $14a^2b^3c^4 = 2^1 \times 7^1 \times a^2 \times b^3 \times c^4 \Rightarrow 20ab^4c^3 = 2^2 \times 5^1 \times a \times b^4 \times c^3$

$35a^5b^3c = 5^1 \times 7^1 \times a^5 \times b^3 \times c$

LCM = Product of all factors with highest powers

$= 2^2 \times 5^1 \times 7^1 \times a^5 \times b^4 \times c^4 = 140a^5b^4c^4$

7. LCM × HCF = Pr oduct of polynomials = $x^2 - 5x + 6$

8. $A = 2^4 \times 3^5 \times x^7 \times y^9 \Rightarrow B = 2^5 \times 3^3 \times x^6 \times y^8 \Rightarrow \text{GCD} = 2^4 \times 3^3 \times x^6 \times y^8$

9. $f(x) = xy + yz + zx + y^2 = xy + y^2 + yz + zx = (y + z)(x + y)$

   $p(x) = x^2 + xy + yz + zx = x(x + y) + z(y + x) = (x + z)(x + y)$

   LCM of $f(x), p(x) = (x + y)(y + z)(z + x)$

10. $x^4 - y^4 = (x^2 - y^2)(x^2 + y^2) = (x - y)(x + y)(x^2 + y^2)$

    $x^6 - y^6 = (x^3 - y^3)(x^3 + y^3) = (x - y)(x^2 + xy + y^2)(x + y)(x^2 + xy + y^2)$

    $\therefore \text{HCF} = (\text{x} - \text{y})(\text{x} + \text{y}) = \text{x}^2 - \text{y}^2$

11. $x^2 - 10x + 16 = (x - 2)(x - 8)$

    $x^2 - 9x + 14 = (x - 2)(x - 7) \Rightarrow x^2 - 10x + 21 = (x - 3)(x - 7)$

    $\text{LCM} = (x - 2)(x - 3)(x - 7)(x - 8)$

12. $x^5 + 2x^4 + x^3$

    $= x^3(x^2 + 2x + 1) = x^3(x + 1)^2 \rightarrow (1) = x^7 - x^5 = (x^2 - 1) = x^5(x - 1)(x + 1) \rightarrow (2)$

    $\text{HCM} = \text{x}^3(\text{x} + 1)$

13. Product of $= x^3 + x^2 - 44x - 84$ two Expressions

    $\text{HCF} = \text{x} + 6 \Rightarrow \text{LCM} \times \text{HCF} = \text{Pr oduct}$

    $$\text{LCM} \times \text{x} + 6 = \text{x}^3 + \text{x}^2 - 44\text{x} - 84 \Rightarrow \text{LCM} = \frac{\text{x}^3 + \text{x}^2 - 44\text{x} - 84}{\text{x} + 6} = \text{x}^2 - 5x - 14$$

14. $x^4 - 11x^2 + 10 = (x^2)^2 - 11x^2 + 10$

    $= a^2 - 11a + 10 = (a - 10)(a - 1) = (x^2 - 10)(x^2 - 1)$

    $= (x^2 - 10)(x + 1)(x - 1) \Rightarrow x^2 - 5x + 4 = (x - 1)(x - 4)$

    $\Rightarrow \text{x}^3 - 3\text{x}^2 + 3\text{x} - 1 = (\text{x} - 1)(\text{x}^2 - 2\text{x} + 1) \ \therefore \text{HCF} = \text{x} - 1$

15. $\Rightarrow x^2 - 3x + 2 = (x - 1)(x - 1)$

    $\Rightarrow \text{x}^3 - 2\text{x}^2 - 3\text{x} = \text{x}(\text{x}^2 - 2\text{x} - 3) = \text{x}(\text{x} - 3)(\text{x} + 1)$

    $\therefore \ \text{LCM} = \text{x}(\text{x} - 1)(\text{x} + 1)(\text{x} - 2)(\text{x} - 3) = \text{x}(\text{x}^2 - 1)(\text{x} - 2)(\text{x} - 3)$

16. Product of two polynomials = LCM × HCF

    $4x^2(x^2 - 3x + 2) \times 12x(x - 2)(x^2 - 4) = \text{LCM} \times 4x(x - 2)$

    $12x^2(x^2 - 3x + 2)(x^2 - 4) = \text{LCM}$

17. $x^3 + 3x^2 + 3x + 1 = (x + 1)^3 \Rightarrow x^2 + 2x + 1 = (x + 1)^2 \Rightarrow x^2 - 1 = (x + 1)(x - 1)$

    $\therefore LCM = (x + 1)^3 \times (x^2 - 1)$

18. $\frac{x^2+y^2}{x^2+2xy+y^2}, \frac{y(x+y)}{x^2-xy} = \frac{\text{HCF of numberators}}{\text{LCM of denominators}} = \frac{\text{HCF of } [x^2+y^2, y(x+y)]}{LCM \text{ of } [(x+y)^2, x(x-y)]}$

    $$= \frac{1}{x(x - y)(x + y)^2} = \frac{1}{x(x^2 - y^2)(x + y)}$$

Level – III

19. A,C 20.A 21.A 22.A 23.D 24.1 25.A

26. a-p; b-q; c-r; d-s 27.A,B 28.A

19. The H.C.F of 14,21,35 is 7 and The H.C.F of x^2y^3, x^2y^2, x^4y^5z is x^2y^2
20. LCM of $2x, p = 4xy \ \therefore p = 4xy$
21. product = HCF × LCM = $4xy \times 12x^2y = 48x^3y^2$
22. product = HCF × LCM ⇒ $(x^2 - 5x + 6)(x^2 - 5x - 14) = 1 \times LCM$
23. product = HCF × LCM = $4x \times 8xy = 32x^2y$
24. $a^3 - b^3 = (a - b)(a^2 + ab + b^2) \Rightarrow a^2 - b^2 = (a - b)(a + b)$
    HCF of $a^3 - b^3, a^2 - b^2 = a - b \Rightarrow \therefore a - b = 3 - 2 = 1$
25. HCF = X ⇒ LCM = $x^3 - 9x$
    Product of polynomials = LCM × HCF
    $(x^2 + 3x) \times p(x) = x \times (x^3 - 9x) \Rightarrow (x^2 + 3x) \times p(x) = x \times x \times (x^2 - 9)$
    $p(x) = \dfrac{x \times x \times (x + 3)(x - 3)}{x(x + 3)} = x^2 - 3x$
26. a) $p(x) = x^2(x - 4) + (x - 4) = (x^2 + 4) \Rightarrow g(x) = x^2 - 12x + 32 = (x - 4)(x - 8)$
    $\therefore$ HCF = $x - 4$
    b) $p(x) = x^2 - 4x + 3 = (x - 1)(x - 3)$
    c) $q(x) = x^2 - 7x + 6 = (x - 1)(x - 6) \Rightarrow HCF = x - 1 \Rightarrow LCM = (x - 1)(x - 3)(x - 6)$
    d) $p(x) = x^2 - 10x + 24 = (x - 4)(x - 6)$
    $g(x) = x^2 - 12x + 32 \Rightarrow LCM = (x - 4)(x - 6)(x - 8)$
27. $x^3 - 343 = (x - 7)(x^2 + 7x + 49) \Rightarrow (x)^3 - (7)^3 \Rightarrow x^2 - 9x + 14 = (x - 7)(x - 2)$
    $x - 7 = 1(x - 7) \Rightarrow$ LCM $= (x - 7)(x - 2)(x^2 + 7x + 49) = (x^2 - 9x + 14)(x^2 + 7x + 49)$
28. LCM of polynomials = $x^2 - 4$
    HCF of polynomials = $x - 2$
    LCM = $(x - 2)(x + 2) \Rightarrow$ HCF = $x - 2$
    HCF of LCM and HCF = $x - 2$

WORKSHEET-4

Level – I

1.B	2.C	3.A	4.D	5.A	6.D	7.B
8.D	9.B	10.D	11.C	12.C		

3. $x^2 + 6x + 5 = x^2 + 5x + x + 5 = (x + 1)(x + 5)$
7. $a^3 + b^3 = (a + b)(a^2 - ab + ab^2)$
9. $a^3 - b^3 = (a - b)(a^2 + ab + ab^2)$

Level – II

1.A	2.D	3.B	4.C	5.A	6.C	7.C
8.A	9.C	10.C	11.A	12.D	13.B	14.A
15.C	16.D	17.B	18.D	19.A	20.C	21.D

1. $(a+b+c)^2+(a-b-c)^2=a^2+b^2+c^2+2ab+2bc+2ca+a^2+b^2+c^2-2ab+2bc-2ca$
   $=2(a^2+b^2+c^2+2bc)$
2. $(a+b+c)^2-(a-b-c)^2=a^2+b^2+c^2+2ab+2bc+2ca-a^2-b^2-c^2+2ab-2bc+2ca=4ab+4ca$
3. $a^2+9a+20=a^2+5a+4a+20=a(a+5)+4(a+5)=(a+5)(a+4)$
4. $y^2+4y-32=y^2+8y-4y-32=y(y+8)-4(y+8)=(y+8)(y-4)$
5. use $x^3-y^3=(x+y)(x^2-xy+y^2)$
6. $8a^3-27=(2a)^3-3^3=(2a-3)(4a^2+6a+9)$
7. $1^3-\left(\frac{1}{x}\right)^3=\left(1-\frac{1}{x}\right)\left(1+\frac{1}{x}+\frac{1}{x^2}\right)$
8. $x^6-y^3=(x^2)^3-y^3=(x^2-y)(x^4+x^2y+y^2)$
9. $x+y+z=0\Rightarrow x^3+y^3+z^3=3xyz$
10. $(a-b)(a^2+ab+b^2)=a^3-b^3$
    $\Rightarrow(2x-3y)(4x^2+6xy+9y^2)=(2x)^3-(3y)^3=8x^3-27y^3$
11. $ab+bc+ca=-\frac{(a^2+b^2+c^2)}{2}\Rightarrow\frac{-20}{2}=-10$
12. $\frac{1}{x^3}-\frac{1}{27}=\left(\frac{1}{x}\right)^3-\left(\frac{1}{3}\right)^3=\left(\frac{1}{x}-\frac{1}{3}\right)\left[\left(\frac{1}{x}\right)^2+\left(\frac{1}{x}\right)\left(\frac{1}{3}\right)+\left(\frac{1}{3}\right)^2\right]$
13. $(a+7)(a-10)+16=a^2-10a+7a-70+16=a^2-3a-54$
    $=a^2-9a+6a-54=a(a-9)+6(a-9)=(a-9)(a+6)$
14. $x^3+9x^2+27x+27=x^3+3(x^2)(3)+3(x)(3^2)+3^3=(x+3)^3$
15. $(3x+4)^3-8=(3x+4)^3-2^3=(3x+4-2)[(3x+4)^2+(3x+4)2+2^2]$
    $=(3x+2)(9x^2+30x+28)$
16. $x^2+11xy+24y^2=x^2+8xy+3xy+24y^2=x(x+8y)+3y(x+8y)=(x+8y)(x+3y)$
17. $a^4+4a^2b^2+4b^4-a^2b^2=(a^2+2b^2)^2-(ab)^2=(a^2+2b^2+ab)(a^2+2b^2-ab)$
18. $2x^3+5x^2y-12xy^2=2x^3+8x^2y-3x^2y-12xy^2=2x^2(x+4y)-3xy(x+4y)$
    $=(2x^2-3xy)(x+4y)=x(2x-3y)(x+4y)$
19. $x^2-2ax+(a+b)(a+b)=x^2-(a+b)x-(a-b)x+(a+b)(a-b)$
    $=x[x-(a-b)]-(a-b)[x-(a-b)]=[x-(a-b)][x-(a+b)]$
20. Put $x^2+x=y$ and factorise the equation
21. $2m^3-0.016n^3=2[m^3-0.008n^3]=2[(m)^3-(0.2n)^3]$
    $=2[m-0.2n][m^2+(m)(0.2)n+(0.2n)^2]\Rightarrow=2[m-0.2n][m^2+0.2mn+0.04n^2]$

Level – III

22.A,B 23.B 24.B 25.C 26.A 27.A,B,C,D
28.A 29.a-r,b-p,c-s,d-q, 30. A,D

22. $8x^3+27y^3+18xy-1=(2x)^3+(3y)^3+(-1)^3-3(2x)(2y)(-1)$

23. $(x^3 - y^3) = (x - y)(x^2 + xy + y^2)$
24. $(2a + 3b)[(2a)^2 - (2a)(3b) + (3b)^2] = (2a)^3 + (3b)^3 = 8a^3 + 27b^3$
25. $(a^3 - 2)[(a^3)^2 + (2)(a^3) + (2)^2] = (a^3)^3 - (2)^3 = a^9 - 8$
26. $\frac{8a^3+27}{4a^2-6a+9} = \frac{(2a+3)(4a^2-6a+9)}{4a^2-6a+9} = 2a + 3\Big]$
27. $a^6 - b^6 = (a^3 - b^3)(a^3 + b^3) = (a - b)(a^2 + ab + b^2)(a + b)(a^2 - ab + b^2)$
28. Conceptual
29. a) $8a^3 + 27 = (2a + 3)(4a^2 - 6a + 9)$

    b) $(a^2 - ab + b^2)(a + b) = a^3 + b^3$

    c) $(a - b)(a^2 + ab + b^2) = a^3 - b^3$

    d) $125x^3 - 8 = (5x)^3 - (2)^3$

    $= (5x - 2)[(5x)^2 + (5x)(2) + (2)^2] = (5x - 2)(25x^2 + 10x + 4)$
30. $(1 - x^2)(1 - y^2) + 4xy = 1 - y^2 - x^2 + x^2y^2 + 4xy = 1 - y^2 - x^2 + x^2y^2 + 2xy + 2xy =$ $(xy + 1)^2 - (x - y)^2$

WORKSHEET-5

Level – I

1.1	2.1	3.3	4.4	5.2	6.2	7.2
8.3	9.2	10.4	11.2	12.1	13.4	14.4
15.2	16.1	17.1				

1. $p(x) = x^4 + 2x^3 + 3x^2 + x - 1 \Rightarrow g(x) = x - 2$

   According to remainder theorem remainder $== p(2)$

   $= (2)^4 + 2(2)^3 + 3(2)^2 + 2 - 1 = 16 + 16 + 12 + 2 - 1 = 45$
2. $= 4x^3 + 4x^2 - x - 1 = 4x^2(x + 1) - 1(x + 1) = (x + 1)(4x^2 - 1) = (x + 1)(2x - 1)(2x + 1)$
3. $\mathrm{p(x)} = \mathrm{x^2 - ax + 5}$

   According to remainder theorem, Remainder $= p(a) = (a^2) - a(a) - 5 = -5$
4. $p(x) = 2x^3 + 7x^2 - 24x - 45 \Rightarrow p(3) = 2(3)^3 + 7(3)^2 - 24(3) - 45 = 54 + 63 - 72 - 45$
5. $p(x) = x^2 + 3x + k$ Given that $p(2) = 0$

   $(2)^2 + 3(2) + k = 0$

   $k = -10$
6. Quadratic Polynomial $\Rightarrow (x - 1)(x - 2) = x^2 - 3x + 2$
7. According to factor theorem,

   $\mathrm{p}(-2) = 0 \Rightarrow 2(-2)^4 + 3(-2)^3 + 2\ \mathrm{m}(-2)^2 + 3(-2) + 6 = 0$

   $\Rightarrow 32 - 24 + 8m - 6 + 6 = 0 \Rightarrow 8m = -8$

   $\therefore \mathrm{m} = -1$
8. According to factor theorem $p(1) = 0$

   $\Rightarrow 2(1)^3 + 9(1)^2 + 1 + \mathrm{k} = 0 \Rightarrow 12 + \mathrm{k} = 0$

$\therefore k = -12$

9. $p(x) = 2\sqrt{2}x^2 + 5x + \sqrt{2} \Rightarrow g(x) = x - \sqrt{2}$

According to remainder theorem the temainder $= p = (\sqrt{2})$

$= 2\sqrt{2}(\sqrt{2})^2 + 5\sqrt{2} + \sqrt{2} = 4\sqrt{2} + 5\sqrt{2} + \sqrt{2} = 10\sqrt{2}$

10. According to factor theorem $f(4) = 0$

$2(4)^3 - 3(4)^2 - 18(4) + p = 0$

$128 - 48 - 72 + p = 0$

$p = -8$

11. According to reminder theorem remainder $= p(+\sqrt{2})$

$= 7(-\sqrt{2})^2 - 4\sqrt{2}(+\sqrt{2}) + 6 = 7(2) - 8 + 6 = 12$

12. $p(x) = x^3 - 10x^2 + ax + b$ given that $x - 1$ is a factor $\therefore p(1) = 0$

$(1)^3 - 10(1)^2 + a(1) + b = 0$

$\therefore a + b = 9$

$(x - 2)$ is also a factor

$\therefore p(2) = 0$

$(2)^3 - 10(2)^2 + a(2) + b = 0$

$8 - 40 + 2a + b = 0$

$\therefore 2a + b = 32$

$\underline{a + b = 9}$

$a = 23$

13. Given that $f(x) = x^3 - 3x + 4x + 63$

According to remainder theorem

$f(b) = -3 \Rightarrow (b)^3 - 3(b)^2 + (4b) + 63 = -3$

$b^3 - 3b^2 + 4b + 66 = 0$

$\therefore b = 3$

14. $p(x) = x^5 - a^2x^3 + 2x + a - 3$ According to factor theorem $p(a) = 0$

$a^5 - a^2(a)^3 + 2(a) + a - 3 = 0 \Rightarrow a^5 - a^5 + 3a - 3 = 0$

$\Rightarrow 3a = 3$

$a = 1$

15. $p(x) = 2x^3 + ax^2 + 3x - 5 \Rightarrow f(x) = x^3 + x^2 - 2x + a$

According to remainder theorem

$$p(2) = f(2) \Rightarrow 2(2)^3 + a(2)^2 + 3(2) - 5 = (2)^3 + (2)^2 - 2(2) + a$$

$4a + 17 = 8 + a \Rightarrow 3a = -9$

$$\therefore a = -3$$

16. According to factor theorem $(-2) = 0$ and $p(-3) = 0$

$(-2)^4 + a(-2)^3 - 7(-2)^2 - 8(-2) + b = 0 \Rightarrow 16 - 8a - 28 + 16 + b = 0$

$b - 8a = -4$

$(-3)^4 + a(-3)^3 - 7(-3)^2 - 8(-3) + b = 0 \Rightarrow 81 + 27a - 63 + 24 + b = 0$

$b - 27a = -42$

solving (1)and (2)

$b - 8a = -4$

$\underline{b - 27a = -42}$

$19a = 38$

$a = 2$

from(1) $b - 8a = 4 \Rightarrow b - (8)2 = 4 = b = 12$

17. Let 'k' be added to $p(x)$ so that $(x + 3)$ is a factor $p(x) = x^2 + 6x + 7 + k$

According to factor theorem $p(-3) = 0$

$(-3)^2 + 6(-3) + 7 + k = 0 \Rightarrow 9 - 18 + 7 + k = 0 \ \because k = 2$

Level – II

18.A,B,D	19.B	20.C	21.D	
22.a-p,b-r,c-q,d-s		23.0	24.A,D	25.C
26.a-p,t; b-p,q; c-r,s; d-q,s			27.3	28.a-p; b-q; c-r,s; d-t

18. Conceptual

19. According to remainder theorem remainder

$\Rightarrow p(1) = (1)^3 - 6(1)^2 + 9(1) + 3 = 13 - 6 = 7$

20. $p(-2) = (-2)^3 - 6(-2)^2 + 13(-2) + 60 = -8 - 24 - 26 + 60 = 2$

21. $2^3 - 6(2)^2 + 11(2) - 6 = 8 - 24 + 22 - 6 = 0$

22. Given that $p(x) = x^3 - 6x^2 + 13x + 60$

a) $p(4) = 4^3 - 6(4)^2 + 13(4) + 60 = 64 - 96 + 52 + 60 = 80$

b) $p(3) = 3^3 - 6(3)^2 + 13(3) + 60 = 27 - 54 + 39 + 60 = 72$

c) $p\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^3 - 6\left(\frac{1}{2}\right)^2 + 13\left(\frac{1}{2}\right) + 60 = \frac{1}{8} - \frac{6}{4} + \frac{13}{2} + 60 = \frac{1-12+52+480}{8}$

d) $p(0) = 0^3 - 6(0)^2 + 13(0) + 60 = 60$

23. $p(x) = x^3 - 6x^2 + 11x - 6 \Rightarrow p(1) = 1^3 - 6(1)^2 + 11(1) - 6 = 0$

24. $p(x) = 2x^3 - 5x^2 + 9x - 8 \Rightarrow p(3) = 2(3)^3 - 5(3)^2 + 9(3) - 8 = 28$

$\Rightarrow p(1) = 2(1)^3 - 5(1)^2 + 9(1) - 8 = -2, p(2) = 2(2)^3 - 5(2)^2 + 9(2) - 8 = 6$

25. $p(x) = x^2 - 3x = x(x - 3) = (x - 0)(x - 3) \ \therefore$ as $x - a$ is factor theorem

$p(a) = 0 \Rightarrow a^2 - 3a = 0 \Rightarrow a = 0,3$

If $a = 0, b = 3 \Rightarrow$ If $a = 3, b = 0 \ \therefore a + b = 3$

26. (a) $x^2 + 21x + 98 = x^2 - 7x - 14x + 98 == (x - 7)(x - 14)$

$p(7) = 0$ and $p(14) = 0 \ \therefore (x - 7)$ and $(x - 14)$ are factor

(b) $x^2 - 18x + 77 = x^2 - 11x - 7x + 77 = x(x - 11) - 7(x - 11) = (x - 11)(x - 7)$

$(x - 11)(x - 7)$ are factor

(c) $x^2 - 19x + 60 = x^2 - 15x - 4x + 60 = x(x-15) - 4(x-15) = (x-15)(x-4)$

$\therefore (x-4)(x-15)$ arefactors

(d) $x^2 - 15x + 44 = x^2 - 11x - 4x + 44 = (x-11)(x-4)$

27. $p(x) = x^2 + ax + 2b \Rightarrow$ According to factor theorem

$p(-2) = 0 \Rightarrow (-2)^2 + a(-2) - 2\,b = 0$

$-2a + 2b = -4$

Given $a + b = 4$

(2) that on solving (1)&(2)

$-2a + 21b = -4$

$2a + 2b = 8$

$4b = 4 \Rightarrow b = 1$

from $(2)(a+b) = 4 \Rightarrow a + 1 = 4$

$\therefore a = 3$

28. a) $p(x) = x^3 - 3x^2 + 3x - 1$

According to factor theorem

$p(1) = (1)^3 - 3(1)^2 + 3(1) - 1 = 1 - 3 + 3 - 1 = 0$ $\therefore x - 1$ is a factor

b) $p(x) = 2x^3 + 11x^2 + 13x + 4$

According to factor theorem $p(-1) = 2(-1)^3 + 11(-1)^2 + 13(-1) + 4$

$= 2(-1) + 11(1) - 13 + 4 = 0x + 1$ is a factor

c) $x^4 - 16 = (x^2-4)(x^2+4) \Rightarrow (x-2)(x+2)(x^2+4)$

$\therefore x - 2$ and $x + 2$ are factors

d) $p(x) = 2x^3 + 7x^2 - 24x - 45 \Rightarrow p(3) = 2(3)^3 + 7(3)^2 - 24(3) - 45$

$= 54 + 6 - 72 - 45 \Rightarrow 117 - 117 = 0 \Rightarrow x - 3 =$ is a factor

---

WORKSHEET-6

Level – I

1) A	2) B	3) D	4) C	5) B	6) D	7) C
8) D	9) A	10) B	11) C	12) B	13) A	

1.

-3	1	-3	-7	27	-18
	0	-3	18	-33	18
	1	-6	11	-6	0

$\therefore Q(x) = x^3 - 6x^2 + 11x - 6$

2.

$$\begin{array}{c|ccccc} \frac{1}{2} & 16 & -8 & 4 & -8 & 4 \\ & 0 & 8 & 0 & 2 & -3 \\ \hline & 16 & 0 & 4 & -6 & \underline{1} \end{array}$$

3. $(x^2-4)(x^2-9) = 1x^4 - 13x^2 + 36 = 1x^4 + 0x^3 - 13x^2 + 36$

   coefficients are $1, 0, -13, 36$.

4.

$$\begin{array}{c|cccc} 1 & 1 & -2 & -19 & 49 & -30 \\ & 0 & 1 & 0 & -19 & 30 \\ \hline 2 & 1 & 0 & -19 & 30 & \underline{0} \\ & 0 & 2 & 4 & -30 & \\ \hline 3 & 1 & 2 & -15 & \underline{0} & \\ & 0 & 3 & +15 & & \\ \hline & 1 & 5 & \underline{0} & & \end{array}$$

$\therefore Q(X) \Rightarrow x+5$

5. $\begin{array}{c|ccc} \frac{-1}{3} & 9 & -6 & 6 \\ & 0 & -3 & 3 \\ & 9 & -9 & 9 \end{array}$ Remainder $= 9$

6. $\begin{array}{c|cccc} \frac{2}{1} & 1 & -9 & 2 & m \\ & 0 & 2 & -14 & -24 \\ & 1 & -7 & -12 & \underline{m-24} \end{array}$ as $x-2$ is a factor, $m - 24 = 0, m = 24$

7. $\begin{array}{c|ccc} \frac{1}{5} & & & \\ \hline 100 & 20 & 25 & \\ 0 & 20 & 8 & \end{array}$ Remainder $= 33$

$$\begin{array}{c|cccc} 1 & 1 & 8 & -1 & -8 \\ & 0 & 1 & 9 & 8 \\ \hline -1 & 1 & 9 & 8 & 0 \\ & 0 & -1 & & -8 \end{array}$$

other factors are $x+1$ and $x+8$

$$\begin{array}{c|cccc} -2 & a & b & c & d \\ & 0 & -2a & -2(b-2a) & 4b-2c-8a \\ \hline & a & b-2a & 4a-2b+c & \underline{d+4b-2c-8a} \end{array}$$

$a = 4$

$b - 2a = 5 \Rightarrow b - 8 = 5 = b = 13$

$4a - 2b + c = 6 \Rightarrow 4(4) - 2(13) + c = 6 \Rightarrow c = 16$

As Remainder $= 0 \Rightarrow d + 4b - 2c - 8a = 0 \Rightarrow d + 4(13) - 2(16) - 8(4) = 0$

$d + 52 - 32 - 32 = 0 \Rightarrow d = 12$

$$\therefore 4x^3 + 13x^2 + 16x + 12$$

10. $(x+3)(x+4)(x+5)$

    $= x^3 + 12x^2 + [(3)(4) + (4)(5) + (3)(5)]x + (3)(4)(5) = x^3 + 12x^2 + 47x + 60$

11.

$$\begin{array}{r|cccc} \frac{3}{2} & 4 & 0 & -3 & 9 \\ & 0 & 6 & 9 & 9 \\ \hline & 4 & 6 & 6 & \underline{18} \end{array} \qquad \therefore Q(x) = 2x^3 + 3x + 3$$

12.

$$\begin{array}{r|cccc} 1 & 3 & -10 & 5 & 10 & -8 \\ & 0 & 3 & -7 & -2 & 8 \\ \hline -1 & 3 & -7 & -2 & 8 & \underline{0} \\ & 0 & -3 & 10 & -8 & \\ \hline & 3 & -10 & 8 & \underline{0} & \end{array}$$

$3x^2 - 10x + 8 = 3x^2 - 6x - 4x + 8 = 3x(x-2) - 4(x-2) = (3x-4)(x-2)$

13.

$$\begin{array}{r|cccc} 3 & 1 & -12 & 17 & -9 & 7 \\ & 0 & 3 & -27 & -30 & -117 \\ \hline 3 & 1 & -9 & -10 & -39 & \underline{-110 = Q_0} \\ & 0 & 3 & -18 & -84 & \\ \hline 3 & 1 & -6 & -28 & \underline{-123 = Q_1} & \\ & 0 & 3 & -9 & & \\ \hline 3 & 1 & -3 & \underline{-37 = Q_2} & & \\ & 0 & 3 & & & \\ \hline & 1 & \underline{O = Q_3} & & & \end{array} \qquad \therefore f(X+3) = x^4 - 37x^2 - 123x - 110$$

$1\ Q_4$

LEVEL – II

14) A,B,C 15) A 16) A 17) C 18) B

19) a - p,r;b-q,r; c-p; d-q,s 20) 3 21) A,C 22) C

23) a -q; b-s;c-r; d-s. 24) B 25) C 26) D 27) A,B

28) a -q; b -p; c - r; d- s 29)7

14. $(2)^3 - 9(2)^2 + 2(2) + m = 0 \Rightarrow 8 - 36 + 4 + m = 0 \Rightarrow m = 24$

15.

$$\begin{array}{r|cccc} -2 & 4 & 13 & 16 & 12 \\ & 0 & -8 & -10 & -12 \\ \hline & 4 & 5 & 6 & \underline{0} \end{array}$$

16. $P(2) = (2)^2 - 9(2) + 18 = 4$

17. $q(4) = (4)^2 + 5(4) - 4 = 22$

18. $P(x) + q(x) - 2 = x^2 - 9x + 18 + x^2 + 5x - 14 - 2$

$2x^2 - 4x + 2 = 2(x^2 - 2x + 1) = 2(x-1)^2$

20. $f(x) = 0 \Rightarrow (3)^3 - a(3)^2 + 4(3) - 12 = 0 \Rightarrow 27 = 9a \Rightarrow a = 3$

21. $f(3) = 0 \Rightarrow K^2(3)^3 - K(3)^2 + 3k(3) - k = 0$

$27k^2 - 9k + 9k - k = 0 \Rightarrow 27k^2 = k \Rightarrow k(27 - 1) = 0 \Rightarrow k = \frac{1}{27}, 0$

22. Conceptual

23. (a) $\frac{x^2}{yz} + \frac{y^2}{zx} + \frac{z^2}{xy} = \frac{x^3+y^3+z^3}{xyz} = \frac{3xyz}{xyz} = 3$

(b) $f(a) = 0 \Rightarrow a^4 - a^2(a)^2 + 3(a) - a = 0 \Rightarrow 2a = 0\ a = 0$

(c) $x^3 + y^3 + z^3 - 3xyz = (x + y + z)[(x + y + z)^2 - 3(xy + yz + zx)]$

$p[(9)^2 - 3(23)] = 9[81 - 69] = 108$

(d) Remainder $= (1)^{2008} - 1 = 0$

24,25,26

$ax^2 + bx + c = (x - 1)(x - 2) = x^2 - 3x + 2 \Rightarrow a = 1, b = -3, c = 2$

27. $x^2 - 6x + 11 \left| \frac{x^2+7x+6}{x^4+x^3-25x^2+41x+66} \right.$

$x^4 - 6x^3 + 11x^2$

(1) $(+)(-)$

$7x^3 - 36x^2 + 41x + 66$
$7x^3 - 42x^2 + 77x$
$(-)(+)\ (-)$

$6x^2 - 36x + 66$

$6x^2 - 36x + 66$

$\underline{(-)(+)(-)}$
$\underline{0}$

$(x^2 + 7x + 6) = (x + 1)(x + 6)$

28.

$$
\begin{array}{r|cccccc}
3 & 1 & 0 & 4 & -1 & 0 & 11 \\
 & 0 & 3 & 9 & 39 & 114 & 342 \\
\hline
\end{array}
$$

$$
\begin{array}{r|ccccc|l}
3 & 1 & 3 & 13 & 38 & 114 & 353 = A_5 \\
 & 0 & 3 & 18 & 93 & 393 & \\
\hline
\end{array}
$$

$$
\begin{array}{r|cccc|l}
3 & 1 & 6 & 31 & 131 & 507 = A_4 \\
 & 0 & 3 & 27 & 174 & \\
\hline
\end{array}
$$

$$
\begin{array}{r|ccc|l}
3 & 1 & 9 & 58 & 305 = A_3 \\
 & 0 & 3 & 36 & \\
\hline
\end{array}
$$

$$
\begin{array}{r|cc|l}
3 & 1 & 12 & 94 = A_2 \\
 & 0 & 3 & \\
\hline
\end{array}
$$

$1 = A_0 \,|\, 15 = A_1$

29. Remainder $= (3)^4 - 4((3)^3 + 4(3)^2 - 2$

$81 - 108 + 36 - 2 = 7$

6. LINEAR EQUATIONS

WORKSHEET-1

Level – I

1.B	2.D	3.C	4.B	5.A	6.C	7.C
8.B	9.C	10.B				

1. Mathematical statement
2. The simple equation has only one variable
3. $\frac{x}{2}+\frac{x}{3}=5 \Rightarrow \frac{3x+2x}{6}=5 \Rightarrow 5x=6\times 5 \Rightarrow x=6$
4. $5x-8=2x-2 \Rightarrow 5x-2x=-2+8 \Rightarrow 3x=6 \Rightarrow x=2$
5. $\frac{5x-3}{1-2x}=\frac{-12}{5} \Rightarrow 25x-15=-12+24x \Rightarrow 25x-24x=-12+15 \Rightarrow x=3$
6. $x+6y=6$

   Let $x=12 \Rightarrow 12+6y=6 \Rightarrow 6y=6-12 \Rightarrow 6y=-6 \Rightarrow y=-1 \Rightarrow \therefore x=12, y=-1 \rightarrow (1)$

   Let $x=6 \Rightarrow 6+6y=6 \Rightarrow 6y=6-6=0 \Rightarrow y=0$

   $\therefore x=6, y=0 \rightarrow (2)$

   $\therefore$ Both (1) and (2)
7. Let, three consecutive natural numbers are $x, x+1, x+2$

   Given $x+x+1+x+2=114 \Rightarrow 3x+3=114 \Rightarrow x+1=38 \Rightarrow x=37, x+2=39$

   $$\therefore 37+38+39=114$$

8. $4x+2=x+8 \Rightarrow 4x-x=8-2 \Rightarrow 3x=6 \Rightarrow x=2$ 9. $x+4=0 \Rightarrow x=-4$

Level – II

1. C	2.B	3.B	4.B	5.C	6.C	7.D
8.B	9.B	10.A	11.B	12.D	13.B	14.A
15.A	16.A	17.B	18.B	19.C	20.D	21.B
22.A	23.C	24.B				

1. $\frac{3}{4}(x+1)-\frac{1}{2}=x-1 \Rightarrow \frac{3x}{4}+\frac{3}{4}-\frac{1}{2}=x-1 \Rightarrow \frac{3x}{4}-x=-1+\frac{1}{2}-\frac{3}{4}$

   $$\Rightarrow \frac{3x-4x}{4}=\frac{-4+2-3}{4} \Rightarrow -x=-5 \Rightarrow x=5$$
2. $(5x-1)(ax+1)=3a(3x-1)$, When $x=1$

   $\Rightarrow 4(a+1)=3a(2) \Rightarrow 4a+4=6a \Rightarrow 2a=4 \Rightarrow a=2$
3. $7\frac{1}{9}x-12=-8\frac{1}{9}-8\frac{2}{3}x \Rightarrow \frac{64}{9}x-12=\frac{-73}{9}-\frac{26}{3}x$

   $$\Rightarrow \frac{64x}{9}+\frac{26x}{3}=\frac{-73}{9}+12 \Rightarrow \frac{64x+78x}{9}=\frac{-73+108}{9} \Rightarrow 142x=35 \Rightarrow x=\frac{35}{142}$$
4. $\frac{2x+1}{3x-2}=\frac{9}{10} \Rightarrow 10(2x+1)=9(3x-2) \Rightarrow 20x+10=27x-18$

   $\Rightarrow 10+18=27x-20x \Rightarrow 7x=28 \Rightarrow x=4$

5. $5x - 7 = 63 \Rightarrow 5x = 63 + 7 \Rightarrow 5x = 70 \Rightarrow x = 14$

6. Given, (2,0) is solution of $2x + 3y = k \Rightarrow 2(2) + 3(0) = k \Rightarrow k = 4$

7. $2x - 7 = 45 \Rightarrow 2x = 45 + 7 \Rightarrow 2x = 52 \Rightarrow x = 26$

8. $\frac{0.3}{2x} = \frac{0.4}{2.4} \Rightarrow 0.3 \times 2.4 = 0.4 \times 2x \Rightarrow 0.72 = 0.8x \Rightarrow x = \frac{0.72}{0.8} \Rightarrow x = 0.9$

9. $3(5 - y) = 4(3y + 2) + 27 \Rightarrow 15 - 3y = 12y + 8 + 27 \Rightarrow 15 - 35 = 12y + 3y$

   $\Rightarrow 15y = -20 \Rightarrow y = \frac{-20}{15} \Rightarrow y = \frac{-4}{3}$

10. Cross multiplication

11. $3x + 5 = 50 \Rightarrow 3x = 45 \Rightarrow x = 15$

12. $5x - 2y = 13$

    Solution that satisfies the above equation

    A/ c to option verification

(1) (3,1)	2) (7,11)	3) (5,6)
$5(3)-2(1)=13$	$5\times7-2\times11=13$	$5\times5-2\times6=13$
$15-2=13$	$35-22=13$	$25-12=13$
$13=13$	$13=13$	$13=13$
$\therefore(3,1)$ is a solution	$(7,11)$ is a solution	$(5,6)$ is a solution

13. $\frac{x}{4} + \frac{x}{3} + \frac{x}{6} = \frac{-3}{4} \Rightarrow \text{LCM}(4,3,6) = 12 \Rightarrow \frac{3x+4x+2x}{12} = \frac{-3}{4} \Rightarrow 9x = -9 \Rightarrow x = -1$

14. $\frac{x}{8} + \frac{x}{6} - x = \frac{5}{6} - \frac{x}{2} \Rightarrow \frac{x}{8} + \frac{x}{6} + \frac{x}{2} - x = \frac{5}{6} \Rightarrow \text{LCM}(2,6,8) = 24$

    $\frac{3x + 4x + 12x - 24x}{24} = \frac{5}{6} \Rightarrow 19x - 24x = 5 \times 4 \Rightarrow -5x = 5 \times 4 \Rightarrow x = -4$

15. $\frac{3x-4}{5} = 4 \Rightarrow 3x - 4 = 20 \Rightarrow 3x = 24 \Rightarrow x = 8$

16. $\frac{x-5}{10} + \frac{x+5}{5} = 5 \Rightarrow \frac{x-5+2(x+5)}{10} = 5 \Rightarrow x - 5 + 2x + 10 = 50 \Rightarrow 3x + 5 = 50 \Rightarrow 3x = 45$

    $\Rightarrow x = 15$

17. $\frac{4x}{3} - \frac{3x}{4} = 28 \Rightarrow \frac{16x-9x}{12} = 28 \Rightarrow 7x = 28 \times 12 \Rightarrow x = 48$

18. $x + x + 1 + x + 2 + x + 3 = 46 \Rightarrow 4x + 6 = 46 \Rightarrow 4x = 40 \Rightarrow x = 10$

19. $0.5x - 2.5 = 77.5 - 0.3x \Rightarrow 0.5x + 0.3x = 77.5 + 2.5 \Rightarrow 0.8x = 80$

    $\Rightarrow x = \frac{80}{0.8} \times \frac{10}{10} \Rightarrow x = \frac{800}{8} \Rightarrow x = 100$

20. $\frac{x}{3} + 4 = \frac{x}{2} + 3\frac{1}{2} \Rightarrow \frac{x}{3} - \frac{x}{2} = \frac{7}{2} - 4 \Rightarrow \frac{2x-3x}{6} = \frac{7-8}{2} \Rightarrow \frac{-x}{3} = -1 \Rightarrow -x = -3 \Rightarrow x = 3$

21. $x = 3, y = 1 \Rightarrow 7(3) + 4(1) = 25;\ x = 2, y = 1 \Rightarrow 7(2) + 4(1) = 18$

    $x = 4, y = 4 \Rightarrow 7(4) + 4(4) = 44;\ x = 5, y = 2 \Rightarrow 7(5) + 4(2) = 43$

22. $\frac{3x-1}{4} - \frac{2x+5}{3} = \frac{5}{2} - 2x \Rightarrow \frac{3(3x-1)-4(2x+5)}{12} = \frac{5-4x}{2}$

    $\Rightarrow 9x - 3 - 8x - 20 = 30 - 24x \Rightarrow 25x = 53 \Rightarrow x = 53/25$

23. $(y+2, x+3) = \left(\frac{y}{2}, \frac{x}{3}\right) \Rightarrow y+2 = \frac{y}{2}, x+3 = \frac{x}{3}$

$\Rightarrow 2y+4 = y, 3x+9 = x \Rightarrow 2y-y = -4, 3x-x = -9$

$\Rightarrow y = -4, x = \frac{-9}{2}$ $(-x, -y) = \left(\frac{9}{2}, 4\right)$

24. The ratio of boys and girls = 4: 1

Number of boys = 4k

Number of girls = k

Given, $4k + k = 60 \Rightarrow 5k = 60 \Rightarrow k = 12 \Rightarrow 4k = 12 \times 4 = 48 \Rightarrow 48,12$

Level – III

25.A 26.A 27.D 28.A,B 29.B,C 30.a-p,q,r,s,t;b-q,r;c-r,d-p,s,t

31.A 32.a-s,b-pc-qd-t 33.a-q,b-s,c-t,d-p

25. $7.5x + 4 = 6.5x + 2 \Rightarrow 7.5x - 6.5x = 2 - 4 \Rightarrow x = -2$

26. $0.8x + 7 = 0.5x + 10.5 \Rightarrow 0.8x - 0.5x = 10.5 - 7 \Rightarrow 0.3x = 3.5 \Rightarrow x = \frac{3.5}{0.3}$

27. $0.3x + 0.4 = 0.28x + 1.16 \Rightarrow x = \frac{1.16-0.4}{0.3-0.28} \Rightarrow x = \frac{0.76}{0.02} \times \frac{100}{100} \Rightarrow x = \frac{76}{2} = 38 \Rightarrow x = 38$

28. $a = b \Rightarrow$ multiply ' c ' on both sides $\Rightarrow ac = bc$

Again , $a = b \Rightarrow$ Multiply ' d ' on B.S $\Rightarrow ad = bd$

Also, $a = b \Rightarrow$ multiply $\frac{1}{c}$ on B.S ($c \neq 0$) $\Rightarrow \frac{a}{c} = \frac{b}{c}$

29. $\frac{x+3}{4} = 5 \Rightarrow x + 3 = 20 \Rightarrow x = 17 \Rightarrow \frac{y-2}{3} = 6; y - 2 = 18 \Rightarrow y = 20$

$\therefore x + y = 17 + 20$

$x + y = 37$ or $x + y = \frac{74}{2}$

30. Conceptual

31. Conceptual

32. Matrix match

a) $\frac{y}{4} + \frac{y}{6} = \frac{y}{12} + 12 \Rightarrow \frac{3y+2y}{12} = \frac{y+144}{12} \Rightarrow 5y - y = 144 \Rightarrow 4y = 144 \Rightarrow y = 36 \rightarrow (4)$

b) $\frac{0.6z+8}{0.8z-6} = 7 \Rightarrow 0.6z + 8 = 5.6z - 42 \Rightarrow 8 + 42 = 5.6z - 0.6z \Rightarrow 5z = 50 \Rightarrow z = 10 \rightarrow (1)$

c) $\frac{7.4a+8}{7.5a+18} = 4 \Rightarrow 7.4a + 8 = 30a + 72 \Rightarrow 7.4a - 30a = 72 - 8 \Rightarrow -22.6a = 64 \Rightarrow a = \frac{-64}{22.6}$

$a = \frac{-32}{11.3} \rightarrow (2)$

d) $\frac{b}{8} + 7 = \frac{b}{6} - 4 \Rightarrow \frac{b}{8} - \frac{b}{6} = -4 - 7 \Rightarrow \frac{3b-4b}{24} = -11 \Rightarrow \frac{-b}{24} = -11 \Rightarrow -b = -24 \times 11$

$b = 264 \rightarrow (s)$

$3 = 3 \ \therefore x = a + b + c$

33. a) $(x + 4, 3y - 2) = (7, -5) \Rightarrow x + 4 = 7, 3y - 2 = -5 \Rightarrow x = 7 - 4, 3y = -5 + 2$

$\Rightarrow x = 3, 3y = -3 \Rightarrow y = -1 \Rightarrow \therefore (x, y) = (3, -1) \rightarrow (2)$

b) $\left(a + 2, b + \frac{1}{2}\right) = (3,4) \Rightarrow a + 2 = 3, b + \frac{1}{2} = 4 \Rightarrow a = 3 - 2, b = 4 - \frac{1}{2} = \frac{8-1}{2}$

$\Rightarrow a = 1, b = 7/2 \Rightarrow \therefore (a, b) = (1, 7/2)$

c) $\left(\frac{p}{3}, q + 5\right) = (-1, -2) \Rightarrow \frac{p}{3} = -1, q + 5 = -2 \Rightarrow p = -3, q = -2 - 5 = -7$

$(p, q) = (-3, -7)$

d) $\left(\frac{a}{2} + 1, b + \frac{1}{2}\right) = \left(2, \frac{1}{2}\right) \Rightarrow \frac{a}{2} + 1 = 2, b + \frac{1}{2} = \frac{1}{2} \Rightarrow \frac{a}{2} = 2 - 1, b = \frac{1}{2} - \frac{1}{2}$

$\Rightarrow \frac{a}{2} = 1, b = 0 \Rightarrow a = 2 \Rightarrow \therefore (a, b) = (2, 0)$

WORKSHEET-2

Level – I

1. A 2.B 3.A 4.D 5.A 6.C 7.B

8.D 9.B 10.D

2. $x + y = 3$

$y = 3 - x$ 1

$4x - 3y = 26 \Rightarrow 4x - 3(3 - x) = 26$

$\Rightarrow 4x - 9 + 3x = 26 \Rightarrow 7x = 35 \Rightarrow x = 5$

3. $2x + y = 7$ (1) $\times 1$

$x + 3y = 11$

$2x + y = 7$

$2x + 6y = 22$

$-5y = -15$

$y = 3$

4. $x + y = 3$ (1) $\times 4$

$\Rightarrow 4x + 4y = 12$

$4x - 3y = 26 \quad y = -2$

$7y = -14$

5. Given $2x + y = 7 \Rightarrow y = 7 - 2x$ (1) $\Rightarrow$ Also given,

$x + 3y = 11 \Rightarrow x + 3(7 - 2x) = 11 \Rightarrow x + 21 - 6x = 11 \Rightarrow -5x = 11 - 21 \Rightarrow -5x = -10$

$x = \frac{10}{5} = 2$

6. Let $\frac{1}{x} = u, \frac{1}{y} = v$

$8u - 5v = 34$

$3u - 2v = 13$

$(2) \times 8$

$24u - 15v = 102$

$24u - 16v = 104$

$v = -2$

$\frac{1}{y} = -2 \Rightarrow y = -\frac{1}{2}$

7. $59x + 47y = 631$

   $47x + 59y = 535$

   $106(x + y) = 1166$

   $x + y = \frac{1166}{106} = 11$

8. Given, $x - y = 0 \Rightarrow x = y \Rightarrow$ also $2x - 2y = 0 \Rightarrow 2x = 2y \Rightarrow x = y$

   $\therefore (0,0)(1,1)(3,3) \Rightarrow\therefore$ All of these

9. $x + y = 4$

   $\frac{x - y = 6}{2x = 10}(2)$

   $x = 5$

   from $(1) \Rightarrow 5 + y = 4 \Rightarrow y = 4 - 5 \Rightarrow y = -1 \Rightarrow\therefore (5, -1)$

10. $2x + 3y = 17\ (1) \Rightarrow 3x - 2y = 6$

    A/C to Options $\Rightarrow (4)(4,3) = (x_1 y) \Rightarrow$ from $2(4) + 3(3) = 17 \Rightarrow 17 = 17$

    from $3(4) - 2(3) = 6 \Rightarrow 6 = 6 \Rightarrow\therefore (4,3)$ is the solution of (1) & (2)

Level – II

1.C	2.A	3.A	4.B	5.D	6.A	7.C
8.B	9.A	10.B	11.D	12.C	13.A	14.B
15.A	16.A	17.B	18.A	19.D	20.A	

1. $2x + 3y = 0 \ldots\ldots (1); 3x - 2y = 0$

   $4x + 6y = 0$

   $\frac{9x - 6y = 0}{13x = 0}; x = 0; \therefore$ from 1

   $2(0) + 3y = 0;\ y = 0$

2. $2x + y = 32 \ldots\ldots (1);\ 3x + 4y = 68 \ldots\ldots (2)$

   $5x + 5y = 100;\ 5(x + y) = 100$

   $x + y = 20$

3. (1,3) is the solution of $3x + 5y = b$

   $3(1) + 5(3) = b;\ b = 18$

4. $x - 2y = 4 \ldots\ldots (1) \Rightarrow$ option verification

   $(0,2)(x);\ 0 - 2(2) = 4$

   $-4 \neq 4;$ option 2(40)

   $4 - 2(0) = 4; 4 = 4 \Rightarrow\therefore (4,0)$ is the solution of (1)

5. $4x - y = 3 - - - (1)$

   let, $x = 1, y = 1; 4(1) - 1 = 3; 3 = 3 \therefore (1,1)$ is the solution of equation (1)

6. $3x + 2y = 13 \ldots . (1) \times 3$

$\underline{5x - 3y = 9 \ldots \ldots (2) \times 2}$

$9x + 6y = 39$

$\underline{10x - 6y = 18}$

$19x = 57$ $\Rightarrow x = \frac{57}{19} = 3$

7. $11x - 6y = 28 \ldots$

$\underline{2x + 3y = 1 \ldots . (2) \times 2}$

$11x - 6y = 28$

$\underline{4x + 6y = 2}$

$15x = 30$

$x = 2 \Rightarrow \text{From}(1) \Rightarrow 11(x) - 6y = 28 \Rightarrow -6y = 28 - 22 \Rightarrow -6y = 6 \Rightarrow y = -1$

8. $3x + 2y = 13 \ldots (1) \times 5$

$5x - 3y = 9 \ldots . (2) \times 3$

$15x + 10y = 65$

$15x - 9y = 27$

$+ \ +$

$19y = 38$

$y = 2$

9. $x + 2y = 4$

$(1) \times 2 \Rightarrow 7x + 4y = 18$

$(2) \times 1$

$2x + 4y = 8$

$7x + 4y = 18$

$- \ - \ -$

$-5x = -10$

$x = 2 \Rightarrow \text{From}(1) \Rightarrow 2 + 2y = 4$

$2y = 2; \ y = 1$

$\because x = 2, y = 1$

10. $3x + 4y = 17$

$\underline{5x - 2y = 11 \ldots . (2)}$

$3x + 4y = 17$

$\underline{10x - 4y = 22}$

$13x = 39$

$x = 3 \Rightarrow \text{from}(1) \Rightarrow 3(3) + 4y = 17 \Rightarrow 4y = 8 \Rightarrow y = 2 \Rightarrow \therefore x - y = 3 - 2 = 1 \Rightarrow x - y = 1$

11. $x - 4y = 5 \ldots . (1) \times 2$

$\underline{2x + y = 7 \ldots \ldots (2) \times 1}$

$2x - 8y = 10$

$2x + y = 7$ $; y = \frac{-1}{3}$

$\underline{- \ - \ -}$

$-9y = 3$

12. $10x + 3y = 75 \ldots\ldots (1) \times 5$

$6x - 5y = 11 \ldots\ldots (2) \times 3$

$50x + 15y = 375$

$18x - 15y = 33$

$68x = 408$

$x = \frac{408}{68}; x = 6$

From $(1) \Rightarrow 10(6) + 3y = 75 \Rightarrow 3y = 75 - 60 \Rightarrow 3y = 15; y = 5$

$\therefore x + y = 6 + 5 = 11 \Rightarrow x + y = 11$

13. $4x + 3y = 7$

$2x + 3y = -1$

$- \quad +$

$2x = 8$

$x = 4$

14. $3x - y = 5$

$3x + 3y = 21$

$- - \quad -$

$y = 4$

15. $x + y = 7 \ldots\ldots (1) \times 4$

$4x + 5y = 30 \ldots. (2) \times 1$

$4x + 4y = 28$

$4x + 5y = 30$

$- \quad - \quad -$

$-y = -2$

$y = 2 \Rightarrow x + y = 7; x + 2 = 7; x = 5$

16. Let $\frac{1}{x} = u \Rightarrow \frac{1}{y} = v \Rightarrow \frac{2}{x} + \frac{2}{3}\frac{1}{y} = \frac{1}{6} \Rightarrow 2u + \frac{2}{3}v = \frac{1}{6} \Rightarrow 6u + 2v = \frac{3}{6} = \frac{1}{2}$

$12u + 4v = 1 \ldots\ldots (1) \Rightarrow \frac{3}{x} + \frac{2}{y} = v \Rightarrow 3u + 2v = 0 \ldots\ldots (2) \Rightarrow (1)$ and (2)

$12u + 4v = 1$

$6u + 4v = 0$

$- \quad -$

$6u = 1$

$u = \frac{1}{6} \Rightarrow \frac{1}{x} = \frac{1}{6} \Rightarrow x = 6$

17. Substitute $x = 6$ in $\Rightarrow \frac{2}{x} + \frac{2}{3y} = \frac{1}{6} \Rightarrow \frac{2}{6} + \frac{2}{3y} = \frac{1}{6} \Rightarrow \frac{2}{3y} = \frac{1}{6} - \frac{2}{6} \Rightarrow \frac{2}{3y} = \frac{-1}{6}$

$\frac{2}{y} = \frac{-1}{2} \Rightarrow y = -4$

18. $5x + 3y = 11$

$3x + 5y = 13$

$8x + 8y = 24$

$8(x+y)=24$

$x+y=3$

19. $\frac{4}{x}+3y=14 \Rightarrow \frac{3}{x}-4y=23 \Rightarrow$ Let $\frac{1}{x}=u$

$4u+3y=14 \ldots\ldots(1)\times 4 \Rightarrow 3u-4y=23 \ldots\ldots(2)\times 3$

$16u+12y=56$

$9u-12y=69$

$25u=125$

$u=5 \Rightarrow \frac{1}{x}=5; x=\frac{1}{5}$

Level – III

21.A	22.A	23.A	24.A,B	25.A,C,D	26.a-s,b-r,c-qd-p
27.5	28.1	29.a-r,b-s,c-p,d-q		30.A,C,D	31.B
32.A	33.C				

20. $2u+3v=uv$

$3u-4v=2uv \ldots\ldots(2)\times 3$

$8u+12v=4uv$

$9u-12v=6uv$

$17u=10uv$

$v=\frac{17}{10}$

21. $2x+3y=200$

$3x+2y=250$

$5x+5y=450$

$5(x+y)=450$

$x+y=90$

22. $2x+3y=200.$

$3x+2y=250 \ldots\ldots(2)\times 3$

$4x+6y=400$

$9x+6y=750$

$-5x=-350$

$x=70 \Rightarrow$ From$(2) \Rightarrow 70+y=90 \Rightarrow y=20 \Rightarrow \therefore x-y=70-20 \Rightarrow x-y=50$

23. The value of $x=70$

24. $3x+2y=-25$

$-2x-y=10 \ldots.(2)\times 2$

$3x+2y=-25$

$-4x-2y=20$

$-x=-5$

$x=5$

From $(1) \Rightarrow 3(5)+2y=-25 \Rightarrow 2y=-40 \Rightarrow y=-20 \Rightarrow \therefore x^2+y=5^2+(-20)$

$=25-20=5=x$

25. $2x+3y=63$

$3x+2y=62$

$5x+5y=125$

$5(x+y)=125$

$x+y=25=5^2=\sqrt{625}$

26

.a) $2x+3y=22$ …. $(1)\times 2$

$3x-2y=7$ … … $(2)\times 3$

$4x+6y=44$

$9x-6y=21$

$13x=65$

$x=5$

b) From(1) 2(5)+3y=22 $\Rightarrow 3y=12\Rightarrow y=4$

c) $x+y=1$

$2x-y=-4$

$3x=-3$

$x=-1$

d) From (1) $\Rightarrow -1+y=1\Rightarrow y=2$

Integer type

27. $x-y=3$ … $(1)\times 2$

$3x-2y=0$ ….. (2)

$2x-2y=6$

$3x-2y=10$

$-\ \ -\ \ -$

$-x=-4$

$x=4$

From(1)

$4-y=3;-y=-1;\ y=1$

$\therefore x+y=4+1=5$

28. $10x+3y=25$ … … $(1)\times 5$

$6x-5y=11$ … …. $(2)\times 3$

$50x+15y=395$

$18x-15y=33$

$68x=408$

$x=6$

From $(1)\Rightarrow 10(6)+3y=75\Rightarrow 3y=15\Rightarrow y=5\Rightarrow\therefore x-y=6-5=1$

29

.a) $x+4y=2$ … …. $(1)\times 1$

$4x-y=-9$ … … $(2)\times 4$

$x+4y=2$

$16x-4y=-36$

$17x=-34$

$x=-2$

b) from $(1)\Rightarrow -2+4y=2;4y=4;y=1$

c) $2x+2y=2$

$2x - y = -4 \ldots\ldots (2) \times 2$

$2x + 2y = 2$

$4x - 2y = -8$

$6x = -6$

$x = -1$

d) From(1)

$2(-1) + 2y = 2$

$2y = 4; y = 2$

30. $4x - 3y = 6$.....

$5x + y = 17 \ldots\ldots (2) \times 3$

$4x - 3y = 6$

$15x + 3y = 51$

$19x = 57$

From(1) $4(3) - 3y = 6; -3y = -6; y = 2 \Rightarrow \therefore x + y = 3 + 2 = 5 = \sqrt{25} = \sqrt[3]{125}$

31. $3x + 2y = 5 \ldots. (1) \times 1$

$x - y = 1 \ldots\ldots (2) \times 2$

$2x - 2y = 2$

$5x = 7$; $x = \frac{7}{5}$

32. From(1)

$3\left(\frac{7}{5}\right) + 2y = 5; \frac{21}{5} + 2y = 5; 2y = 5 - \frac{21}{5}; 2y = \frac{25-21}{5}; \ y = \frac{4}{10}; y = \frac{2}{5}$

33. $x^2 - y^2 = \frac{49}{25} - \frac{4}{25} = \frac{45}{25} = \frac{9}{5}$

WORKSHEET-3

Level – I

1.B	2.A	3.A	4.D	5.A	6.D
7. A	8. A	9.A	10.B	11.A	12.B

1. Given $\Rightarrow x + y = \mathbf{1000} \rightarrow (1) \Rightarrow$ Also given,

$x^2 - y^2 = 256000$

$(x - y)(x + y) = 256000$

$1000(x - y) = 256000$

$x - y = 256$.....(2)

(1) + (2)

$x + y = 1000$

$x - y = 256$

$2x = 1256$

$(x = 628)$

from (1)

$y = 1000 - 628$

$(y = 372)$

2. Let cost of pen is ' x ' cost of pencil is ' y '

$59x + 47y = 631$

$47x + 59y = 535$

$106x + 106y = 1166$

$(x + y = 11) \rightarrow (1)$

$59x + 47y = 631$

$47x + 59y = 535$

$12x - 12y = 96$

$12(x - y) = 96$

$(x - y = 8) \rightarrow 2$

$x + y = 11$

$x - y = 8$

$2x = 19$

$(x = 9.5)$rs

Sub

$x = 9.5$ in eq(1)

$(y = 1.5)$

3. Let cost of pen is x

cost of pencil is y

$37x + 53y = 320$

$53x + 37y = 400$

$90x + 90y = 420$

$90(x + y) = 720$

$(x + y) = 8$

sum of costs of pen & pencils are Rs. 8

4. A two digit number is represented by $10x + y$

or $= 10y + x$

5. Sum of the angles in triangle = 180°

6. The sum of the angle in a quadrilateral = 360°

7. Distance = speed × time

8. Speed of down stream = $(x + y)$km/ph

9. Speed of up stream = $(x - y)km/ph$

10. Three digit number is represented by = $100y + 10z + x$

11. In a Square each angle = 90°

12. $x + y = 90$

$x - y = 40$

$2x = 130$

$x = 65$

Level – II

1.A	2.C	3.A	4.A	5.A	6.C	7.A
8.C	9.A	10.A	11.A	12.A	13.B	14.D
15.D	16.C	17.B	18.B	19.B		

1. 1 Let present age of shobha = xy years

   13 years ago, shobha age = $x - 13$ after 13 years,

   shobha age = $x + 13 \Rightarrow$ given, $x - 13 = \frac{1}{2}(x + 13) \Rightarrow 2x - 26 = x + 13 \Rightarrow 2x - x = 13 + 26 \Rightarrow (x = 39)$

   $\therefore$ present age of shobha x = 39 years

2. let, The first consecutive numbers multiple of 3 = $3x$

   second consecutive number = $3x + 3 \Rightarrow$ Given, $3x + 3x + 3 = 39$

   $6x = 66 \Rightarrow (x = 11) \Rightarrow \therefore 3x = 3 \times 33 = 33$

   $3x + 3 = 33 + 3 = 36 \Rightarrow \therefore$ 33,36

3. Let, Two numbers be $x, y \Rightarrow$ given $\Rightarrow 2x + 3y = 89 \rightarrow (1)x2$ and

   $4x = 5y + 13$

   $4x - 5y = 13 \rightarrow (2)$

   $(1) \Rightarrow 4x + 6y = 178$

   $(2) \Rightarrow \frac{4x-5y=13}{11y=165}$

   $y = \frac{165}{11} \Rightarrow y = 15 \Rightarrow \text{From}(1) \Rightarrow 2x + 3(15) = 89 \Rightarrow 2x = 89 - 45 \Rightarrow 2x = 44 \Rightarrow x = 22$

4. Let

   man's age present ' x ' $\Rightarrow$ son's age ' y ' $\Rightarrow$ 6 years hence $\Rightarrow x + 6 = 3(y + 6) \rightarrow (1)$

   3 years ago $\Rightarrow x - 3 = 9(y - 3) \rightarrow (2) \Rightarrow (1) \Rightarrow x + 6 = 3y + 18$

   $x - 3y = 12 \Rightarrow (3) \Rightarrow (2)$

   $x - 3 = 9y - 27$

   $x - 9y = 27 \Rightarrow (4)$

   (3)-(4)

   $x - 3y = 12$

   (y = 6)

   $y = 6$ in eq $\Rightarrow 3 \Rightarrow x - 3(6) = 12 \Rightarrow x = 12 + 18 \Rightarrow x = 30 \Rightarrow$ present ages, 30&6

5. Let the digit at ten's place = x

   the digit at one's place = y $\Rightarrow \therefore$ Number = 10x + y

   Given, sum of its digits = 15 $\Rightarrow x + y = 15$

   one reversing the order,

   $\therefore$ Number formed = 10y + x

   Given,

$10y + x = 10x + y + 9 \Rightarrow 10x - x + y - 10y = 9 \Rightarrow 9x - 9y = -9 \Rightarrow 9(x - y) = -9 \Rightarrow x - y = -1$

Solving (1)&(2)

$x + y = 15$

$x - y$

$2x = 14$

$x = 7$

From (1)

$7 + y = 15$

$y = 15 - 7$

$y = 18$

6. required number $= 10x + y = 10(7) +$

Let the two digit number is $10x + y$,

given

$y - x = 3$

By problem ,

$7(10x + y) = 4(10y + x) \Rightarrow 70x + 7y - 40y - 4x = 0 \Rightarrow 66x - 33y = 0 \Rightarrow 66x = 33y$

$2x = y \rightarrow (2 \Rightarrow$ substitute (2) in (1)

$2x - x = 3 \Rightarrow x = 3 \Rightarrow x = 3$ in $(2) \Rightarrow 2(3) = y \Rightarrow y = 6$

$\therefore$ the number is $10(3) + 6 = 30 + 6 = 36$

7. $x + x + 50 + x - 20 = 180 \Rightarrow 3x = 150 \Rightarrow x = 50$

8. speed of current is $= \frac{1}{2}(p - q)kmph$

9. Let, At present

cousin age = x years

Then Raju age = $x - 19$ years

After 5 years,

cousin age = $x + 5$

Raju age = $x - 19 + 5 = x - 14$

Given, $\frac{X-14}{X+5} = \frac{2}{3} \Rightarrow 3X - 42 = 2X + 10 \Rightarrow X = 52$

$\therefore$ Present age of cousin $(x) = 52$ years

Present age of Raju $(x - 19) = 52 - 19 = 33$ years

10. Let unit digit ' x '

Ten's digit ' y '

By problem $x = 2y \rightarrow (1)$

given number $\rightarrow 10y + x$

reverse number $\rightarrow 10x + y$

According problem

$10x + y = 10y + x + 27$

$10x + y - 10y - x = 27$

$9x - 9y = 27$

$x - y = 3 \rightarrow (2)$

from (1)&(2) $\Rightarrow 2y - y = 3 \Rightarrow y = 3 \Rightarrow x = 2(3) = 6$

original number $= 10(3) + 6 = 30 + 6 = 36$

11. Given $\angle A = (2x - 1), \angle B = (y + 5) \Rightarrow \angle C = (2y - 1), \angle D = (4x - 7)$

and $ABCD$ is a cyclic quadrilateral

then, $\angle A + \angle C = 180° \Rightarrow \angle B + \angle D = 180°$

If $\angle A + \angle C = 180° \Rightarrow 2x - 1 + 2y + 15 = 180°$

$2x + 2y = 180 - 14 \Rightarrow 2(x + y) = 166 \Rightarrow (x + y) = 83 \rightarrow (1)$

If $\angle B + \angle D = 180° \Rightarrow y + 5 + 4x - 7 = 180°$

$4x + y = 182 \rightarrow (2)$

(2)-(1)

$4x + y = 182$

$x = 33$

From(1)

$33 + y = 83$

$y = 50 \ \therefore \angle B = y + 5 = 50 + 5 = 55°$

12. Let speed of stream ' x '

speed of man 18kmph

upstream speed $= 18 - x$

downstream speed $= 18 + x$

By problem

speed of down stream = 3 (speed of upstream)

$18 + x = 3(18 - x) \Rightarrow 18 + x = 54 - 3x \Rightarrow 4x = 36 \Rightarrow x = 9$

speed of stream = 9kmph

13. given $\angle C = \angle B = 2(\angle A + \angle B)$

Let $\angle C = 3\angle B = 2(\angle A + \angle B) = x$

$\angle C = x, \angle B = \frac{2}{3}x \Rightarrow 2\angle A + \frac{2x}{3} = x \Rightarrow 2\angle A = \frac{x}{3} \Rightarrow \angle A = \frac{x}{6}$

In $\triangle ABC \angle A + \angle B + \angle C = 180 \Rightarrow \angle A = \frac{x}{6} = 20° \Rightarrow \angle B = \frac{x}{3} = 40°$

$\angle C = x = 120° \Rightarrow \frac{x}{6} + \frac{x}{3} + x = 180$

$x + 2x + 6x = 180 \times 6 \Rightarrow 9x = 180 \times 6 \Rightarrow x = 120$

14. Let present age of A& present age of B are x&y

3 years ago

A was thrice as old as B

$x - 8 = 3(y - 8) \rightarrow (1)$ ten years later

$x + 10 = 2(y + 10) \rightarrow (2)$

1. $\Rightarrow x - 8 = 3y - 24 \Rightarrow x - 3y = -24 + 8 \Rightarrow x - 3y = 16$

2. $\Rightarrow x + 10 = 2y + 20 \Rightarrow x - 2y = 10 \rightarrow (4)$

by 3&4

$x - 3y = -16$

x - 2y = -10

———————

$y = 26$

4. $\Rightarrow x - 3(26) = -16 \Rightarrow x - 78 = -16 \Rightarrow x = 62$

15. Let units digit x

ten's digit y

number $\rightarrow 10y + x \Rightarrow y = x + 2 \rightarrow (1)$

$x + y = \frac{1}{7}(10y + x) \rightarrow (2)$

2. $\Rightarrow 7x + 7y = 10y + x \Rightarrow 7x + 7y - 10y - x = 0 \Rightarrow 6x - 3y = 0 \Rightarrow 6x = 3y \Rightarrow y = 2x \rightarrow (3)$

From 1&2 $\Rightarrow 2x = x + 2 \Rightarrow x = 2$

3. $\Rightarrow y = 2(2) \Rightarrow y = 4$

the number is $= 10(y) + x = 10(4) + 2 = 40 + 2 = 42$

16. Let two numbers x&y

$x + y = 58$

$x = 35$

$y = 23$

17. Let fraction $= \frac{a}{b}$

given

$a + b = 8 \rightarrow (1)$

$\frac{a+3}{b+3} = \frac{3}{4}$

$4a + 12 = 3b + 9$

$4a - 3b = -3 \rightarrow (2)$

$(1) \times 4 \Rightarrow 4a + 4b = 32$

$\underline{(2)} \Rightarrow \frac{4a - 3b = -3}{7b = 35}$

$b = 5$

$b = 5 \Rightarrow a + b = 8$

$a + 5 = 8$

$a = 3$

$\therefore$ the fraction $\rightarrow \frac{3}{5}$

speed of the stream = 2kmph

18. speed of boat ' u '

Speed of stream ' v '

upstream speed ' u - v '

down stream speed ' $u + v$ '

case(i)

$\frac{12}{u-v}+\frac{40}{u+v}=8 \Rightarrow \frac{3}{u-v}+\frac{10}{u+v}=2$

case(ii)

$\frac{16}{u-v}+\frac{32}{u+v}=8 \Rightarrow \frac{2}{u-v}+\frac{4}{u+v}=1 \rightarrow (2)$

take $\frac{1}{u-v}=a, \frac{1}{u+v}=b$

$3a + 10b = 2 \rightarrow (3)$

$2a + 4\ b = 1 \rightarrow (4)$

$(3) \times 2 = 6a + 20b = 4$

$\underline{(4) \times 3 = 6a + 12b = 3}$

$8b = 1$

$b = \frac{1}{8}$

if $b = \frac{1}{8}$

then $2a + 4\left(\frac{1}{8}\right) = 1 \Rightarrow 2a = \frac{1}{2} \Rightarrow a = \frac{1}{4} \Rightarrow a = \frac{1}{4}, b = \frac{1}{8} \Rightarrow \frac{1}{u-v} = \frac{1}{4}, \frac{1}{u+v} = \frac{1}{8}$

$u - v = 4\ u + v = 8$

$u - v = 4$

$\underline{u + v} = 8$

$u = 6 \Rightarrow v = 2$

19. Conceptual

Level – III

20.a-s,b-r,c-p,t;d-q 21.6 22.4 23.B 24.A 25.B

26.A,C,D 27.A,B,D 28.a-q,t;b-p,r;c-r,p;d-s 29.B 30.A,B

31.A,B,C,D

20. Conceptual

21. $3x + 6 = 24 \Rightarrow 3x = 24 - 6\ 3x = 18 \Rightarrow x = 6$

22. $7x + 8x = 60 \Rightarrow x = 4$

23. By adding ' 3 ' to 5,9,17,27

8,12,2027, will be in proportion $\therefore x = 3$

24. Second number = 9 + 3 = 12

Third number = 17 + 3 = 20

Sum of these numbers = 12 + 20 = 32

25. Thrice of second number added to 4 is = 3(12) + 4 = 36 + 4 = 40

26. The sum of opposite angles in cyclic

Quadrilateral is 180° = 2 × 90° = Straight angle

27. Three angles = $1:2:3 \Rightarrow x + 2x + 3x = 180 \Rightarrow 6x = 180° \Rightarrow x = 30°$

= right angle = $\frac{1}{2}$(straightangle) = $\frac{1}{4}$(completeangle)

28. a) In a cyclic Quadrilateral ⇒ Sum of angles = 360° = 4 right angles

b) In a triangle sum of angle is 180° = 2 right angles

c) The sum of apposite angles in cyclic

Quadrilaterals = 180° = 2 right angles

d) In rectangle, each angle is right angle.

29. Let speed of boat ' u '

speed of stream ' v '

upstream = u − v down stream = u + v

According to problem $\frac{u-v}{u+v} = \frac{3}{4} \Rightarrow 4u - 4v = 3u + 3v \Rightarrow 4u - 3u = 7v \Rightarrow \frac{u}{v} = \frac{7}{1}$

u: v = 7: 1

30. Let ' A ' along complete work in ' x ' days

' B ' along complete work in ' y ' days

capacity of ' A ' =→ $\frac{1}{x}$

'B' =→ $\frac{1}{y}$

A & B together complete the work in '10 days'

capacity of $A\&B \rightarrow \frac{1}{10}$

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{10}$$

By problem

$$12\left(\frac{1}{x}\right) + 6\left(\frac{1}{y}\right) = 1$$

do (ii) −(i) × 12

$$\frac{12}{x} + \frac{6}{y} = 1$$

$$\frac{12}{x} + \frac{12}{y} = \frac{12}{10}$$

$$-\frac{6}{y} = \frac{-2}{10}$$

y = 30days

$\frac{1}{x} + \frac{1}{30} = \frac{1}{10} \Rightarrow \frac{1}{x} = \frac{1}{10} - \frac{1}{30} \Rightarrow \frac{1}{x} = \frac{2}{30} = \frac{1}{15} \Rightarrow x = 15$ days

31. Four angles $ABCD$ quadrilateral

$\angle A = 2x - 1,\ \angle B = y + 5 \Rightarrow \angle C = 2y + 15 \angle D = 4x - 7$

$2x - 1 + y + 5 + 2y + 15 + 4x - 7 = 360 \Rightarrow 6x + 3y = 348$

$2x + y = 116$

$\angle A + \angle C = 180^\circ \Rightarrow 2x + 2y + 14 = 180^\circ \Rightarrow 2x + 2y = 166$

$x + y = 83$

$(1) - (2)$

$2x + y = 116$

$x + y = 83$

$x = 33$

$x + y = 83$

$\Rightarrow y = 83 - 33 = 50^\circ$

$\therefore \angle A = 2x - 1 = 2(33) - 1 = 66 - 1 = 65^\circ$

$\angle B = y + 5 = 50 + 5 = 55^\circ$

$\angle C = 2y + 15 = 2(50) + 15 = 100 + 15 = 115^\circ$

$\angle D = 4x - 7 = 4(33) - 7 = 125^\circ$

WORKSHEET-4

Level – I

1. A	2. B	3. C	4. A	5. B	6. B
7. C	8. D	9. A	10. B		

1. $x = k$
2. Conceptual

Level – II

1. A	2. D	3. D	4.A	5. A	6.C	7. C
8. D	9. C	10. D	11. C	12. A	13.B	14. C
15. D	16. A	17. A	18. D	19.A	20. B	21. C
22. D	23. B	24. C	25. D	26. B,C	27. A	28. A
29. B	30. A	31.a-r,b-s,c-p,d-q		32. a-r,t,q b-r,t,c-p,d-q		
33.a-s,b-q,c-u,d-u,e-t		34. B	35.B			

3. $N = \{1,2,3,4 - - - - - - - n\}$

If $X = 1 \Rightarrow y = 0 \notin N$

$x = 2 \Rightarrow y = -1_{\notin N}$

4. $(0,0)$
5. $(x, y) = (-2,3)$
6. $8(0) + 9(0) + K = 9$

$k = 9$

7. $x + y = 100$

   x- y= 50

   2x= 150

   $\Rightarrow x = 75 \& y = 25$

8. Both lines are coincident lines, so infinite solutions.

9. $\frac{12}{K} = \frac{-6}{-2} \Rightarrow$ K = 4

10. Conceptual

11. $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

12. $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$

13. $\frac{2}{4} = \frac{3}{6} = \frac{1}{2}$, infinite solutions.

14. $4(3) - 3(4) = 12 - 12 = 0$

15. $\frac{2}{K} = \frac{5}{10} \Rightarrow$ K = 4

16. $2(2) + 3$ K $= 7 \Rightarrow 3$ K $= 3 \Rightarrow$ K $= 1$

17. $y = mx + c$ on y-axis $\Rightarrow y = c$

18. $ax + by + c = 0$ on x-axis $\Rightarrow ax = -c \Rightarrow x = -\frac{c}{a} \Rightarrow y = 0$

19. $\frac{2}{a} = \frac{3}{b} \Rightarrow \frac{a}{b} = \frac{2}{3}$

20. $\frac{2}{K-1} = \frac{3}{K+2} \Rightarrow 2K + 4 = 3K - 3$

    $K = 7$

21 $46x + 54y = 254$ 1

   $54x + 46y = 246 - - \cdots - - - - 2$

   $1 + 2 \Rightarrow x + y \Rightarrow 5 \Rightarrow 1 - 2 \Rightarrow x - y = -1 \Rightarrow (x, y) = (2,3)$

22. Passes through (0,0)

23. Let the three numbers

    $x, 2x, 3x$

    There sum $= 60 \Rightarrow x = 10$

    Smallest number = 10

24. Let breadth = b

    length $\Rightarrow +4$

    $\therefore 2(1 + h) = 40$

    $\ddot{b} = 8$ m $\Rightarrow$ I = 12 m

25. $xy + 1 = y$

    $xy + 1 = x$

    - - .

    y-x =0

26. Conceptual

27. Conceptual
28. parallel to X axis
29. parallel to Y axis
30. (-1,1)
31. Conceptual
32. Conceptual
33. Conceptual
34. Add 1&2 $\Rightarrow x + y = 10$

    Sub 1&2 $\Rightarrow x - y = 2$

    $x = 6\ y = 4 \Rightarrow x - y = 2$

WORKSHEET-5

Level - I

1. B	2.A	3.A	4.B	5.C	6.A	7.D
8.D	9.A	10.C	11.A			

1. linear Inequation
2. solution set
3. does not change
4. $3x - 2 < 10 \Rightarrow 3x < 12 \Rightarrow x < 4$
5. $3x < 15, x \in N \Rightarrow x < 5 \Rightarrow \{1,2,3,4\}$
6. Inequation
7. $2 + x > -8 \Rightarrow x > -10 \Rightarrow \{-9, -8, -7, ...$
8. $x > -4 - 4 < x - 1$

   given, $x < 4 - 2$

   from 1&2

   $-4 < x < 4 \Rightarrow \{3, -2, -1,0,1,2,3,\}\ \therefore x = -3$
9. $9x - 2 \geq 16 \Rightarrow 9x \geq 18 \Rightarrow x \geq 2$
10. $3x - 2 < 4$
11. $-2x < -6 \Rightarrow 2x > 6 \Rightarrow x > 3$

Level – II

1.C	2.A	3.A	4.B	5.A	6.D	7.C
8.B	9.A	10.A	11.C	12.A	13.B	14.A
15.A	16.C	17.A	18.C	19.B		

1. $-2x < 2 \Rightarrow -x < 1 \Rightarrow x > -1$
2. $5x - 9 < 15, x \in w \Rightarrow 5x < 24 \Rightarrow x < 4.81 x \in w \Rightarrow x = \{0,1,2,3,4\}$
3. $\frac{2x}{3} > 12 \Rightarrow 2x > 12 \times 3 \Rightarrow x > 18$
4. Transitive
5. $\frac{x}{2} + 3 > 5, x \in N \Rightarrow \frac{x}{2} > 2 \Rightarrow x > 4 \Rightarrow x = \{5,6,7,8,9, ...$

6. $7-\frac{y}{2}>\frac{5y}{3}-6 \Rightarrow 7+6>\frac{5y}{3}+\frac{y}{2}$

$13>\frac{10y+3y}{6} \Rightarrow 13\times 6>13y \Rightarrow 6>y \Rightarrow y<6$

7. $5-4x<x-10 \Rightarrow 5+10<x+4x \Rightarrow 15<5x \Rightarrow 3<x \Rightarrow x>3$

$x=\{4,5,6,7,8,\ldots\} \Rightarrow \therefore x=\{4,5,6,7,8,$

8. $a>0$ and $|x|<$ athen $x<\pm a \Rightarrow -a<x<a$

9. $\frac{7t+5}{6}>\frac{t-1}{2} \Rightarrow 2(7t+5)\geq 6(t-1)$

$14t+10\geq 6t-6 \Rightarrow 14t-6t>-6-10$

$8t\geq -16$

$t>-2$

$t=\{-2,-1,0,1,2,3,\ldots\}\ t=-1$

10. $5x-1<2x+8 \Rightarrow 5x-2x<8+1$

$3x<9 \Rightarrow x<3 \Rightarrow x=\{-2,-1,0,1,2\}$

11. $(2,3)=(x,y)$

A/ C to options ; $x-y-5<0 \Rightarrow 2-3-5<0 \Rightarrow -6<0$

12. a < b and c > 0 then ac < bc $\therefore \frac{a}{c}<\frac{b}{c}$

13. $3(x-1)>2(x+2)-9 \Rightarrow 3x-3>2x+4-9$

$3x-2x>-5+3 \Rightarrow x>-2 \Rightarrow x=\{-1,0,1,2,3,4,\ldots..$

14. $2y<x+4 \Rightarrow A/C$ to options

1. $(1,-3)=(x,y) \Rightarrow 2(-3)<1+4 \Rightarrow -6<5$ $\therefore (1,-3)$ is correct option.

15. $x+2\leq 8 \Rightarrow x<6, \forall x\in N \Rightarrow x=\{1,2,3,4,5,6\}$

16. $-(x-3)+4<5-2x \Rightarrow -x+3+4<5-2x \Rightarrow -x+2x<5-7 \Rightarrow x<-2$

17. $\frac{1}{2}\left(\frac{3}{5}x+4\right)\geq\frac{1}{3}(x-6) \Rightarrow 3\left(\frac{3}{5}x+4\right)\geq 2(x-6)$

$\frac{9x}{5}+12\geq 2x-12 \Rightarrow \frac{9x}{5}-2x\geq -12-12 \Rightarrow \frac{9x-10x}{5}\geq -24 \Rightarrow -x\geq -24\times 5$

$-x\geq -120 \Rightarrow x\leq 120$

18. Consecutive numbers whose sum is less than 25 are (11,12)(10,11)

19. $-15<\frac{3(x-2)}{5}\leq 0 \Rightarrow -15\times 5<3(x-2)<0 \Rightarrow -75<3(x-2)\leq 0$

$-\frac{75}{3}<\frac{(x-2)}{3}\leq\frac{0}{3} \Rightarrow -25<x-2\leq 0 \Rightarrow -23<x\leq 2$

Level - III

20.C	21.C	22.C	23.a-q,b-p,q,s;c-s,d-p	24.9	25.A,C
26.A,B,C	27.B	28.a-r,b-p,s;c-p,s;d-q	29.D	30.A	31.A
32.B	33.C	34.C,D	35.a-s,b-sc-p,r:d-r		

20. $x>y$ and $y>z \Rightarrow$ relation b/wx and $z \Rightarrow x>z$

21. Z < 0then xz < yz

22. Z < 0then x + z > y + z

1. Inequation → 3
2. Inequation → 1,2,4
3. symbolic form of two number = $x > 0 \rightarrow 4$
4. $3x - 2 < + \rightarrow 1$

24. shortestside = $x \Rightarrow$ longestside = $3x$

Third side = $3x - 2 \Rightarrow$ Givenperimeter = 61

$x + 3x + 3x - 2 = 61 \Rightarrow 7x = 63 \Rightarrow x = 9$ cm

25. $2x + 3 > 7 \Rightarrow 2x + > 4 \Rightarrow x > 2 \Rightarrow 2 < x - 1 \Rightarrow x + 4 < 10 \Rightarrow x < 6 - 2$ from 1&2

$2 < x < 6 \Rightarrow \therefore x = \{3,4,5\}$

26. $\frac{1}{a} < \frac{1}{b} \Rightarrow \frac{a+x}{b+x} < \frac{a}{b} \Rightarrow \frac{b^2}{a^2} < \frac{b}{a}$

27. $2x - 3 < x + 2 < 3x + 5 \Rightarrow 2x - 3 < x + 2 \Rightarrow 2x - 3 < x2 + 3$

X < 5 1

Again consider

$x + 2 < 3x + 5 \Rightarrow x - 3x < 5 - 2 \Rightarrow -2x < 3 \Rightarrow -x < \frac{3}{2}$

$-x < 1.5 \Rightarrow x < -1.5$

$-1.5 < x < 5 \Rightarrow x = \{-1,0,1,2,3,4\}$

28. $y + 3 < 9; y \in N \Rightarrow y < 9 - 3 \Rightarrow y < 6 \Rightarrow y = \{1,2,3,4,5\}$

2. $a > b$ and c < 0 $\Rightarrow \frac{a}{c} < \frac{b}{c}$_1

3. $a < b$ and c > 0 ⇒ ac < bc 4

4. solution set of z if $0 < 5, z \in W \Rightarrow z = \{1,2,3,4\}$

comprehension $-2 \leq$ X ≤ 2

29. First in Equation $-2 \leq x$

30. Second Inequation $x \leq 2$

31. Solution set of $Z = \{2, -1,0,1,2\}$

32. $\left|\frac{2(3-x)}{5}\right| < \frac{3}{5} \Rightarrow \frac{-3}{5} < \frac{6-2x}{5} < \frac{3}{5} \Rightarrow \frac{-3}{5} < \frac{6-2x}{5} < \frac{3}{5} \Rightarrow -3 < 6 - 2x < 3$

$-3 - 6 < -2x < 3 - 6 \Rightarrow -9 < -2x < -3 \Rightarrow \frac{9}{2} > x > \frac{3}{2} \Rightarrow \frac{3}{2} < x < \frac{9}{2} \Rightarrow \left\{\frac{3}{2}, \frac{9}{2}\right\}$

33. $|2x - 3| < 7 \Rightarrow -7 < 2x - 3 < 7 \Rightarrow -4 < 2x < 10 \Rightarrow -2 < x < 5$ $x = \{-1,0,1,2,3,4\}$; No. of solutions = 6

34. Given x + y + z = 1 and $(1 - x)(1 - y)(1 - z)$ is not greater than is nothing but $(1 - x)(1 - y)(1 - z) \leq$

Let $x = \frac{1}{3}, y = \frac{1}{3}, z = \frac{1}{3} \Rightarrow 1 - x = \frac{2}{3} \Rightarrow 1 - y = \frac{2}{3} \Rightarrow 1 - y = \frac{2}{3} \Rightarrow 1 - z = \frac{2}{3}$

$(1 - x)(1 - y)(1 - z) \leq \Rightarrow \left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right) \Rightarrow < \left(\frac{2}{3}\right)^3 \Rightarrow \leq \frac{8}{27} \Rightarrow \leq \left(\frac{64}{729}\right)^{1/2}$

35. 1)$|x| > 7 \Rightarrow -7 > x > 7 \Rightarrow x > 7$ 4

2)$2x + 3 > 17 \Rightarrow 2x > 14 \Rightarrow x > 7_4$

3) $|x + 5| > 3 \Rightarrow -3 > x + 5 > 3 \Rightarrow -3 - 5 > x > 3 - 5 \Rightarrow -8 > x > -2 \Rightarrow -8 > x$

$x < -8$ 3; $x > -2$ 1

4)$3x + 1 < -23 \Rightarrow 3x < -23 - 1 \Rightarrow 3x < -24 \Rightarrow x < -8$ 3